

Cognitive Skills and the Timing of First Births*

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Abstract

In the NLSY79, 63% of lowest-quartile AFQT women have a first birth before 22, against 18% at the top. The gap is in unwanted births, opens in high school, survives family background. A life-cycle model lets ability shift adherence—how reliably a method is used—beyond wages and schooling. Common adherence is rejected: it reproduces the early births, not that they are unintended. Equalizing adherence cuts first births by 21 by 17%, worth 15% of lifetime consumption to the lowest quartile; an equally effective price cut does least for the lowest quartile. A long-acting method at Colorado’s take-up reproduces the program’s estimated effect.

JEL codes: J13, J12, I21, J24, C61.

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1 Introduction

Early childbearing varies sharply with cognitive ability. In the National Longitudinal Survey of Youth 1979 (NLSY79), whose respondents were born between 1957 and 1964, 63% of women in the lowest quartile of the Armed Forces Qualification Test (AFQT) have had a first birth before age 22, against 18% in the highest. The gap is one of *timing*: first-birth hazards fall steeply with ability in the teen years and converge by the late twenties, so mean age at first birth differs by 5.5 years across quartiles, while the share of women who ever become mothers differs far less.

This paper asks whether the standard opportunity-cost channel can account for the gap, and finds that it cannot. I first show that the gap is not an artifact of education sorting or of contraceptive method selection. I then estimate a dynamic life-cycle model of schooling, work, marriage and contraceptive method choice in which cognitive ability can shift *adherence*—how reliably a chosen method is used. The data reject the version of the model in which adherence is common across ability groups, even when every other channel—wages, schooling, marriage and ability-specific preferences for motherhood—is free to adjust. That version can reproduce how many low-ability women have an early birth, but not that these births are unintended: the share of women with an unintended first birth by age 21 is 22 percentage points higher in the lowest ability quartile than in the highest (21 in the estimated model), and without an ability difference in adherence the gap shrinks to 13 points. Matching the data requires an ability shifter in the mapping from a chosen method to realized births—a fertility-control channel absent from existing structural models of fertility—and the welfare gains from acting on it are concentrated among the least advantaged women.

The question matters because early childbearing reduces women’s education, labor supply and lifetime earnings (Adda et al., 2017), delayed motherhood raises children’s human capital (Miller, 2009), and financially constrained young parents under-invest in their children (Caucutt and Lochner, 2020). Because early births concentrate among disadvantaged women, ability-dependent fertility timing is also a channel through which inequality is transmitted across generations (Aizer et al., 2022; Seshadri and Zhou, 2022). About 40% of U.S. pregnancies were unintended as of 2011 (Finer and Zolna, 2016), and falling birth rates

among young women without a high-school degree are the largest contributors to the recent fertility decline (Doepke et al., 2023; Kearney and Levine, 2025; Kearney et al., 2022; Onofri et al., 2026). Whether policy should act on the incentive to avoid a birth or on the ability to do so depends on which constraint binds.

The reduced-form evidence points to fertility control. The gap consists of births women report not having wanted at the time: by age 41, women in the lowest and highest ability quartiles are about equally likely to have had a wanted first birth, and the whole difference is in unwanted ones. Ability predicts the two margins of birth risk a woman controls—whether she uses a method and how well it works—and not the one she does not, the birth rate of non-users. Women in the lowest quartile are 14.7 percentage points less likely than those in the highest to be using any contraceptive method, and among women using the same method their annual birth rate is about 60 percent higher than in the highest quartile (3.6 against 2.2 percent, holding education, parity and age fixed). Among women using no method, by contrast, birth rates do not vary with ability.

The gap is not education sorting: it opens at ages 14–17, when 92 percent of childless girls are still in high school, and it survives holding their grade and expected schooling fixed. And it is not the choice of method: low-ability women who contracept rely more on the pill, not less, and holding the method fixed widens the gap, from 1.2 to 1.4 points in the same sample.

The paper models live births and abstracts from abortion. Differential abortion cannot *generate* the gradient: doing so would require top-quartile women to terminate 2.04 times as many pregnancies as bottom-quartile women by age 25, against an observed ratio of 1.18 (see also Woodley of Menie et al., 2017). Because abortion is if anything more common at the top, the live-birth gap the model targets is somewhat wider than the conception gap beneath it, and a model that maps adherence into live births attributes to fertility control about a fifth more gradient than conception data alone would support.¹

To interpret these patterns, I build a life-cycle model in which cognitive ability is an early,

¹Reported abortions per woman by age 25 *rise* weakly across AFQT quartiles, from 0.160 to 0.189; the income gradient in abortion that Jones and Jerman (2017) document does not transfer to cognitive ability in this cohort.

persistent trait² and women jointly choose schooling, labor supply, experience, marriage and—while childless and at risk—a contraceptive method from a menu of known efficacy. The innovation is in what follows that choice: a method’s failure rate depends on adherence, the reliability with which it is used, which is what separates perfect-use from typical-use failure rates ([Trussell, 2011](#)), and cognitive ability shifts adherence in addition to its effects through schooling and wages. The model is also held to whether each birth was wanted: the value of a birth varies across otherwise identical women, and the estimation targets the share of first births that were wanted by age. This is what gives the test its power: when adherence does not vary with ability, the model reduces to the opportunity-cost explanation, which can still produce early births among low-ability women, but only as wanted births, which the data say they are not.

Three counterfactuals sharpen the policy reading. Equalizing adherence to high-ability levels lowers first births by age 21 by 17% and before 18 by 28%, and its welfare gains accrue to the three lowest quartiles and not at all to the highest. A uniform price cut chosen to lower first births by 21 by the same 17%—89 percent of the way to making both methods free—has gains that go to the top of the ability distribution: most of what it is worth is a transfer to women already buying a method, and even net of that transfer the highest quartile gains more than the lowest and the lowest gains least of all. Adding a long-acting reversible contraceptive (LARC), which removes the need for daily adherence, to the menu at a price that reproduces the take-up of Colorado’s program lowers first births by 21 by 7%, within about a point of the program’s estimated effect ([Lindo and Packham, 2017](#)), with the largest gains in the lowest quartile and the smallest in the highest; offered at the pill’s cost, so that the two clinical methods differ only in the adherence margin, it is taken up by 34 percent of users and lowers first births by 21 by 8%, still short of the 17% from closing the adherence gap. The shortfall is take-up, not technology: the method removes the adherence margin only for the women who adopt it. This is the distribution of effects reported in evaluations of LARC programs, which are concentrated among younger and poorer women ([Lindo and Packham, 2017](#); [Secura et al., 2014](#)). These numbers are for a

²Measured cognitive ability is highly stable from late adolescence through adulthood ([Almlund et al., 2011](#); [Heckman et al., 2006](#)).

cohort that contracepted before long-acting methods were available, and the adherence and free long-acting experiments are upper bounds for it, not forecasts for women choosing among today’s methods. What carries over is the mechanism: wherever effectiveness depends on what the user does, differences in the capacity to control fertility produce differences in birth timing that the price of contraception cannot undo.

The paper contributes to the structural literature that, building on [Becker \(1960\)](#) and [Willis \(1973\)](#), endogenizes birth timing jointly with schooling, work and marriage ([Adda et al., 2017](#); [Caucutt et al., 2002](#); [Francesconi, 2002](#); [Heckman and Walker, 1990](#); [Hotz and Miller, 1988](#); [Keane and Wolpin, 2010](#); [Sheran, 2007](#); [Wolpin, 1984](#)). Closest are the models in which fertility control is imperfect: [Choi \(2017\)](#) adds conception risk and abortion, [Ejrnæs and Jørgensen \(2020\)](#) model family planning under uninsurable income risk, and [Amador \(2017\)](#) the policy environment for abortion and contraception. In all of them the mapping from fertility-control behavior to conception is common across women of the same education and age. Letting cognitive ability shift that mapping is what is new here, and the nested test establishes that the channel is required rather than merely permitted: the career-cost incentives of [Adda et al. \(2017\)](#) and education-specific contraception costs are both present in the estimated model, and neither reproduces the gradient without it.

The paper also joins two literatures that have run in parallel. One documents that cognitive and noncognitive skills predict early parenthood ([Fe et al., 2022](#); [Heckman et al., 2006](#); [Shearer et al., 2002](#)) but treats the channel as a black box: [Fe et al. \(2022\)](#) find that cognitive ability at age eight lowers the probability of being a parent by 25 and expect the effect to run through education. The other studies fertility control: how the price and availability of contraception shape births and human-capital investment ([Bailey, 2006](#); [Bailey et al., 2023](#); [Goldin and Katz, 2002](#); [Kearney and Levine, 2009](#); [Musick et al., 2009](#)), how teen childbearing responds ([Di Nola et al., 2025](#); [Kearney and Levine, 2012, 2015](#)), and, closest to this paper, that more schooled couples use more efficiently the methods that leave room for misuse ([Rosenzweig and Schultz, 1989](#)) and that typical-use failure rates within a method vary with age, income and education ([Kost et al., 2008](#); [Sundaram et al., 2017](#)). Health economics supplies the framing: schooling and ability raise the output obtained from a given health input ([Grossman, 2006](#); [Michael, 1973](#); [Rosenzweig and Schultz, 1989](#)), patient

self-management explains part of the socioeconomic gradient in health, and technologies that depend on the user’s effort widen that gradient while technologies that do not compress it (Goldman and Lakdawalla, 2005; Goldman and Smith, 2002). The two literatures measure different skills, so I ask whether the gradient here is the other one in disguise: the Rotter and Rosenberg scales, which the NLSY79 fields before its contraception modules open, leave 89–105 percent of every reduced-form coefficient standing (Online Appendix OA.5.3). The within-method evidence here shows where the two meet: skills act not only on the incentive to contracept but on how well contraception works, within education, and the model makes that channel a parameter that policy can act on, in the way a regimen that does not depend on the patient acts on adherence.

2 Data and Motivating Evidence

This section describes the data and shows that the ability gap in early first births is a gap in unwanted births, that it opens before schooling diverges, and that it persists within contraceptive method.

2.1 Data Description

The NLSY79 follows a nationally representative cohort born between 1957 and 1964 who were ages 14–22 at the first interview in 1979, with longitudinal records of schooling, employment, earnings, partnerships and fertility that cover their reproductive and working lives. Cognitive ability is proxied by the AFQT, a composite of the Armed Services Vocational Aptitude Battery administered early in the panel (e.g., Heckman et al., 2006; Neal and Johnson, 1996). The analysis uses the 5,820 women with a valid AFQT score whose schooling histories the model can represent.³

The within-method analysis uses the seven waves in which the survey records both contraceptive method and pregnancy wantedness—1982, 1984, 1985, 1986, 1988, 1990 and 1992—when the cohort is aged 18–35. The outcome is a contraceptive failure: a birth in the window counts as one only if the woman reports that she had not wanted the pregnancy at the time,

³Online Appendix OA.1.3 documents the estimation samples and the quartile cut-points.

since a woman who stopped using her method in order to conceive has not had it fail.⁴ Births are measured in a fixed 12-month window from July of year t through June of $t + 1$, constructed from child birth dates; because interviews were fielded between late January and mid-August, the method reported at the wave- t interview precedes the outcome. Methods are grouped into three tiers by typical-use efficacy—oral contraceptives, barrier methods and behavioral methods—with dual users assigned to the most effective tier. LARC and sterilization are excluded.⁵

2.2 Wantedness and the Nature of the Ability Gap

Figure 1 shows what kind of births make up the ability gap. For every woman in the sample it records whether she had a first birth by ages 17, 21 and 41, split by the wantedness she reported for that birth on its own pregnancy record: *unwanted* if it came too soon or was not wanted at all, *wanted* otherwise, including births conceived after stopping contraception in order to become pregnant. Low-ability women are not more likely to ever have a wanted first birth—by age 41, 46.2 percent of women in the lowest quartile and 48.7 percent in the highest have had one—but they are nearly twice as likely to have had an unwanted one, 40.1 against 21.3 percent. The lifetime difference in first births between the bottom and top quartiles is therefore a difference in unwanted births. It is also front-loaded: more than half of the gap in first births by age 21 has opened by 17, and 62 percent of the gap at 17 is unwanted births; at 21 the share is 51 percent, because low-ability women also have their wanted first births sooner.⁶

2.3 Ability and the Three Margins of Birth Risk

Two alternatives could produce the gradient without any difference in how well women use contraception: education sorting and contraceptive method selection. I take them in turn.

⁴Online Appendix OA.1.7 gives the full classification rule and the birth-record and pregnancy-record items it draws on.

⁵IUD uptake was negligible in this cohort after the Dalkon Shield recall, and sterilization removes a woman from the risk pool irrespective of ability. Online Appendix OA.1 gives the full construction.

⁶The figure counts women rather than births. The usual summary, the unintended share of births, divides by births, which also move with ability; among women who have a first birth at a given age the unintended share does not differ by ability (Online Appendix OA.3).

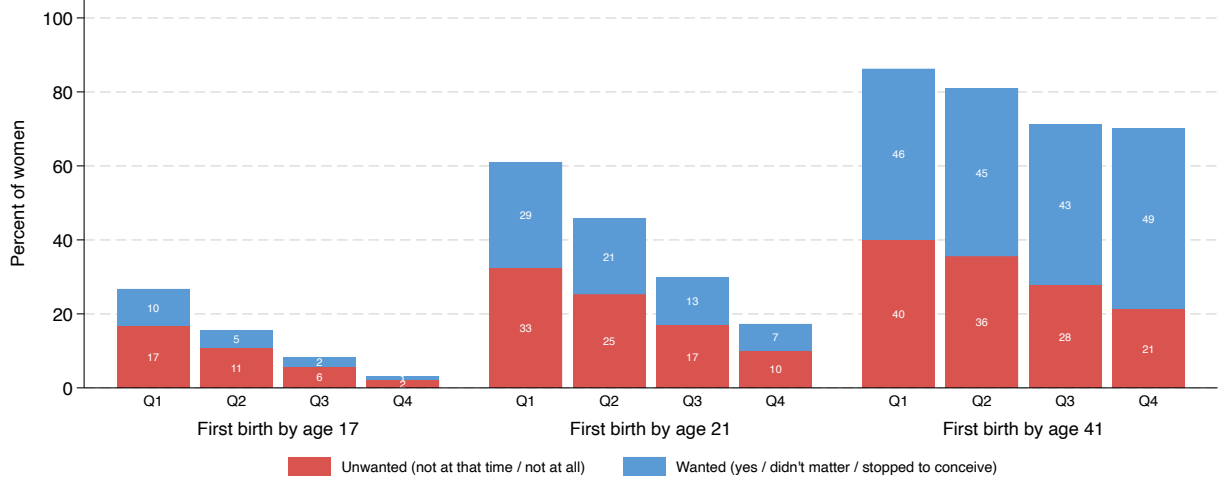


Figure 1. Wanted and unwanted first births by AFQT quartile. Share of women with a first birth by the age shown, by whether she reports having wanted the pregnancy at the time. NLSY79; numbers and standard errors in Online Appendix Table OA.6.

Is the ability gap an artifact of education sorting? AFQT predicts educational attainment, and education is associated with fertility through channels the model will interpret as opportunity costs.⁷ Comparing women of different ability with the same completed schooling does not work for the births that matter most: more than half of the gap in first births by age 21 has opened by 17 (Figure 1), while girls in every quartile are still in school, and a teen birth lowers the schooling a woman goes on to complete (Ashcraft et al., 2013; Fletcher and Wolfe, 2009; Hotz et al., 2005; Klepinger et al., 1999), so completed schooling is a “bad control” (Angrist and Pischke, 2009). Following the literature on first-birth hazards (Blossfeld and Huinink, 1991; Upchurch and McCarthy, 1990), I measure schooling as it stands before the conception, which the NLSY79’s annual interviews allow from age 14. For each childless woman aged 14–20 at an annual interview t , I estimate

$$F_{it} = \sum_{q=2}^4 \beta_q \mathbb{1}[\text{AFQT}_i = q] + \mathbf{s}'_{it} \boldsymbol{\lambda} + \mathbf{z}'_i \boldsymbol{\gamma} + \alpha_{a(i,t)} + \delta_t + \varepsilon_{it}, \quad (1)$$

⁷The quasi-experimental evidence on the causal effect of schooling on fertility is mixed (Fort et al., 2016; McCrary and Royer, 2011). Nothing below turns on a causal reading; the exercise asks whether the ability gradient survives when schooling is held fixed.

where F_{it} equals one if her first child is born in calendar year $t + 1$, a birth conceived, almost always, after the interview; $\mathbf{s}_{it}$ is her schooling at the interview, enrollment status and highest grade completed, which precedes the conception; $\mathbf{z}_i$ holds what she reported in 1979, before any of these births: the schooling and the number of children she expected, her Rotter locus-of-control score and her parents' schooling; $\alpha_{a(i,t)}$ and δ_t are age and year fixed effects, and standard errors are clustered by woman.

Table 1 reports the estimates. At ages 14–17, 92 percent of the woman-years are girls enrolled in high school: schooling has not yet diverged, and the gradient is already there. Among them, the top-quartile coefficient on the annual probability of a first birth is -9.4 percentage points (s.e. 1.1), against a bottom-quartile rate of 10.4 percent, and -6.8 (s.e. 0.9) for unintended first births. Holding enrollment and grade fixed leaves the gradient where it was (columns 1–2). The 1979 expectations and family background remove the gradient in intended first births altogether (-0.8 , s.e. 0.7, in column 4) but only about a quarter of the gradient in unintended ones, which stays at -5.3 (s.e. 1.0): girls who planned more schooling planned their births later, and family background does not account for the unwanted early births of low-ability girls. Adding race and ethnicity as well leaves that gradient at -3.6 (s.e. 1.0) on a bottom-quartile rate of 7.1 percent (column 5). Only at 18–20, when schooling begins to differ, does schooling to date absorb part of the gradient, from -10.3 to -7.0 (columns 6–7).⁸

Among adults, whose schooling is largely complete, the gradient also survives within education groups, though it is smaller, and what limits it there is not family background but race. Parents' schooling and the family's 1979 income and poverty status leave it where it is; a race indicator removes it.⁹ Race is not a confounder to be partialled out here: the AFQT is measured before labor-market entry, so the between-group component a race indicator removes is what the premarket-skill literature reads as skill (Carneiro and Heckman, 2003; Neal and Johnson, 1996), and the gradient the model is estimated on holds within every

⁸A prospective design in the spirit of Levine and Painter (2003)—the 1,803 girls who were 14–16 and childless in 1979, with only what was measured then as controls—gives the same answer: the gradient in unintended first births by 21 is -22.5 percentage points with age controls alone and -20.1 (s.e. 3.3) with the full 1979 set, while the gradient in intended first births falls from -14.0 to -5.3 (Online Appendix OA.4).

⁹Online Appendix OA.4 reports the within-education estimates; Online Appendix OA.5.2 adds the two blocks of controls separately and decomposes the movement variable by variable; Online Appendix OA.5.5 estimates the gradient within each race and ethnicity group.

Table 1. The Ability Gradient in First Births Before Schooling Is Realized (LPM, 1979–1994)

	Ages 14–17					Ages 18–20	
	All		Enrolled in high school			All	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Any first birth: Q4 minus Q1	-9.5 (1.0)	-9.3 (1.1)	-9.4 (1.1)	-6.1 (1.2)	-5.0 (1.2)	-10.3 (1.0)	-7.0 (1.1)
Q1 mean	11.4	11.4	10.4	10.4	10.4	14.1	14.1
Unintended first birth: Q4 minus Q1	-6.1 (0.8)	-6.6 (0.9)	-6.8 (0.9)	-5.3 (1.0)	-3.6 (1.0)	-4.0 (0.7)	-2.9 (0.8)
Q1 mean	7.3	7.3	7.1	7.1	7.1	6.1	6.1
Intended first birth: Q4 minus Q1	-3.4 (0.6)	-2.7 (0.6)	-2.5 (0.6)	-0.8 (0.7)	-1.3 (0.8)	-6.2 (0.7)	-4.1 (0.8)
Q1 mean	4.1	4.1	3.3	3.3	3.3	8.0	8.0
Enrollment status at t		✓					✓
Highest grade completed at t		✓	✓	✓	✓		✓
1979 expectations, Rotter, parents				✓	✓		
Race and ethnicity					✓		
Woman-years	5,521	5,521	5,055	5,055	5,055	8,617	8,617
Women	2,418	2,418	2,268	2,268	2,268	3,643	3,643

Notes: Top-minus-bottom AFQT quartile difference in the probability of a first birth in the following year, in percentage points, from equation (1) on childless women aged 14–17 (columns 1–5; columns 3–5 enrolled in high school) or 18–20 (columns 6–7). Standard errors clustered by woman in parentheses. Controls are defined in Online Appendix OA.4.

race and ethnicity group in any case. Most of the gap opens in the teen years, where education cannot account for it and where, as column 5 of Table 1 shows, the gradient survives family background and race together. What these regressions cannot say is how much of the gradient schooling would explain if it worked through every channel at once; that is what the structural model is for, and the test of Section 5 asks whether an ability difference in adherence is still needed once schooling has been given every channel the model allows.

Method selection. If high-AFQT women adopted more effective methods, the gradient could reflect the composition of methods rather than per-use performance. They do not: conditional on using a method, low-ability women choose the more effective one. In a multinomial logit on women using reversible contraception, with AFQT quartile, education

group, parity and wave fixed effects as regressors,¹⁰ the predicted probability of using the pill is 76.5% in the bottom quartile and 62.5% in the top; that of a barrier method is 15.5% and 29.4%.¹¹ Low-ability women concentrate on the pill, which fails less often in typical use than a barrier method, and still have more births: method selection works against the ability gradient rather than producing it.

Within-method birth hazard. For each method group $n \in \{\text{pill, barrier, behavioral}\}$, I estimate the linear probability model

$$B_{it} = \alpha_n + \sum_{q=2}^4 \beta_{nq} \mathbb{1}[Q_i = q] + \mathbf{x}'_{it} \boldsymbol{\gamma}_n + \delta_t + \varepsilon_{it}, \quad (2)$$

where B_{it} equals one if woman i had a live birth in the 12-month window following wave t , Q_i is her AFQT quartile (quartile 1 omitted), $\mathbf{x}_{it}$ contains education group, parity (0/1/2+ prior births), age-at-wave indicators, marital status at the previous interview and a window-length indicator,¹² and δ_t are wave fixed effects; the controls are the state variables on which the model's birth probability depends.¹³ Standard errors are clustered by respondent, and a pooled specification adds method-group fixed effects. If higher-ability women have fewer births only because they choose more effective methods, ability should not predict failure within a method, $\beta_{nq} = 0$; if they also use a given method more reliably, $\beta_{nq} < 0$.

The gradient does not disappear within method (panel A of Table 2).¹⁴ In the pooled

¹⁰Online Appendix OA.6.2 reports the full multinomial logit estimates.

¹¹The comparison is among users, not among all women of reproductive age, which is why the gradient runs opposite to NSFG studies that find pill use rising in education (e.g., Sundaram et al., 2017); Online Appendix OA.6.1 reconciles the two (Mosher, 1990; Piccinino and Mosher, 1998).

¹²The controls exclude race and ethnicity: AFQT is measured before labor-market entry and absorbs the part of group differences that runs through skills, so conditioning on race would remove part of the variation in ability rather than a confounder (Carneiro and Heckman, 2003; Neal and Johnson, 1996). Online Appendix OA.5.5 reports the estimates within each group. The within-method estimates below are a statement about the pooled population rather than a within-group fact; the structural test of Section 5.3 does not rest on them, since the eight within-method failure moments carry 6 percent of the test statistic, against 68 percent for the first-birth timing and unwanted-birth moments, whose gradients hold within every group.

¹³Age enters separately from the wave because birth cohorts span eight years within a wave. Marital status is measured at the interview before the wave, so pregnancy-induced marriage cannot move it. Online Appendix OA.5.1 reports the full estimates, the regressions with the NLSY79 sampling weights and without the age and marital-status controls, and Online Appendix OA.5.2 the regressions that add family background and race separately.

¹⁴Online Appendix Table OA.10 reports the full estimates.

specification, women in the second, third and fourth AFQT quartiles are 1.3, 1.1 and 1.4 percentage points less likely to have a birth in a year than women in the first ($p < 0.01$, $p < 0.05$ and $p < 0.01$), against a rate of 3.6 percent in the first quartile: the top quartile’s failure rate is about two-fifths lower. Among pill users the top-quartile coefficient is -1.5 points ($p < 0.01$) on a mean rate of 1.7 percent—notable because daily ingestion is a narrow behavioral margin on which cognitive ability can affect consistency within a single technology. Among barrier users the three coefficients are -2.4 , -2.4 and -2.5 pp ($p < 0.10$, $p < 0.10$ and $p < 0.05$), and among behavioral users none is significant.¹⁵

Where ability enters. The accounting used for teen births (Kearney and Levine, 2012; Santelli et al., 2007) writes a woman’s annual chance of a birth as exposure times the share of the time she uses no method times the non-user birth rate, plus the share she uses one times its failure rate. Table 2 shows where ability enters. Not through fecundity or exposure: among women using no method the birth rate is flat in ability, 2.4 percent a year in the lowest quartile against 2.5 in the highest, with a regression-adjusted difference of $+0.4$ points (panel B), and intercourse is more frequent at the top, not less.¹⁶ It enters through the two margins of fertility control (panel A): low-ability women are less likely to use any method—62.1 percent of woman-waves in the lowest quartile against 75.9 in the highest—and, when they use one, more likely to see it fail, 3.6 births per hundred user-years against 2.2. The first margin is a choice, over the cost of a method and the value of avoiding a birth; the second is adherence. Non-use contributes more to the gradient: 85.1 percent of the lowest quartile’s unintended first births were conceived with no method in use, against 53.0 percent in the highest (panel C). The model carries both margins with different parameters: the price of a method and the value of a birth govern use, and adherence governs failure.

Table 2. Three Margins of Birth Risk by AFQT Quartile

	Q1 level	Q2 difference from Q1	Q3	Q4
<i>A. Fertility control</i>				
Using any method (% of woman–waves)	62.1	6.5 (1.2)	10.7 (1.2)	14.7 (1.3)
Births among users (% per year)	3.6	−1.3 (0.4)	−1.1 (0.4)	−1.4 (0.4)
Pill users	3.1	−1.1 (0.5)	−1.1 (0.5)	−1.5 (0.5)
Barrier users	5.9	−2.4 (1.3)	−2.4 (1.3)	−2.5 (1.3)
Behavioral users	3.9	−0.6 (1.9)	1.5 (2.1)	1.7 (2.3)
<i>B. Exposure and fecundity</i>				
Births among non-users (% per year)	2.4	0.0 (0.6)	0.1 (0.6)	0.4 (0.7)
<i>C. Where unintended first births come from</i>				
Conceived with no method in use (% of unintended first births)	85.1	−5.4 (2.4)	−21.7 (2.9)	−32.1 (3.3)

Notes: The Q1 column reports the lowest quartile’s level; the Q2–Q4 columns report each quartile’s difference from Q1, with robust standard errors clustered by respondent in parentheses. Panel A: share of woman–waves aged 18–35 using any method, with the quartile coefficients of a linear probability model with the controls of equation (2); the within-method birth rate from the pooled specification of equation (2), and, by method, the quartile coefficients of the same specification estimated within pill, barrier and behavioral users, with the raw Q1 birth rate of each group as the level (full estimates in Online Appendix Table OA.10). Panel B: annual first-birth rate among childless non-users aged 18–33, annualized over the interval to the next wave as in the within-method moments; the differences are regression-adjusted. Panel C: share of unintended first births conceived with no method in use; the differences are raw, with heteroskedasticity-robust standard errors. Online Appendices OA.3.1, OA.6.3 and OA.3.

2.4 Life-Cycle Facts

The model must also match the broader life-cycle patterns that are its estimation targets. Table 3 reports them: first-birth probabilities by model period—the four-year age bins on which the model is estimated—and AFQT quartile, among women childless at the start of each bin (Panel A); the cumulative shares that are the estimation targets (Panel A’); and age at first birth, marriage at first birth and completed fertility (Panel B). The remaining descriptive tables are in Online Appendix OA.2. The gradient is steep early and attenuates with age: at

¹⁵Assigning dual users to the most effective tier and within-window method switching both attenuate the gradient toward zero. Across the conventions for classifying a birth without a pregnancy record (Online Appendix OA.5.4) the top-quartile coefficient ranges from -0.014 to -0.022 and is significant under each; unconditional on method it is -3.0 points (Online Appendix OA.5.6).

¹⁶Online Appendix OA.6.4 reports coital frequency by ability quartile and the exposure-margin estimates.

Table 3. Fertility Timing and Outcomes by Ability Quartile

	Ability Quartile			
	Q1 (Low)	Q2	Q3	Q4 (High)
<i>Panel A: Conditional First-Birth Probability by Model Period (%)</i>				
14–17	28	16	9	3
18–21	48	38	25	16
22–25	38	36	28	22
26–29	25	25	26	29
30–33	18	17	16	26
34–37	10	11	10	12
38–41	2	4	4	5
<i>Panel A': Share with a First Birth by End of Period (%)</i>				
by 17	28	16	9	3
by 21	63	48	31	18
by 25	77	67	50	36
by 29	82	75	63	55
by 33	85	79	69	66
by 37	87	81	72	71
by 41	87	82	73	72
<i>Panel B: Age at First Birth and Completed Fertility</i>				
Age at first child	20.17	21.67	23.51	25.63
Married at first birth	0.48	0.66	0.80	0.88
At least one child by age 40	0.87	0.82	0.73	0.72

Notes: NLSY79 estimation sample; age bins are the model's four-year periods. Panel A: probability of a first birth within the period among women childless at its start. Panel A': cumulative share with a first birth by the end of the period, the moment targeted in estimation. Panel B: mean age at first birth, share married at first birth, and share with a child by age 40.

14–17, 28% of women in the lowest quartile have a first birth against 3% in the highest; at 18–21, 48% against 16%; from 22–25 onward the conditional probabilities converge. Mean age at first birth rises from 20.2 to 25.6 across quartiles, and women in the highest quartile are far more likely to be married at their first birth (0.88 against 0.48), yet the share with at least one child by age 40 falls only from 0.87 to 0.72: the pattern is postponement rather than avoidance of motherhood. Ability also predicts schooling steeply—51% of the highest quartile complete college against 3% of the lowest, and by age 21 roughly 73% of high-school dropouts have had a first birth against 8% of college graduates—and higher-ability women face steeper returns to experience and accumulate more of it, so a career interruption costs them more.¹⁷ Premarital childbearing is associated with lower marriage rates and lower

¹⁷The unconditional experience gradient (57% against 24% log wage growth after five years in the highest and lowest quartiles) runs largely through education: within a completed-education group, early-career growth differs little by ability (Online Appendix OA.10). Online Appendix OA.10.5 asks how much larger the within-education gradient would have to be to account for the fertility gradient on its own.

spousal earnings, especially among less educated women.¹⁸ The wage process, experience accumulation and the marriage market must therefore generate ability-graded incentives to delay that are consistent with these patterns before any fertility-control channel is invoked.

3 Model

I develop a dynamic life-cycle model of fertility timing with imperfect fertility control, in the tradition of models that jointly endogenize schooling, labor supply, marriage and fertility (Adda et al., 2017; Attanasio et al., 2008; Ejrnæs and Jørgensen, 2020; Keane and Wolpin, 2010). Its innovation is a fertility-control technology in which women choose among contraceptive methods of known efficacy and cognitive ability shifts adherence—how reliably the chosen method is used.

Costs and benefits of childbearing. Each woman can have at most one child, which saves state space while retaining the margin that drives the ability gradient, the timing of the *first* birth. A birth delivers a “warm glow” from parenthood and utility from child quality, which depends on a one-time monetary investment. It imposes three costs: a *goods cost*, through the household’s consumption-equivalence scale and the investment; an *opportunity cost*, because a birth during the schooling years raises the psychic cost of continuing and thereafter raises the disutility of work, reducing labor supply and slowing experience accumulation; and a *fertility-control cost*, because avoiding a birth requires choosing and using a method, which is costly and imperfectly effective. Assets supply the intertemporal margin under a no-borrowing constraint.¹⁹

Because ability raises wages and returns to experience, high-ability women face larger opportunity costs of a birth, and education raises the baseline effectiveness of fertility control (Rosenzweig and Schultz, 1989). To these two channels the model adds ability-dependent adherence and an ability-specific intercept in the utility of a child, so that heterogeneous preferences for early motherhood compete with fertility control on equal terms.

¹⁸Online Appendix OA.2.2 reports the marriage and spousal-earnings estimates.

¹⁹The one-period child could in principle reward delay artificially. Diminishing returns in the child-quality technology, discounting of the utility of parenthood and the finite fertility window (ages 14–41), which makes excessive delay risk involuntary childlessness, limit that force.

Time is discrete and each period is four years. Women enter at age 14 with cognitive ability θ and assets $a_1 = 0$. They remain fertile through age 41, work until 65, and are retired from 66 to 81, receiving Social Security income that depends on education. The decision unit is the household: a single adult before marriage and a two-adult unit that pools income after it; marriage is absorbing. The model abstracts from income uncertainty.

3.1 State variables, choices, and timing

The household state at the beginning of period t is

$$\Omega_{it} = \{a_t, \theta_i, e_t, x_t, m_t, k_t, m_k\},$$

where a_t denotes assets; $\theta_i \in \{1, 2, 3, 4\}$ is the cognitive-ability quartile (1 lowest, 4 highest); $e_t \in \{HSD, HS, C\}$ is education attainment/status; x_t is accumulated labor-market experience; $m_t \in \{0, 1\}$ is marital status; $k_t \in \{1, 2, 3\}$ records childbearing status (1: no prior birth, 2: first birth occurs in period t , 3: first birth occurred in an earlier period); and $m_k \in \{0, 1\}$ records marital status at the first birth (only relevant when $k_t \neq 1$).

Within a fertile period, single women may meet and accept a partner, childless women choose a contraceptive method n_t and then face stochastic conception, and households choose labor supply l_t , consumption c_t , saving a_{t+1} and, if a birth occurs, child investment i_t . In the two schooling periods the continuation decision—continue high school or drop out at $t = 1$, graduate from college or drop out at $t = 2$ —comes *after* the fertility outcome, so that a birth can raise the cost of staying in school; high-school graduates also choose whether to enter college at the end of $t = 1$. Enrolled women do not marry; college students do not work and receive a student allowance net of tuition. After fertility ends the problem is a labor–savings problem, and in retirement labor supply is zero.²⁰

²⁰Online Appendix [OA.7.1](#) shows the life cycle and the within-period sequencing (Figures [OA.3](#) and [OA.4–OA.5](#)) and gives the complete Bellman equations.

3.2 Dynamic Household Problem and Value Functions

I present the parts of the problem that matter most for the paper; Online Appendix [OA.7](#) states the full recursion stage by stage.²¹

Marriage. Marriage pools resources: spousal income offsets the earnings a mother forgoes, so a married woman values avoiding a birth less than an otherwise identical single woman. If single ($m_t = 0$), the woman meets a potential husband with probability $\mu(e_t, t)$, which depends on her education and takes one value within each age group,²² and accepts if the continuation value under marriage is higher. Writing $\Omega_{it}(m)$ for the state with m_t set to m ,

$$V_t^1(\Omega_{it}) = \begin{cases} \mu(e_t, t) \max\{V_t^2(\Omega_{it}(1)), V_t^2(\Omega_{it}(0))\} + (1 - \mu(e_t, t)) V_t^2(\Omega_{it}(0)), & \text{if } m_t = 0, \\ V_t^2(\Omega_{it}), & \text{if } m_t = 1. \end{cases}$$

Contraception and first-birth risk. Only childless women make a fertility-control decision, i.e. when $k_t = 1$ and $t \leq T_F$. They choose a contraceptive method $n_t \in \{0, 1, 2\}$ —no method, a low-efficacy method (barrier or behavioral), or the pill—each of which implies a probability of a first birth in period t , $p_n(\theta_i, e_t, t)$, that depends on age, education and, through adherence, ability; Section [3.4](#) specifies it. Method n carries a utility cost κ_n ($\kappa_0 = 0$) and a type-I extreme-value taste shock $\sigma \varepsilon_{nt}$, and the value of a birth carries a transitory shock $\sigma_\omega \varepsilon_{it}^\omega$, $\varepsilon_{it}^\omega \sim N(0, 1)$ i.i.d. across women and periods, realised before the method is chosen. Only a woman who is sexually active is at risk of conception: with probability $a_{\theta t}$, the exposure share, she chooses a method, and otherwise she continues childless. Hence

$$V_t^2(\Omega_{it}) = V_t^{3,j=1}(\Omega_{it}) + a_{\theta t} \mathbb{E}_{\varepsilon, \varepsilon^\omega} \max_{n \in \{0, 1, 2\}} \left\{ p_n(\theta_i, e_t, t) [\Delta V_t(\Omega_{it}) + \sigma_\omega \varepsilon_{it}^\omega] - \kappa_n + \sigma \varepsilon_{nt} \right\},$$

$$\Delta V_t(\Omega_{it}) \equiv V_t^{3,j=2}(\Omega_{it}) - V_t^{3,j=1}(\Omega_{it}),$$

where ΔV_t is the value of a first birth relative to remaining childless and ε_{it}^ω is the idiosyncratic component of that value—a partner, a circumstance, a change of mind—that the survey’s

²¹Superscripts on the value function (V^1, V^2, V^3) index within-period sub-stages, not time periods.

²²The groups are 18–21, 22–25, 26–33 and 34–41 for women without college, and 22–25, 26–33 and 34–41 for college women, who do not marry while enrolled.

wantedness answer reflects. A woman who values a birth ($\Delta V_t \geq 0$) chooses no method with high probability; a woman who does not trades each method's efficacy, $p_n |\Delta V_t|$, against its cost κ_n . If $k_t \neq 1$, the household skips this stage and $V_t^2(\Omega_{it}) = V_t^{3,j=3}(\Omega_{it})$.

Working stage. After the fertility realization the household solves a labor–consumption–saving problem indexed by child status j : $j = 2$ if a first birth occurs in t , $j = 1$ if the woman remains childless, $j = 3$ if she gave birth earlier. The discrete labor choice $l_t \in \{0, 1\}$ carries a Type-I extreme value shock, so $V_t^{3,j}(\Omega_{it}) = \mathbb{E}_\varepsilon \max_{l \in \{0,1\}} \{v_t^{3,j}(\Omega_{it}, l) + \sigma_l \varepsilon_t(l)\}$ with

$$v_t^{3,j}(\Omega_{it}, l) = \max_{a_{t+1}, c_t, i_t \geq 0} \left\{ u(c_t) + [\psi_l(t, e) + \psi_{lk}(e) \mathbb{1}_{\{j=2\}}] \mathbb{1}_{\{l=1\}} + u_k(i_t) \mathbb{1}_{\{j=2\}} \right. \\ \left. + \beta \mathbb{E}[V_{t+1}(\Omega_{i,t+1}) \mid \Omega_{it}, l, a_{t+1}, j] \right\} \\ \text{s.t. } \phi_c(m_t, \mathbb{1}_{\{j=2\}}) c_t + a_{t+1} = (1 + r) a_t + y_t(\Omega_{it}, l) - i_t \mathbb{1}_{\{j=2\}}, \quad x_{t+1} = x_t + \mathbb{1}_{\{l=1\}}.$$

Two parameters govern the disutility of work. The baseline $\psi_l(t, e)$ varies by education and over four life stages, with a retirement term that raises the cost of work at 58–65.²³ The second, $\psi_{lk}(e)$, applies only while the child is present and is the child penalty on labor supply (Adda et al., 2017).

College. At the end of $t = 1$, teens who complete high school draw a Type-I extreme value shock and choose whether to enroll at $t = 2$, comparing the beginning-of-period values under each choice net of a non-pecuniary entry cost $\kappa_C(\theta, j)$ that varies with ability and with birth status.²⁴

3.3 Preferences, technologies and transfer system

Preferences over effective consumption. Utility is CRRA over effective consumption, $u(c_t) = c_t^{1-\gamma}/(1-\gamma)$. Household composition enters through the equivalence scale in the budget constraint,

$$\phi_c(m_t, k_t) = 1 + \phi_{ca} \mathbb{1}_{\{m_t=1\}} + \phi_{ck} \mathbb{1}_{\{k_t=2\}},$$

²³The stages are ages 14–29, 30–49, 50–53 and 54–65; the retirement term $\psi_{ret} \cdot \max\{t - 11, 0\}$ is common across education groups.

²⁴Online Appendix OA.7.6 states the college-age problem in full.

so that $\phi_c(m_t, k_t) c_t$ is the expenditure required to attain c_t , with $\phi_{ca} \geq 0$ the additional needs of a second adult and $\phi_{ck} \geq 0$ those of a child present in the household.

Preferences over child quality. If a first birth occurs, parents choose a one-time monetary investment i_t that increases child “quality.” Parental altruism enters as utility from child outcomes:

$$u^k(i_t) = \omega_{0,\theta} + \omega_1 i_t^{\omega_2},$$

where $\omega_{0,\theta}$ is the utility from having a child for a woman of ability θ , so that preferences over motherhood may differ by ability and compete with fertility control in explaining the timing gradient (Section 3.4), ω_1 scales the marginal value of investment, and $\omega_2 \in (0, 1)$ imposes diminishing returns and ensures an interior investment choice.

College psychic cost. College choices are disciplined by a student allowance w_C , tuition TC , and an ability-dependent psychic cost. Following the structural education literature, I model the psychic cost of college attendance as

$$\kappa_C(\theta, k_t) = \frac{\xi_C}{\theta^{\omega_C}} + \xi_\theta (\theta - 1) + \mathbb{1}_{\{k_t=2\}} \phi_{k,BAC},$$

where the first term captures the standard finding that psychic costs decline with cognitive ability, the second gives the estimation more freedom in how the cost falls across quartiles (as in [Abbott et al., 2019](#)), and the third is the additional burden of attending college while raising a child.²⁵ Three further terms make an early birth costly for schooling: a woman with a child pays an additional utility cost to stay in high school ($\phi_{k,HSD}$), to attend college ($\phi_{k,BAC}$), and to graduate rather than drop out ($\phi_{k,D}$); graduating also carries a cost ϕ_{grad} for everyone. The student allowance is $w_C a_\theta$, where w_C is estimated and a_θ scales it by the mean family income of quartile θ at ages 14–17 relative to the sample mean, measured in the NLSY79 and held fixed.²⁶ The model has no parental sector; the allowance is a reduced-form transfer rule, and indexing it on ability keeps the correlation between family resources and

²⁵Psychic costs of schooling are a standard ingredient in structural models of educational choice (e.g., [Abbott et al., 2019](#); [Keane and Wolpin, 2010](#)); unmeasured monetary costs are observationally equivalent to them.

²⁶Online Appendix [OA.1.3](#) documents the sample and the construction of every externally measured input.

ability from being loaded onto other parameters.²⁷

Progressive taxes and transfers. I adopt the parametric U.S. tax-and-transfer schedule in [Daruich and Fernández \(2024\)](#): gross annual household income $\tilde{y}^0$ (labor earnings, spousal earnings if married, schooling allowances when enrolled, or Social Security in retirement) maps to disposable income $\tilde{y} = \lambda(\tilde{y}^0)^{1-\tau} + T(m_t)$. Progressivity ($\tau > 0$) reduces the sensitivity of after-tax resources to gross income, and $T(m_t)$ is a reduced-form transfer floor varying by marital status.

3.4 Fertility control technology

Fertility is controlled imperfectly through the choice of a discrete method, whose efficacy is a property of the technology, used with an adherence that is a property of the user. The specification is motivated by the gap between “perfect use” and “typical use” failure rates, which is driven by inconsistent or incorrect use and varies systematically with user characteristics ([Black et al., 2010](#); [Rosenzweig and Schultz, 1989](#); [Trussell, 2011](#)).²⁸

Exposure. Not every childless woman is at risk of conception, and sexual debut comes later for high-ability women. Rather than let adherence absorb this margin, I measure it: $a_{\theta t}$ is the share of women in quartile θ who are childless at the start of period t and have had intercourse by its end, from the NLSY79 age-at-first-intercourse questions.²⁹ With probability $1 - a_{\theta t}$ the woman is not exposed, conceives with probability zero and uses no method. Exposure is data rather than a choice because it is not where the gradient is (Section 2.3) and because it is conservative for the estimate of adherence: the later debut of high-ability girls is credited to exposure, not to $c_{\theta g}$, whereas an abstinence margin responding

²⁷The rule follows [Keane and Wolpin \(2001\)](#) and [Johnson \(2013\)](#). Indexing the transfer rather than adding a borrowing constraint follows the evidence that short-run credit constraints explain little of the income gradient in attendance ([Carneiro and Heckman, 2002](#)), while family resources conditional on ability do matter ([Belley and Lochner, 2007](#)). With $a_{\theta} \equiv 1$ the model reduces to the common-allowance specification.

²⁸Related life-cycle models with imperfect fertility control use a continuous effort or a logit-type mapping for conception risk ([Choi, 2017](#); [Ejrnaes and Jørgensen, 2020](#)). Here the control object is a discrete method with an externally measured efficacy, and ability shifts the reliability with which the method is used rather than the cost of using it or the baseline risk.

²⁹Online Appendix [OA.1.3](#) reports the exposure shares by quartile and age.

to ΔV would compete with adherence at exactly the ages where no within-method moment exists. The cost is that the counterfactuals hold sexual activity fixed.³⁰

Methods and adherence. A childless woman of ability θ , education e and age group $g = g(t)$ chooses $n \in \{0, 1, 2\}$: no method, a low-efficacy method (condoms and other barrier methods, withdrawal, rhythm), or the pill. Let p_{be}^0 be the four-year probability of a first birth with no method,

$$p_{be}^0 = \left[\lambda_{\max} \frac{\lambda_{be}}{1 + \lambda_{be}} \right]_{\lambda_{\min}}^{\lambda_{\max}},$$

where $\lambda_{be} > 0$ are the odds that parameterize this *baseline risk*, indexed by education and by an age band b (14–25, 26–33, 34–41), and the truncation keeps the risk in $[\lambda_{\min}, \lambda_{\max}]$.³¹

A woman who uses method $n \geq 1$ perfectly faces the fraction ρ_n of that risk, the ratio of the method’s perfect-use failure rate to the no-method rate in Trussell (2011): $\rho_1 = 0.03/0.85$ for the low tier and $\rho_2 = 0.003/0.85$ for the pill. Two objects are estimated. The first is *adherence*, $c_{\theta g} \in (0, 1)$: the share of the time a woman of ability θ and age group g uses her method as prescribed.³² The second is the failure rate of a user who never adheres, $\rho_n^0 \in [\rho_n, 1]$, one per method and common to all women. The probability of a first birth for a user of method n is

$$p_n(\theta, g, e) = p_{be}^0 [(1 - c_{\theta g}) \rho_n^0 + c_{\theta g} \rho_n], \quad n \in \{1, 2\}, \quad p_0(\theta, g, e) = p_{be}^0. \quad (3)$$

The anchor ρ_n^0 is the counterpart of the typical-use rate, but it is estimated rather than imported, because the NLSY79 failure measure is a birth rate per year of *reported* use and not a first-year typical-use rate.³³ Self-reported use also understates lapses: with electronic monitoring, the share of pill users missing three or more pills a month is 30–51

³⁰For a model in which teen sexual activity is a choice, see Di Nola et al. (2025).

³¹The odds are ordered by education, $\lambda_{b,HSD} \geq \lambda_{b,HS} \geq \lambda_{b,C}$; the dropout–high-school gap is estimated at 14–25 only. The bands b are coarser than the age groups g that index adherence because no within-method moment exists before 18 to identify a separate teen baseline risk.

³²The term is the medical literature’s, where adherence is the extent to which a regimen is followed as prescribed (Osterberg and Blaschke, 2005; World Health Organization, 2003); in the taxonomy of Vrijens et al. (2012), which separates initiation, implementation and persistence, $c_{\theta g}$ is implementation.

³³Online Appendix OA.6.1 compares the NLSY79 failure measure with typical-use benchmarks and documents the method mix.

³⁴A missed pill and an unused condom are not the same exposure, so the ratio of pill to low-tier failure among imperfect users is also left to the data (Online Appendix OA.9).

percent against 10–14 percent by self-report (Potter et al., 1996). Adherence is indexed by the age groups $g \in \{14\text{--}17, 18\text{--}25, 26\text{--}33, 34\text{--}41\}$, with the last set equal to 26–33 because the NLSY79 contraception waves end when the cohort is 35. Within each group adherence is parameterized on the logistic scale,

$$c_{\theta g} = \frac{1}{1 + \exp\{-(a_g + \sum_{r \leq \theta} \Delta_{rg})\}}, \quad \Delta_{1g} \equiv 0, \quad \Delta_{rg} \geq 0,$$

so that the intercept a_g sets the bottom-quartile level freely and the cumulative, non-negative increments make $c_{\theta g}$ weakly increasing in ability. The nine increments $\{\Delta_{rg}\}$ are what the nested test restricts to zero.

Adherence is measured, not residual: the within-method birth rates of Section 2.3 are targets of the estimation, and equation (3) maps them into $c_{\theta g}$ given the external ρ_n . Two limitations remain. Because the model maps method use into live births, $c_{\theta g}$ absorbs any ability-correlated difference in abortion.³⁵ And because the NLSY79 first asked about methods in 1982, when its youngest respondents were already 18, $c_{\theta,14\text{--}17}$ is identified from the teen first-birth hazards and the wantedness of teen first births alone. The behaviors adherence requires also depend on non-cognitive traits correlated with the AFQT (Heckman et al., 2006), so $c_{\theta g}$ is best read as *fertility-control capacity* rather than cognition alone, which is why Section 6 studies interventions that remove the need for adherence rather than ones that raise skill.³⁶

Method choice. Given the value of a first birth ΔV (Section 3.2), which from here on includes the realised transitory component $\sigma_\omega \varepsilon^\omega$, the payoff of method n is $p_n \Delta V - \kappa_n + \sigma \varepsilon_n$, with $\kappa_0 = 0$, $\kappa_1 \geq 0$ the utility cost of the low tier, the same for every woman, and $\kappa_2(e) = \kappa_2 + \Delta \kappa_{2,e}$ the cost of the pill. Only the pill carries an education shifter ($\Delta \kappa_{2,HSD} = 0$), unrestricted in sign. It is not an access assumption: it gives education a route into the cost of the more effective method, so that education-specific contraception costs can compete with adherence as an explanation of the gradient, and Section 5 tests whether they do. The

³⁵Online Appendix OA.6.5 bounds the abortion channel.

³⁶Online Appendix OA.9 discusses the interpretation and scope of the fertility-control channel.

choice probabilities are

$$P(n \mid \Omega) = \frac{\exp\{(p_n \Delta V - \kappa_n)/\sigma\}}{\sum_{n'} \exp\{(p_{n'} \Delta V - \kappa_{n'})/\sigma\}},$$

and the expected probability of a first birth is $\bar{p} = \sum_n P(n) p_n$. The κ_n are effective costs in units of ΔV , not prices: they stand for whatever makes a method costly to obtain and to use, and a policy that changes the effective price of a method is a change in κ_n . Choice is where incentives live: within an education–age cell, women of different ability face different ΔV because they face different wages, and they choose different methods for that reason alone.

Unintended births. In the model a first birth is *unintended* if, at the start of the period in which it occurs, the woman would have preferred to remain childless, $\Delta V + \sigma_\omega \varepsilon^\omega < 0$, whether the method she chose failed or she chose none. The second case is the common one: a woman for whom a birth is unwelcome but $|\Delta V|$ is small often chooses no method, and if she conceives she is the survey’s “never used, not wanting” respondent. The definition matches the survey’s,³⁷ and the targeted moment counts women, not births: the share of a quartile’s women whose first birth comes by age 21 and is unintended.

The model nests the opportunity-cost explanation: if $c_{\theta g} = c_g$ for all θ , the ability gradient in first births has to be produced by wages, schooling, marriage and $\omega_{0,\theta}$ alone. Section 5 tests this restriction.

4 Estimation

Parameters are disciplined in three steps: some are set externally, the earnings process is estimated outside the structural model, and the remaining structural parameters are estimated by the Simulated Method of Moments.

³⁷Online Appendix [OA.1.7](#) gives the survey rule.

4.1 Externally Set Parameters and Earnings Process

Externally set parameters. The discount factor (0.959 a year) and the coefficient of relative risk aversion (1.98) are [Adda et al.’s \(2017\)](#) estimates. Tuition, taxes and transfers follow established calibrations ([Darulich and Fernández, 2024](#); [Vandenbroucke, 2023](#)), method efficacies follow [Trussell \(2011\)](#), and bounds on conception probabilities rule out both perfect control and deterministic fecundity.³⁸ The exposure shares $a_{\theta t}$ are data: at ages 14–17 they are 0.63, 0.57, 0.46 and 0.36 from the lowest to the highest quartile, 0.82–0.89 at 18–21, and above 0.9 thereafter.

Externally estimated earnings process. Predicted annual real earnings are the model’s deterministic earnings opportunities. For women, $\tilde{w}_t^f = X_t^f \hat{\beta}^f$, where X_t^f contains age and its square, education and ability-quartile indicators, experience, the interactions education×experience, ability×experience and education×ability, and indicators for marriage and a nonmarital first birth. Husbands’ earnings, $\tilde{w}_t^h = X_t^h \hat{\beta}^h$, are a reduced-form function of age and its square, the wife’s education, and their interactions with an out-of-wedlock first birth, so that assortative mating and marriage selection are captured.³⁹ Estimating this process outside the structural step ensures that wage–experience profiles by ability are disciplined by labor-market data alone ([Eisenhauer et al., 2015](#)).⁴⁰

4.2 Estimation and Identification

Estimation. I estimate the remaining parameters by Simulated Method of Moments (SMM) ([Gourieroux et al., 1993](#); [McFadden, 1989](#); [Pakes and Pollard, 1989](#)), choosing Θ to minimize the distance between an empirical moment vector $m^{data} \in \mathbb{R}^{138}$ and its model analogue:

$$\hat{\Theta} = \arg \min_{\Theta} (m^{sim}(\Theta) - m^{data})' W (m^{sim}(\Theta) - m^{data}),$$

³⁸Online Appendix Table [OA.29](#) lists every externally set parameter and its source.

³⁹Modeling the husband’s earnings as a reduced-form function of the wife’s characteristics, rather than as a separate decision problem, is standard in life-cycle models of women’s choices ([Eckstein et al., 2019](#); [Keane and Wolpin, 2010](#); [van der Klaauw, 1996](#)).

⁴⁰Online Appendix [OA.10](#) gives the samples, specifications and estimates.

where W is diagonal and *fixed*, with $W_{jj} = 1/\hat{\sigma}_j^2$ and $\hat{\sigma}_j$ the bootstrap standard error of data moment j from 1,000 replications resampling women.⁴¹ The model is overidentified—138 targeted moments against 76 structural parameters—and standard errors follow from the usual minimum-distance sandwich, inflated for simulation noise (McFadden, 1989); Online Appendix OA.11 reports them and the distribution theory for the nested test of Section 5.3.

Moment blocks. The 138 targeted moments⁴² are organized into blocks aligned with the model’s mechanisms: 10 schooling and early-fertility moments, 2 child-investment ratios by mother’s education, 28 first-birth shares by age bin and ability quartile, 17 marriage shares and 36 employment rates by age bin and education, and 25 contraception-use rates by age bin, education and ability half. Three further blocks discipline the fertility-control technology directly: the share of women whose first birth came by age 21 and was reported as unintended, by quartile (4); the unintended share of first births by age at first birth, for the lowest and highest quartiles (8);⁴³ and the annual birth rate within pill users and within low-tier users, by quartile (8).⁴⁴ The pill share among users (8) is not targeted; it is reported as an untargeted check in Section 5.1.⁴⁵ All parameters are disciplined jointly through the model’s cross-equation restrictions; the blocks are a guide to the main sources of identifying variation, not a claim of one-to-one identification (Adda et al., 2017; Keane and Wolpin, 1997, 2010).

Identification: separating opportunity costs from fertility control. The same ability type that matches schooling, labor supply and marriage must also rationalize fertility timing, so explanatory burden cannot be reallocated across margins without degrading the

⁴¹I do not use the asymptotically efficient $W = \Omega^{-1}$: it carries severe small-sample bias (Altonji and Segal, 1996), and $\hat{\Omega}$ is numerically singular here (Online Appendix OA.11).

⁴²The estimation code carries 146 moments; the eight pill-share moments enter with zero weight and are held out as an untargeted check (Online Appendix OA.11.5).

⁴³Online Appendix OA.3 documents the wantedness measures behind both blocks.

⁴⁴The within-method block is compared at an *annual* rate on both sides: the simulated four-year probability p enters the criterion as $1 - (1 - p)^{1/4}$, the convention under which the estimates compare with the annual typical-use rates of Trussell (2011).

⁴⁵The model has no channel for the method mix (Online Appendix OA.6.1), so targeting it would move the method costs without informing the mechanism. The contraception block is measured at the education $\times$ age $\times$ ability level, so that within-cell variation in use is disciplined by the data rather than attributed to adherence by construction.

others (Keane and Wolpin, 2010). Four restrictions keep within-education birth gradients from being absorbed elsewhere. The earnings process is estimated outside SMM. Schooling costs are disciplined by schooling outcomes, including schooling conditional on early fertility. Exposure $a_{\theta t}$ and the allowance scale a_{θ} , the two ability-indexed inputs that are neither preferences nor technology, are measured and held fixed. And baseline risk p_{be}^0 , the probability of a first birth with no method, is common within age–education cells, so within-cell ability gradients load on adherence, preferences or method choice.⁴⁶ The last restriction is what lets the within-method birth rates identify adherence, and it is a restriction the data support rather than an assumption of convenience. Baseline risk is the chance of conceiving when no method is used, which depends on fecundity and on how often a woman has intercourse, not on which method she chooses: method choice is already held fixed when failure is compared within a method. Had baseline risk been allowed to vary with ability, equation (3) could not have told the two apart: pill users of high ability could have fewer births either because they take the pill more reliably or because they would conceive less often even without it. Women who use no method rule the second channel out: for them the birth rate is the baseline risk itself, so an ability gradient in baseline risk would appear there. It does not. Among childless women using *no* method the first-birth rate over the interval to the next wave is 4.2 percent in the lowest quartile and 4.7 percent in the highest, and the top-quartile gradient within education, age and wave is 0.0071 (s.e. 0.0122), slightly *positive*, where a baseline-risk explanation requires it to be negative.⁴⁷ Coital frequency runs the same way: intercourse is *more* frequent at higher ability, by 1.92 acts a month between the bottom and top quartiles (s.e. 0.44), so conditioning on it moves the within-method gradient from -0.0224 to -0.0239, away from zero.⁴⁸

Identifying fertility-control parameters. Each object of the technology has a block of moments that speaks to it most directly.⁴⁹ Baseline risk p_{be}^0 (through the odds λ_{be} that

⁴⁶Method choice is *not* restricted to be constant across ability groups within an education–age cell: the choice probabilities depend on $-\Delta V$, which varies with ability through wages and continuation values, and the within-cell variation in method use by ability is itself targeted (25 moments).

⁴⁷Online Appendix OA.6.3 reports the non-user birth-rate estimates.

⁴⁸Online Appendix OA.6.4 reports the coital-frequency estimates.

⁴⁹Online Appendix Table OA.28 maps the objects to the moments and the variation used.

parameterize it) is disciplined by the level and age profile of first births, given the education composition of each quartile that the schooling block pins down. Holding the method fixed, equation (3) makes the ability gradient in within-method failure a gradient in $c_{\theta g}$, while the level of failure and its pill-to-low-tier ratio, common across quartiles, pin ρ_n^0 . The method costs κ_1, κ_2 and the dispersion σ are disciplined by the share using any method by education, age and ability half, given the ΔV the rest of the model implies. The preference intercepts $\omega_{0,\theta}$ are disciplined by the unintended-birth moments: given how many women have a first birth by 21, which the timing moments pin down, the block says how many of those births came to women who would rather have waited. Raising $\omega_{0,\theta}$ turns unintended births into intended ones at given adherence and method costs, while adherence and method costs move the share only through births to women trying to avoid one. The scale of the transitory component, σ_ω , is disciplined by the composition of first births by age at first birth: the preference intercepts move the quartiles' *levels*, and σ_ω moves how far, within a quartile and an age, the classification departs from all-or-nothing. Without the two wantedness blocks the shock is disciplined only by the timing moments (Section 5.3).⁵⁰

What the moments rule out. Consider two childless women of the same education and age but different ability. If both use the pill, their birth rates differ only through adherence, and the within-pill moment measures that difference directly. If they choose their method, the higher-ability woman's larger $-\Delta V$ makes her more likely to use one, and to use the pill—an ability gradient in use and in births with no difference in adherence at all. The any-method moments discipline how strongly choice responds to ΔV , and the unintended-birth moments discipline $\omega_{0,\theta}$, so the estimator cannot attribute to adherence a gradient that incentives or preferences produce. Each channel is thus pinned down by a block of moments other than first-birth timing: adherence by within-method failure, incentives by method use, preferences by wantedness. Whether the three together reproduce the ability gradient in first-birth timing is the question the nested test of Section 5 answers.

⁵⁰Online Appendix OA.3 reports the scale of the shock against the dispersion of ΔV that the state produces.

5 Results

This section presents the estimation results in three steps. Section 5.1 shows that the model fits the data well: it reproduces the ability gradient in the timing of first births and its narrowing by the end of the fertile years, the wantedness of early births, and the two margins of fertility control, while matching schooling, employment, marriage and contraceptive use over the life cycle. Section 5.2 presents the estimated fertility-control technology: adherence rises steeply with ability among teenagers and young adults and converges across quartiles by the late twenties, so the ability gradient in fertility control is a young-adult phenomenon. Section 5.3 tests whether opportunity costs and preferences alone can account for the data, by re-estimating the model with adherence common across ability groups and every other parameter free. The restriction is rejected: the restricted model can reproduce how many low-ability women have an early birth, but not how many of those births were unwanted, and it distorts the employment and marriage profiles in the attempt. Section 6 then uses the estimated model to compare closing the adherence gap with lowering the price of contraception.

5.1 Model fit

The model fits the data well. Figure 2 compares it with the NLSY79 on the four outcomes that matter for the paper—the timing of first births by ability quartile, the share of early births that were unwanted, and the two margins of fertility control, use and failure—and it reproduces the level and the ability gradient of each. It also matches the remaining targets, schooling by ability, child investment and the life-cycle profiles of employment, marriage and contraceptive use by education; Online Appendix OA.12 plots those profiles and reports all 138 moments with their contribution to the criterion.

The model captures the ability gradient in the timing of first births and its narrowing with age (Figure 2a). By age 21, 58 percent of the lowest quartile and 23 percent of the highest have had a first birth in the model, compared with 63 and 18 percent in the data, and by age 41 the gap is 17 percentage points in the model and 15 in the data. The middle quartiles fall in between, in the same order as in the data, and 27 of the 28 timing moments

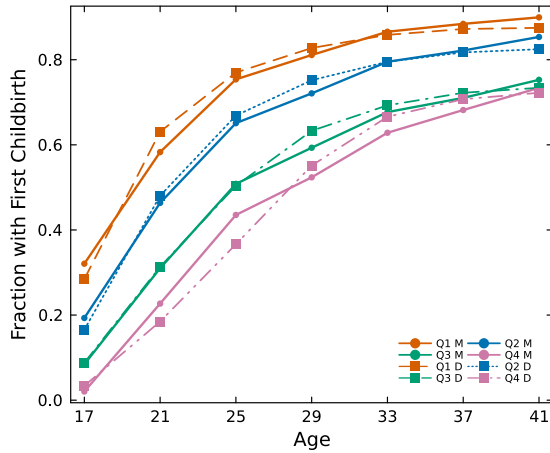
are within five percentage points of their targets. The model matches the ability gradient in unintended first births almost exactly (Figure 2b): the share of women with an unintended first birth by 21 falls from 31.9 percent in the lowest quartile to 10.6 percent in the highest, compared with 32.5 and 10.2 percent in the data. Panel C of Table 6 shows that the model also reproduces the unintended share of first births within each age at first birth.

The model captures the two margins of fertility control of Section 2.3. Use of any method is higher at the top of the ability distribution than at the bottom in the model as in the data, from 54 to 64 percent of childless women against 47 to 72 percent (Figure 2c), although the gradient is flatter in the model, which targets use by education and ability half rather than by quartile. Within method, the annual first-birth rate declines with ability for both the pill and the low-tier methods, as in the data (Figure 2d); these are the moments that identify adherence. The third margin, fecundity and exposure, is held common within education and age, as the flat birth rate among non-users in Table 2 supports.

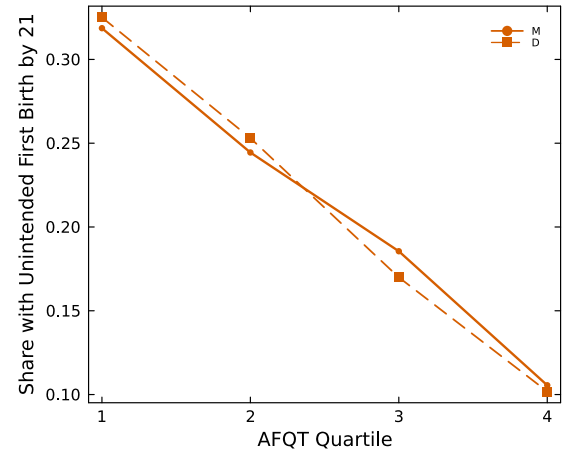
Two features of the fit deserve comment. First, the model places a few more high-ability first births in the early twenties than the data: 44 percent of the highest quartile have had a first birth by 25 in the model against 37 percent in the data, with correspondingly fewer between 26 and 33. This reflects the transitory wantedness shock, whose dispersion is common across quartiles; it does not change the ordering of the quartiles at any age. Second, pill failure in the two lowest quartiles is somewhat higher in the model than in the data, 2.8 and 1.6 percent a year against 2.0 and 1.0 percent, a difference of under one percentage point that is the price of matching the lowest quartile’s early births. Among the moments not shown, the only one the model misses by a wide margin is contraceptive use among high-school dropouts at ages 22–25 (48 percent in the model against 30 in the data). The model matches these women’s first births, so the gap is in how the survey’s use question maps into the model’s use margin, and it has no bearing on the ability gradient, which is identified within education.

As an untargeted check, in the spirit of Keane and Wolpin (2007), I compare the model’s pill share among method users with the data. If the ability gradient in first births reflected which method women choose rather than how consistently they use it, the pill share would rise steeply with ability. It does not: among childless women aged 18–33 it is flat in the data,

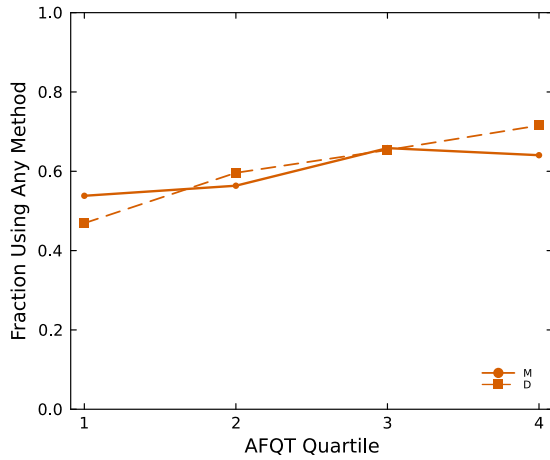
Figure 2. Model Fit: First-Birth Timing, Wantedness, and Fertility Control



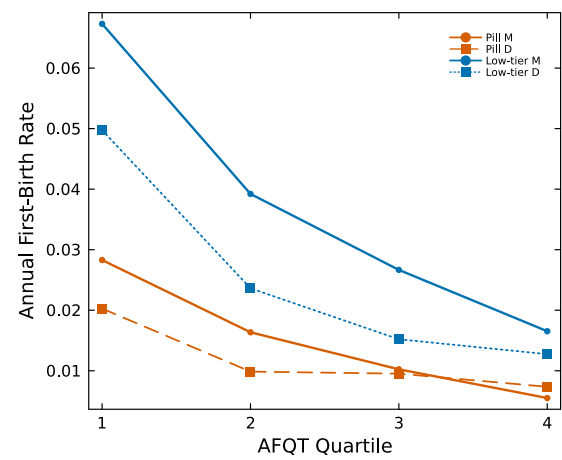
(a) First birth by age, by ability quartile



(b) Unintended first birth by 21, share of women



(c) Use of any method, childless women 18–33



(d) Annual first-birth rate within method, ages 18–33

Notes: Panel (a): cumulative fraction of women with a first birth by the age shown, by AFQT quartile (Q1 = lowest). Panel (b): share of women in each quartile with a first birth by 21 that they reported as mistimed or unwanted. Panel (c): share of childless women at risk who use any method, ages 18–33, by quartile. Panel (d): annual first-birth rate among childless women using the pill and among those using a low-tier method, ages 18–33, by quartile. Panels (a), (b) and (d) are targeted; the use share by quartile in panel (c) is not (the estimation targets use by education and ability half). Solid lines with circular markers are the model, dashed lines with square markers the NLSY79 data.

0.835 in the lowest quartile and 0.824 in the highest, and the model, which places no weight on these moments, predicts it flat as well. The model does predict a lower pill share than the data (0.394 to 0.414)⁵¹: a two-method technology with one adherence parameter per ability group and age cannot match at once how many first births among users are unintended, how often each method fails, and which method women choose, and the estimation gives the first two priority. The results do not hinge on this. Re-estimating the method costs to match the pill share changes the fall in first births by 21 from equalizing adherence from 17 to 15 percent and from the long-acting method at Colorado’s take-up from 7 to 8 percent, and moves the lowest quartile’s welfare gain from that method from 5.8 to 5.3 percent of consumption. What the check establishes is that the model does not manufacture a gradient in method choice where the data have none: in the data the lowest quartile uses the pill, the more effective method, slightly more than the highest and still fails more often.

5.2 The Estimated Fertility-Control Technology

Table 4 reports the estimated adherence parameters $c_{\theta g}$ by ability quartile and age group, with standard errors, together with the differences from the lowest quartile (panel C).⁵²

Adherence rises steeply with ability among teenagers and young adults and converges across quartiles by the late twenties. At ages 14–17, adherence is 0.28 in the lowest quartile, 0.60 in the second and 0.76 in the third; at 18–25 it rises from 0.58 in the lowest quartile to 0.90 in the highest. These levels are the ones the adherence literature measures directly: 47 percent of pill users miss at least one pill per cycle (Rosenberg et al., 1998), women aged 18–20 followed weekly use contraception consistently in 72 percent of weeks (Stidham Hall et al., 2013), and 40 percent are always consistent against 11 percent never (Wu et al., 2016). By 26–33 adherence is essentially complete in every quartile (0.99 to 1.00), and no difference between quartiles is distinguishable from zero. The age profile is also the one that literature finds: failure of user-dependent methods is nearly twice as high under 21 as above, while the failure of long-acting methods does not vary with age (Bradley et al., 2019;

⁵¹Online Appendix OA.12.2 reports the pill share by quartile and age group in the data and the model, and the re-estimation that targets it.

⁵²Online Appendix OA.13 reports the full parameter vector for the estimated and the restricted model and plots adherence against the restricted model (Figure OA.7).

Winner et al., 2012). Since perfect-use failure is close to zero, the ratio of non-adherence, $(1 - c_{Q1,g})/(1 - c_{\theta g})$, is the ratio of the failure risks two users of the same method face: a pill user in the lowest quartile faces about 4.3 times the failure risk of a user in the highest at ages 18–25, and about 3.0 times that of a third-quartile user at 14–17.⁵³

The ability differences are precisely estimated where they matter. The gap between the lowest and the third quartile at 14–17 is 0.48 (standard error 0.20) and the gap between the lowest and the highest quartile at 18–25 is 0.32 (0.12); the third- and fourth-quartile differences are significant at the five percent level at both ages, and the second-quartile differences have the same sign, are about half as large, and sit at the margin of that level. The narrowing with age is less precisely estimated: the gap between the lowest and the highest quartile at 18–25 exceeds the one at 26–33 by 0.31 (0.23). What the estimates establish, then, is a gradient in fertility control between the bottom quartile and the top half of the ability distribution that is concentrated in the years before 26. These are the years in which an unintended birth is most costly—schooling is unfinished and the returns to experience are steepest—and they are also, in Figure 2a, the years in which the ability gradient in first births opens up.

A natural question is whether the adherence gradient could instead reflect a misspecified earnings process: if the ability gradient in wages were steeper than estimated, the opportunity cost of a birth would rise faster with ability and the model would need less of a gradient in adherence to fit the data. To assess this, I re-estimate the fertility-control parameters with the ability gradient in earnings scaled up by a factor of up to two, well beyond what measurement error in the AFQT could justify. The adherence estimates are robust: at one and a half times the estimated gradient they move by less than five percent, and at twice it the teenage gap falls by 6 percent and the young-adult gap not at all, while the fit of the model worsens. The reason is that adherence is identified by the within-method birth rates,

⁵³Adherence for the highest quartile at 14–17 is estimated at 0.998, where the logistic is nearly saturated, because near-perfect use is the only margin on which the model can bring these women’s birth rate at 14–17 down to the 3.4 percent in the data; its logit standard error is unstable, so the teen gradient is read off the third quartile. 3 parameters carry no standard error: the three ability increments at 26–33, where adherence is essentially complete in every quartile and the criterion is flat in them; Online Appendix OA.13 lists them. The utility cost of the low-tier method, κ_1 , is estimated at 0.002 and is not distinguishable from zero, so the pill’s cost κ_2 is in effect identified relative to a free alternative.

Table 4. Estimated adherence $c_{\theta g}$, by AFQT quartile and age group

Ages	Q1 (lowest)	Q2	Q3	Q4 (highest)
<i>A. Estimated model: adherence $c_{\theta g}$</i>				
14–17	0.276 (0.272)	0.603 (0.173)	0.756 (0.102)	0.998 (0.019)
18–25	0.584 (0.142)	0.743 (0.090)	0.823 (0.064)	0.904 (0.037)
26–33	0.988 (0.175)	0.996 [‡]	0.998 [‡]	0.999 [‡]
<i>B. Restricted model: $c_{\theta g} = c_g$</i>				
14–17	0.973 (0.162)	0.973 (0.162)	0.973 (0.162)	0.973 (0.162)
18–25	0.492 (0.368)	0.492 (0.368)	0.492 (0.368)	0.492 (0.368)
26–33	1.000	1.000	1.000	1.000
<i>C. Ability gaps, estimated model: $c_{\theta g} - c_{Q1,g}$</i>				
14–17	–	0.327** (0.164)	0.480** (0.200)	0.722*** (0.266)
18–25	–	0.159** (0.077)	0.239** (0.097)	0.320*** (0.117)
26–33	–	0.008 [‡]	0.010 [‡]	0.011 [‡]

Notes: $c_{\theta g}$ is the probability that a woman of AFQT quartile θ in age group g uses her chosen method as intended. Panel A: estimated model; Panel B: the restricted model with $c_{\theta g} = c_g$ (Section 5.3); Panel C: the gap relative to Q1. Standard errors in parentheses, from the sandwich of Section 4 by the delta method; in Panel C they are the standard errors of the difference, which account for the correlation between cells. Adherence at ages 34–41 is set equal to its 26–33 value and not estimated, because the NLSY79 contraception waves end when the cohort is 35. ***, ** and *: significant at the 1, 5 and 10 percent levels. [‡] marks a cell (Q2 at ages 26–33, Q3 at ages 26–33 and Q4 at ages 26–33) that contains a parameter in which the criterion is flat—the logistic is saturated near one, or the increment is not separately identified—so no meaningful standard error can be computed.

which condition on the method in use and do not respond to wages.⁵⁴

5.3 Nested Specification Test: Is Ability-Dependent Fertility Control Necessary?

The test asks whether opportunity costs and preferences alone can account for the ability gradient in first births. The restricted model imposes

$$H_0 : c_{\theta g} = c_g \quad \text{for all } \theta,$$

⁵⁴Online Appendix OA.10.5 reports the sensitivity analysis.

by fixing the nine adherence increments at zero: adherence may vary with age but not with ability. Every other parameter is re-estimated on the same 138 moments, weights and bounds—schooling costs, marriage, the method costs κ_n and their dispersion σ , the non-adherent failure rates ρ_n^0 , the scale of the transitory wantedness component σ_ω , and, most importantly, the ability gradient in preferences $\omega_{0,\theta}$. The restricted model is therefore the best model without the adherence channel that the data admit, not the estimated model with the mechanism switched off; in particular it remains free to reproduce the ability gradient in whether a method is used, since method choice responds to $-\Delta V$ and to the costs κ_n .

The minimized criterion is 382.3 for the estimated model and 530.7 for the restricted one. Because the models are nested, the difference $D = L_R - L_U = 148.5$ is a test statistic for the nine restrictions, and under the null it is asymptotically distributed as χ_9^2 , whose one-percent critical value is 21.7. The hypothesis of common adherence is rejected: the statistic is 7 times the critical value.⁵⁵

Table 5 decomposes the increase in the criterion under the restriction by block of moments. Two blocks account for most of it. The first-birth shares by age and quartile deteriorate by 66 points and the two wantedness blocks by 74 points, together 94 percent of the total: without ability differences in adherence, the model can neither place the early births in the right quartiles nor reproduce how many of them were unwanted. The employment and marriage profiles, which the restriction does not touch directly, deteriorate by a further 22 points, because the restricted model distorts them to keep births where the timing moments want them. The contraception-use moments work the other way: the restricted model fits whether a method is used better (25 against 60 points), so the statistic is net of a block the restriction improves. The within-method failure moments, whose ability gradient is a pooled fact rather than a within-group one (footnote 12), contribute 8 points; the rejection rests on

⁵⁵The χ_9^2 distribution assumes the optimal weighting matrix and a null in the interior of the parameter space; neither holds exactly here, and neither affects the conclusion. With the diagonal weighting matrix the statistic is a weighted sum of chi-squares, and the mean-matched version that corrects for this is $D_{NW} = 182$; because the null fixes the adherence increments at the boundary of their parameter space, the asymptotic p -value is conservative. Nor does it depend on the wantedness moments, which account for nearly half of the increase in the criterion: re-estimating both models on the 126 moments that exclude them gives $D = 134.1$, 6 times the one-percent critical value. Online Appendix OA.11.4 gives the distribution theory and verifies, with 16 starts of the restricted model and 16 of the estimated one, that each reported criterion is the lowest found.

the timing and wantedness of first births and on the life-cycle profiles the restricted model distorts to fit them.

Table 6 compares the two models on the moments that identify the fertility-control technology. The estimated model reproduces the ability gradient in unintended first births by 21 (panel A). The restricted model recovers only 60 percent of it, and the shortfall is at the bottom: it understates the lowest quartile’s unintended share by 7 percentage points and overstates the highest quartile’s by 2. The within-method failure rates (panel B) show the same pattern: they decline with ability in the estimated model, as in the data, and are essentially flat in the restricted model, where adherence is common across quartiles.

Panel D shows how the restricted model compensates. With one adherence level per age group, an early birth to a low-ability woman can come only from non-use, so the restricted model makes a child less attractive to the highest quartile (the increment $\omega_{0,Q4}$ falls from -0.08 to -0.27), raises the dispersion of the transitory wantedness component (from 0.140 to 0.325), re-sets the method costs, and pushes the common adherence level toward perfect use at 14–17 and 26–33 to hold down births in the top quartile. These adjustments have a cost elsewhere, which is why preferences and adherence are separately identified: it understates the unintended share of first births to the lowest quartile at 14–17 and 18–21 and overstates it for the highest quartile at 14–17 (panel C), and it fits the employment and marriage profiles worse (Table 5). In sum, a model without ability differences in adherence can reproduce how many low-ability women have an early birth, but not how many of those births were unwanted: the ability gradient in unintended first births by 21 is 22 percentage points in the data, 21 in the estimated model and 13 under common adherence.

Table 5. Nested specification test: the increase in the criterion under common adherence, by block of moments

Moment block	# moments	Criterion		Share of increase
		Estimated	Restricted	
First births by age $\times$ quartile	28	153	219	44%
Unintended first birth by 21	4	3	38	23%
Unintended share by age at first birth	8	48	87	26%
Contraception use	25	60	25	-24%
Employment	36	74	94	14%
Within-method failure	8	20	28	6%
Marriage	17	11	13	1%
Schooling	10	11	24	9%
Child investment	2	2	3	0%
Total	138	382.3	530.7	148.5

Notes: Weighted criterion by block of moments for the estimated model and for the restricted model with $c_{\theta g} = c_g$; the last column is each block's share of the increase in the total, $D = L_R - L_U$, which is the test statistic of Section 5.3 (nine restrictions; χ^2_9 one-percent critical value 21.7). Both models use the same 138 moments, weights and bounds; the restricted model fixes the nine adherence increments at zero and re-estimates every other parameter. Shares sum to 100 up to rounding; a negative share is a block the restricted model fits marginally better.

Table 6. Where the Two Models Separate: Fertility-Control Moments and the Parameters the Restriction Moves

		Model		% of own criterion	
	Data	Estimated	Restricted	Estim.	Restr.
<i>A. Unintended first birth by 21, share of women</i>					
Q1	0.325	0.319	0.254	0.1	5.7
Q2	0.253	0.244	0.237	0.1	0.4
Q3	0.170	0.186	0.174	0.6	0.0
Q4	0.102	0.106	0.120	0.1	1.0
<i>B. Annual first-birth rate within method, ages 18–33</i>					
Pill, Q1	0.020	0.028	0.017	0.6	0.1
Pill, Q2	0.010	0.016	0.016	1.4	0.8
Pill, Q3	0.010	0.010	0.015	0.0	1.0
Pill, Q4	0.007	0.005	0.014	0.2	2.5
Low tier, Q1	0.050	0.067	0.018	0.4	0.8
Low tier, Q2	0.024	0.039	0.022	1.0	0.0
Low tier, Q3	0.015	0.027	0.019	1.3	0.1
Low tier, Q4	0.013	0.017	0.014	0.3	0.0
<i>C. Unintended share of first births by age at first birth, percent</i>					
14–17, Q1	62.6	64.8	52.7	0.2	3.0
14–17, Q4	69.6	84.0	81.1	1.2	0.5
18–21, Q1	45.6	42.2	32.5	0.5	5.9
18–21, Q4	56.4	42.8	42.7	4.0	2.9
22–25, Q1	29.4	35.9	30.6	0.9	0.0
22–25, Q4	33.0	32.3	36.8	0.0	0.3
26–41, Q1	25.6	14.1	18.1	2.4	0.7
26–41, Q4	14.2	19.9	20.5	3.3	2.9
		Estimated	Restricted		
<i>D. Parameters the restriction moves (standard errors in parentheses)</i>					
Child-utility increment, Q4, $\omega_{0,Q4} - \omega_{0,Q1}$		-0.081 (0.024)	-0.266 (0.042)		
Transitory wantedness component, σ_ω		0.140 (0.031)	0.325 (0.032)		
Cost of the low-tier method, κ_1		0.002 (0.017)	0.236 (0.065)		
Cost of the pill, κ_2		0.042 (0.036)	0.220 (0.026)		
Dispersion of method taste shocks, σ		0.047 (0.013)	0.022 (0.008)		
Adherence intercept, 14–17 (logit)		-0.96 (1.36)	3.58 (6.13)		
Adherence intercept, 26–33 (logit)		4.41 (14.73)	10.00 (–)		
Child penalty, high school, ψ_{lk}		-0.030 (0.006)	-0.070 (0.058)		

Notes: Panels A–C: the twenty targeted moments that identify the fertility-control technology (Table OA.28), in the data, in the estimated model and in the restricted model that imposes $c_{\theta g} = c_g$; the last two columns are each moment’s share of its own model’s weighted criterion, in percent. Panel A is the share of women in each AFQT quartile with a first birth by 21 that they reported as mistimed or unwanted. Panel B is the annual first-birth rate among childless women using each method, ages 18–33, annualized as $1 - (1 - p)^{1/4}$ on both sides. Panel C is the percent of first births in each age range that were unintended, lowest and highest quartiles; in the model a first birth is unintended if $\Delta V + \sigma_\omega \varepsilon^\omega < 0$ at the start of its period (Section 3.4). Panel D: the parameters that move under the restriction, with sandwich standard errors; in the restricted model the adherence intercept is common to all quartiles. The full vectors, with standard errors, are in Online Appendix OA.13.

6 Counterfactual Experiments and Policy Implications

Low-ability women have more early, unwanted births because they use contraception less reliably, not only because they use it less or value children differently. The distinction matters for policy: a subsidy lowers the price of choosing a method, whereas a long-acting reversible method removes the need for adherence, so if the gradient is in adherence the two instruments differ not only in how much they reduce early births but in whom they reach. Three experiments quantify that difference, each standing for a policy. The first gives every woman the adherence of the highest ability quartile at her age, leaving method costs, preferences and wages at their estimates: the benchmark for policies that act on how reliably women use the methods they choose—counseling, adherence support, or a method that does not depend on daily use—and an upper bound on what any of them could deliver, since it closes the entire estimated gap at no cost to the recipient. The second cuts the price of both methods for every woman by the same proportion, set so that it lowers first births by 21 by as much as the first experiment: the general subsidy of the Affordable Care Act’s contraceptive mandate and of the Medicaid family-planning expansions studied by [Kearney and Levine \(2009\)](#), so that the two instruments are compared at equal effect on births.⁵⁶ The third adds a long-acting reversible method to the menu, with a failure rate that does not depend on adherence, at the utility cost that reproduces the take-up of Colorado’s Family Planning Initiative, 25 percent of users at ages 18–25: the policy of the state programs that offered intrauterine devices and implants through public clinics ([Lindo and Packham, 2017](#); [Secura et al., 2014](#)).⁵⁷

Two findings stand out. A price cut that reduces early births by as much as closing the adherence gap reaches the opposite end of the ability distribution, and most of its value is a

⁵⁶Formally $\kappa_n \mapsto (1 - \delta)\kappa_n$; the matched cut is $\delta = 0.89$, most of the way to making both methods free. I also report the smaller cut, $\delta = 0.58$, that lowers first births by 21 by ten percent.

⁵⁷The long-acting method’s first-birth risk is $p_{be}^0 \rho_L$ with $\rho_L = 0.005/0.85$, the midpoint of the IUD and implant failure range in [Trussell \(2011\)](#) over the no-method rate. It shares one taste draw with the pill, so the most birth-averse pill users switch first, as in Colorado, where the rise in long-acting use was mirrored by a fall in pill use. Two variants bound the technology channel: the method at the pill’s cost, and every method in use made adherence-proof. Because adherence at 14–17 is identified from the teen birth and wantedness moments alone and is saturated for the top quartile, every experiment holds each quartile’s 14–17 adherence at its 18–25 value, which the within-method moments identify; the unrestricted version raises the headline magnitudes by between a half and a factor of two but leaves the comparison between instruments unchanged.

windfall to the women who already controlled their fertility well. And when the long-acting method is added at Colorado’s take-up, the model predicts a fall in early births close to the one the program’s evaluation found, which it was not estimated to match, with the largest gains for the lowest-ability women, who have most of the unwanted births.

Fertility. Table 7 reports the effects of the three experiments. Closing the adherence gap lowers the share of women with a first birth by age 21 from 0.378 to 0.314, a fall of 17 percent. The matched price cut lowers it to 0.314, a fall of 17 percent by construction, and a long-acting method at Colorado’s take-up to 0.351. The two main policies also reach opposite ends of the ability distribution. Closing the adherence gap acts on the bottom: first births by 21 in the lowest quartile fall from 0.532 to 0.389, while the highest quartile is untouched by construction. The price cut does the reverse, lowering first births by 21 by 12 percent in the lowest quartile and by 24 percent in the highest: a subsidy is taken up most by the women who already controlled their fertility well. The reason is that more use is not more effective use. The price cut raises the share of childless women using any method at ages 18–25 from 0.712 to 0.788, far more than closing the adherence gap does (0.718), and the pill’s share among users rises from 0.389 to 0.577 because the cut is largest in dollars for the dearer method, but the marginal users are the poorest adherers, so the share of first births that are unintended does not fall: it moves from 0.381 to 0.387, whereas closing the adherence gap lowers it to 0.315.

Welfare. Table 8 reports the welfare gains as consumption-equivalent variation, the permanent proportional change in consumption that gives a woman at age 14, before any shock is drawn, the value she attains under the experiment.⁵⁸ Two features govern how it should be read. First, the utility flow from a child is an estimated parameter and the measure counts the mother’s utility only, so the numbers say how the gains from each experiment are distributed across ability quartiles—who gains and who does not—rather than how much any experiment is worth in absolute terms. Second, a price cut benefits a woman in two ways. Every woman already using a method receives the price cut on what she was buying

⁵⁸The measure is ex ante and integrates over every taste shock, so it credits a new option only where it is chosen for its payoff.

anyway, a windfall she collects without changing any decision; and some women respond to the lower price by using a method, or a better one, and avoid births they did not want. Only the second is the value of the policy as fertility control, and only it is comparable with closing the adherence gap, which changes no price. The lower panel of Table 8 separates the two. The windfall to existing users is 74 percent of the lowest quartile’s total gain from the price cut and 59 percent of the highest’s; it is large because the estimated cost of the pill, which rationalizes how many women at risk use no method, is high relative to the marginal utility of consumption. The windfall is smallest for the lowest quartile, 18.4 percent of lifetime consumption against 28.9 for the highest, because high-ability women use methods, and the pill, more; its share of the gain falls across quartiles only because the top of the distribution also gains more net of it. Net of it, the price cut is worth 4.6 percent of lifetime consumption to the lowest quartile and 11.7 percent to the highest. Closing the adherence gap, by comparison, is worth 15.3 percent to the lowest quartile, 10.2 and 9.3 percent to the second and third, and nothing to the highest. The distribution of the gains follows the fertility results: the price cut is worth more to the women who already controlled their fertility well, and closing the adherence gap to the women who did not.

Long-acting reversible contraception. At Colorado’s take-up, first births by age 21 fall by 7.0 percent. [Lindo and Packham \(2017\)](#) estimate that the initiative reduced teen birth rates by 6.4 percent over five years in the counties its clinics reached. The model, estimated on a cohort that contracepted before long-acting methods were widely available and given only the program’s take-up, lands within about a point of that estimate. The result is an instance of a general one: interventions that substitute for the user’s effort close skill gradients that price reductions leave in place, as an intensive diabetes regimen removed the education gap in glycemic control ([Goldman and Smith, 2002](#)) and single-pill regimens narrowed disparities in hypertension ([Goldman and Lakdawalla, 2005](#)). The gains fall among lower-ability women, where the evaluations of these programs find their effects: the gain is 5.8 percent of lifetime consumption for the lowest quartile and 1.2 for the highest, because removing the adherence margin is worth most to the women for whom adherence was binding. Against closing the adherence gap, a long-acting method at program scale does less than

half as much; offered at the pill’s cost it is taken up by 34 percent of users and lowers first births by 21 by 8 percent, and only a menu in which every method in use is adherence-proof matches the targeted experiment. The technology is a substitute for adherence, and at the prices programs have offered it, a partial one.⁵⁹

Schooling and the labor market. The experiments change early fertility far more than they change schooling or careers. Closing the adherence gap lowers first births by age 21 by 17 percent but raises college attendance only from 36.7 to 37.2 percent, and employment and earnings move by less than one percent under every experiment. The main reason is who the policies reach: the births they avert are concentrated among low-ability women, whose schooling and careers were the least sensitive to a birth to begin with, since a teen birth lowers college attendance from 41 to 8 percent in the data, but women in the lowest quartile attend at 12 percent whether or not they have one. The gains in Table 8 are therefore the direct value of avoiding an unwanted birth, not recovered schooling or earnings. A second reason is a limitation of the model: a woman has at most one child, present for a single period, so the career cost of an early birth is one period of reduced labor supply rather than the extended interruption that Adda et al. (2017) measure; on this margin the counterfactuals are a lower bound.

⁵⁹Online Appendix OA.9 discusses what adherence and an adherence-proof technology each remove. The women who switch are pill users, whose share the model under-predicts (Section 5.1), so the price that delivers a given take-up is calibrated on a smaller clinical nest than the data’s.

Table 7. Counterfactual experiments: outcomes

	Baseline	Adherence = Q4	Price cut		LARC menu		
			Matched	−10%	Colorado	Pill’s cost	All users
<i>Panel A: Aggregate</i>							
First birth by 17	0.127	0.091	0.104	0.113	0.110	0.108	0.060
First birth by 21	0.378	0.314	0.314	0.340	0.351	0.346	0.255
First birth by 25	0.577	0.515	0.490	0.523	0.550	0.543	0.447
Unintended share	0.381	0.315	0.387	0.380	0.348	0.342	0.229
Any method, 18–25	0.712	0.718	0.788	0.760	0.719	0.721	0.732
Pill share among users	0.389	0.376	0.577	0.510	0.164	0.089	0.355
LARC share among users	0.000	0.000	0.000	0.000	0.249	0.335	0.000
College attendance	0.367	0.372	0.382	0.376	0.371	0.371	0.379
Employment, 26–45	0.533	0.529	0.532	0.533	0.531	0.531	0.526
Own earnings at 30	16.699	16.531	16.770	16.757	16.652	16.631	16.390
<i>Panel B: First birth by 21, by AFQT quartile</i>							
Q1	0.532	0.389	0.466	0.494	0.485	0.476	0.339
Q2	0.438	0.367	0.373	0.401	0.406	0.400	0.305
Q3	0.301	0.259	0.235	0.263	0.281	0.277	0.194
Q4	0.241	0.241	0.184	0.203	0.234	0.231	0.183
Price cut δ (% of cost)	–	–	89.1%	57.8%	–	–	–

Notes: Experiments as defined in Section 6, run at the estimated parameters with common random numbers. Column headings are abbreviated. ADHERENCE = Q4 gives every quartile the top quartile’s adherence; the two price-cut columns scale both method costs by $1 - \delta$, at the δ that matches the fall in first births by 21 of the adherence experiment and at the δ that would deliver a ten percent fall; the three LARC columns add an adherence-proof method to the menu at the cost that reproduces Colorado’s take-up, at the pill’s cost, and to every method in use. Use, pill and long-acting shares are measured over childless woman-periods at ages 18–25; the unintended share is defined as in the estimation moment (Section 3.4). First births by 17 and the long-acting share, both by ability quartile, are in Online Appendix Table OA.33.

Table 8. Counterfactual experiments: consumption-equivalent variation (percent)

	Adherence = Q4	Price cut		LARC menu		
		Matched	−10%	Colorado	Pill’s cost	All users
Q1	15.3	24.9	13.2	5.8	7.2	24.0
Q2	10.2	32.4	16.7	3.8	5.1	20.5
Q3	9.3	54.3	26.3	4.3	6.4	28.8
Q4	0.0	49.0	24.1	1.2	2.6	12.6
All	8.5	38.9	19.7	3.8	5.3	21.2

Price cuts, decomposed: the windfall to women already using a method, and the remainder

	Matched		−10%	
	Windfall to existing users	Net of windfall	Windfall to existing users	Net of windfall
Q1	18.4	4.6	11.3	1.6
Q2	23.4	5.9	14.0	2.1
Q3	33.7	11.1	19.6	4.6
Q4	28.9	11.7	17.1	5.1

Notes: Permanent proportional change in consumption that makes a woman at age 14 indifferent between the baseline and the experiment, from the value function by ability quartile; the last row weights quartiles by population. Columns are the experiments of Table 7. Lower panel, price cuts only: *windfall to existing users* is the expected discounted saving on the methods a woman chose at baseline when their costs fall, which she collects without changing any choice; *net of windfall* is the total less that amount, the value of the fertility control the lower price induces and everything downstream of it, each converted to consumption units separately. Online Appendix OA.15 reports the full decomposition.

7 Conclusion

This paper asks whether the channels standard life-cycle models emphasize, schooling and opportunity costs, can explain why women with higher cognitive ability delay first births. In the NLSY79 the ability gradient in early fertility is steep and closes with age, and the additional first births of low-ability women are births they did not want at that time. The gradient opens while girls are still in high school and survives holding their schooling to date and their 1979 expectations fixed; in the pooled population it persists within contraceptive method, and method selection runs *counter* to it. What remains is fertility control: ability predicts whether a method is used and, among users, how well it works, and predicts nothing about conception among women using none.

A life-cycle model in which cognitive ability shifts adherence reproduces the gradient and its convergence with age while matching education, marriage, labor supply and contraception. Opportunity costs and preferences alone cannot rationalize the data: common adherence is rejected, and the restricted model reproduces how many low-ability women have an early birth only by making those births wanted: the ability gradient in unintended first births by 21 shrinks to 60 percent of its size in the data, the unintended births move from the lowest quartile to the highest within age at first birth, and the labor-supply and marriage moments are distorted to keep births where the timing moments want them. The estimated adherence gradient is large at ages 14–25 and closed by 26, where the gradient in first births stops widening.

Equalizing adherence lowers first births by age 21 by 17% and is worth 15% of lifetime consumption to the lowest ability quartile against nothing to the highest. A uniform price cut that lowers first births by 21 by the same 17% has gains that accrue to the top of the ability distribution. A long-acting method added to the menu at Colorado’s take-up lowers first births by 21 by 7%, close to what that program’s evaluation found, with the largest gains in the lowest quartile and the smallest in the highest; only a menu in which every user’s method is adherence-proof matches equalizing adherence. For disadvantaged women in this cohort, and within this model, the margin with more room is the capacity to control fertility timing rather than the incentive to do so.

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Appendices

Online Appendix

OA.1 Data Construction and Variable Definitions

This appendix defines the samples and the variables used in the paper.

OA.1.1 Data source and cohort coverage

The NLSY79 follows a nationally representative cohort born 1957–1964, aged 14–22 at the first interview in 1979. Interviews are annual from 1979 to 1994 and biennial thereafter.

OA.1.2 Panel structure and alignment to model time

The data form an annual panel indexed by woman i and calendar year t , with age at interview

$$\text{Age}_{it} = t - \text{BirthYear}_i,$$

where birth year is inferred from age at baseline when age is missing. The model runs on four-year periods from age 14, so annual observations map into

$$\text{AgeBin}_{it} = 14 + 4 \left\lfloor \frac{\text{Age}_{it} - 14}{4} \right\rfloor,$$

the periods 14–17, 18–21, ..., 38–41. Period-level objects (employment, experience, the fertility hazard) aggregate annual measures within the period by the rules below.

OA.1.3 Estimation samples and the model’s age grid

Estimation sample. Every moment the model targets, and every AFQT quartile in the paper, is defined on one sample of women (Table OA.1). Of the 6,283 NLSY79 women, 5,939 have a valid AFQT score; 119 of these hold a college degree without having attended college at ages 18–22, a route the model does not allow, and are dropped, leaving 5,820. AFQT quartiles are assigned on this sample, one observation per woman, with cut-points at 17.2, 38.4 and 65.2 percentile points; a woman in a reduced-form sample who is outside it is assigned a quartile by the same cut-points. Education is the highest grade ever completed.

Conditioning sets. Each block of moments is computed over the women, or woman-years, for whom its object is defined (Table OA.2); differences in the number of observations across

Table OA.1. Estimation Sample

	Women
NLSY79 women	6,283
with a valid AFQT score	5,939
with a valid date of first birth, or no birth	5,939
a college degree implies college attendance at 18–22	5,820

Notes: AFQT quartiles are drawn on the last row, one observation per woman.

blocks reflect these conditioning sets and item non-response, not different samples.

Table OA.2. Targeted Moments: Unit of Observation and Conditioning Set

Block	Unit	Conditioning set
Schooling and early fertility	woman	all women; college graduation among women who attended college and had no birth at 14–17
First births by age	woman	all women; cumulative share with a first birth by the end of each period
Marriage by age	woman	all women, by education; cumulative share married by the end of each period
Employment	woman–year	all women, by education and period
Contraception use	woman–year	childless women aged 22–41 who have had intercourse and are not pregnant
Unintended first birth by 21	woman	all women, except those whose first birth by 21 has no wantedness answer
Unintended share by age at first birth	woman	women with a first birth at 14–41 that has a wantedness answer
Method failure (and pill share)	woman–wave	childless women aged 18–33 at the seven contraception waves, using a reversible method, not trying to conceive

Notes: Every block is restricted to the women of Table OA.1 except the method-failure block, which is built on the within-method sample of Section 2 and uses the same quartile cut-points. Employment follows the definitions of Appendix OA.10.

Standard errors. A cluster bootstrap over women (1,000 replications) resamples the schooling, first-birth, marriage, employment, contraception-use and unintended-first-birth moments jointly, so their covariances across blocks are those of the data. The method-failure moments are bootstrapped on their own sample, the unintended shares by age at first birth use binomial variances, and both blocks enter the covariance matrix as independent of the rest.

Reduced-form samples. The samples of Section 2 are woman–wave panels of the seven waves that field the contraception and wantedness modules (1982, 1984, 1985, 1986, 1988, 1990 and 1992). The unconditional sample keeps women aged 18–35 at the wave with a valid AFQT score and drops woman–waves at which the woman reports a pregnancy she had wanted: 32,091 woman–waves (5,884 women). The within-method sample keeps those

woman–waves, removes planned births from the outcome instead, and contains the 16,087 woman–waves at which a pill, barrier or behavioral method is reported. Education in these samples is the highest grade completed at the wave.

Age grid. Every target is measured on the model’s four-year periods, 14–17 through 38–41. First birth and marriage are absorbing states, so the data moment for period a is the share of women who have had a first birth (married) by age $a + 3$, not an average of a person-year indicator over the bin; employment and contraception use are within-period flows, averaged over the four ages of the bin. The adherence parameters $c_{\theta g}$ are indexed by the coarser groups 14–17, 18–25 and 26–33, with ages 34–41 assigned the 26–33 value; the reduced-form tables of Section 2 and Online Appendix OA.5 use the same groups where they split by age. The contraception block starts at age 22: the NLSY79 question is asked of women who have ever had intercourse and are not pregnant, a universe that differs sharply by education at 18 (Section 4).

Exposure and the allowance scale. Two ability-indexed inputs are measured on the estimation sample and held fixed. The exposure share $a_{\theta t}$ (Section 3.4) is the share of women in quartile θ who are childless at the start of period t and report, in the 1983–85 rounds, a first intercourse by its end. The allowance scale a_{θ} (Section 3) is mean total net family income at interviews at ages 14–17 (1979–82, deflated by the interview-year mean), by quartile, relative to the overall mean and renormalised to average one: 0.63, 0.86, 1.07, 1.45. Both are unweighted, like every target.

OA.1.4 Global cleaning conventions and special codes

NLSY79 variables encode nonresponse and survey routing (refusal, don’t know, valid skip, non-interview) as negative values. Three conventions apply throughout:

1. **Nonresponse codes:** values < 0 are missing unless they have a structural meaning, such as “no spouse” for spouse income.
2. **Structural zeros:** economically meaningful zeros, such as spouse income with no partner present, are set to 0.
3. **Deflation:** dollar amounts are in 2016 dollars, deflated by the CPI by calendar year.

OA.1.5 Cognitive ability

Ability is the AFQT score of the NLSY79 created score files. Its quartiles, $q \in \{1, 2, 3, 4\}$, drawn once per woman on the estimation sample of Online Appendix OA.1.3, are the ability measure throughout.

OA.1.6 Education

Education is highest grade completed (HGC), taken as the woman’s maximum over the panel to remove year-to-year reporting variation, and grouped as

$$\text{HSD} : \text{HGC} < 12, \quad \text{HSG} : 12 \leq \text{HGC} < 16, \quad \text{COL} : \text{HGC} \geq 16.$$

College attendance at ages 18–22 is the indicator

$$\mathbb{1} \{ \exists t \text{ s.t. } 18 \leq \text{Age}_{it} \leq 22 \text{ and } \text{HGC}_{it} > 12 \},$$

equal to one if she reports more than 12 completed years at any interview at those ages.

OA.1.7 Fertility and pregnancy histories

First birth timing. Age at first birth is the year of the first child’s birth, from the created child birth-date variables, minus the woman’s birth year,

$$\text{AgeAtFirstBirth}_i = \text{BirthYearChild1}_i - \text{BirthYear}_i,$$

set to 99 for women with no recorded birth in the observation window.

Wantedness and contraception. Pregnancy-level indicators from the fertility and contraception modules are aggregated to the model’s four-year periods.

(i) *Wantedness.* For each pregnancy the fertility module records whether the respondent was using a method when she became pregnant, whether she had stopped it or was not using one in order to become pregnant, and whether she wanted to become pregnant at that time (Q9-81: yes; didn’t matter; not at that time; not at all). The birth is *unintended* if the answer is “not at that time” or “not at all”, and intended if it is “yes” or “didn’t matter” or if she had stopped, or was not using, a method in order to conceive, in which case the question is not asked. The targeted moment uses the record of the pregnancy that produced the woman’s *first* birth: for a first birth in or before 1982, the first pregnancy of the 1982 retrospective history; for a later birth, the first pregnancy reported at the first interview in the year of the birth or the two years after it, since after 1982 the roster starts at the previous interview. Of the 4,580 women in the estimation sample with a first birth at ages 14–41, 4,136 have a matched record and 4,128 of them answer the wantedness question. The moment is, for each quartile, the share of its women whose first birth came at ages 14–21 and was unintended; a woman whose first birth by 21 has no wantedness answer is dropped

from the denominator as missing at random.⁶⁰

(ii) *Contraception at conception.* For each pregnancy p ,

$$\text{NoContraception}_{ip} \equiv \mathbb{1}\{\text{no contraceptive method reported at the time of conception}\},$$

where a method is any reported method, including rhythm and withdrawal.

(iii) *Contraceptive method (Q9-65 battery).* The within-method analysis of Section 2.3 uses the methods a woman reports currently using at each wave, in three tiers ordered by typical-use efficacy: the pill; barrier methods (condom, foam, jelly or cream, suppository, diaphragm, sponge, cervical cap); and behavioral methods (rhythm, withdrawal, natural family planning). IUD and sterilization are excluded, for the reasons given in Section 2. A woman reporting several methods is assigned to the most effective tier. Women whose *only* method is abstinence are excluded, since they have no exposure; abstinence combined with rhythm or withdrawal remains, as a fallible backup.

Classifying a window birth. The within-method outcome is built from child birth-date records rather than from the self-reported contraceptive-failure item (Q9-79), which classifies most imperfect-use failures as deliberate cessation and yields annual failure rates an order of magnitude below typical-use benchmarks. A birth is flagged for a woman–wave if any child’s birth date falls in the 12-month window from July of year t through June of $t + 1$, which starts after the fielding period, so that the method reported at wave t precedes the outcome. The pregnancy record at the next interview classifies the birth. It is *planned* if she had stopped using, or was not using, a method in order to become pregnant (Q9-80), or if she wanted the pregnancy; it is *unintended* if it came too soon or was not wanted at all, and also if it “didn’t matter” while she was using a method: the mistimed-or-unwanted versus intended dichotomy has no place for ambivalence (Santelli et al., 2003), and the failure literature reads use itself as the intention to avoid a birth (Kost et al., 2008; Sundaram et al., 2017). Only unintended births count, so the outcome measures the model’s birth probability $p_n(\theta, g, e)$; the same classification defines the unintended first birth of Section 3.4 and Online Appendix OA.3. A birth whose pregnancy has no record is missing rather than a failure: the literature counts a conception during use (Kost et al., 2008; Sundaram et al., 2017), and a birth that cannot be tied to a record cannot be shown to be one. Of the 911 window

⁶⁰A person-level flag, the maximum over waves of the answer for the first pregnancy *listed at that wave*, is a different object: after 1982 the first listed pregnancy is the first since the previous interview, and such a flag disagrees with the first birth’s own report for 14.7 percent of women (Online Appendix OA.3). The reduced-form outcome “unintended birth” of Section 2 is built the same way for every birth: a live birth in the twelve-month window whose pregnancy record, at the first of the next two interviews that has one, says “not at that time” or “not at all”. Counting any new pregnancy reported at the next interview instead would mix in pregnancies that did not end in a birth in the window and would lose a quarter of them to the same indexing problem.

births in the within-method sample, 144 have no record; 409 are planned, wanted, or “didn’t matter” to a woman using nothing, and are coded zero; 358 are failures. Treating unrecorded births as missing assumes they are missing at random within cells; counting them as failures would put 18 percent of the lowest quartile’s window births and 13 percent of the highest quartile’s into the numerator by default, and the two conventions bound the gradient. The within-method sample conditions only on the method reported at t and keeps the woman-wave itself; the sample unconditional on method drops woman-waves at which the interview reports a pregnancy the woman had wanted, since she is not at risk.

Mapping to the model. Childless women exposed to the risk of conception choose a method each period. The targeted contraception moments are use rates,

$$\Pr\left(\text{AnyMethod}_{it} = 1 \mid \text{childless, ever had intercourse, not pregnant, period } b, \text{Educ}_{it}, \mathbb{1}\{\text{AFQT}_i > \text{median}\}\right),$$

for periods b covering ages 22–38, with the high-school and college groups split at the median AFQT and dropouts pooled (Section 4). In the model the same object is the probability of choosing a method among childless women at risk, $1 - P(0)$, averaged over the corresponding cells; exposure enters the birth probabilities, not this conditional share.

OA.1.8 Marriage and partner outcomes

Marital status. Marital status is defined annually from marriage start and end dates,

$$\text{Married}_{it} = \mathbb{1}\{t \in [\text{MarriageStart}_i, \text{MarriageEnd}_i)\},$$

with a marriage that has a valid start date and no end date treated as ongoing.

Partner earnings and work. Partner wage-and-salary income comes from the spouse/partner earnings modules, with “no spouse” codes set to 0. Partner weeks and hours worked define partner employment in the earnings-process estimation.

OA.1.9 Labor market outcomes: hours, earnings, employment, experience

Annual hours. Total annual hours and annual weeks worked come from the Work History and Weekly Status files.

Annual earnings. Earnings are annual wage-and-salary income of the respondent and of the spouse or partner, in 2016 dollars.

Interpolation and internal consistency checks. Two checks precede estimation: (1) earnings are set to 0 in years with zero hours; (2) earnings recorded as zero in years with low but positive hours are set to missing and linearly interpolated within woman across years with valid neighbouring values. The checks remove measurement-error spikes while preserving low earnings that low hours corroborate.

Employment and experience. A woman-year is employed if it has (i) at least 26 weeks worked, (ii) average weekly hours above 20 and (iii) real annual wage-and-salary income of at least \$10,500 (2016 dollars). Annual experience is $\text{ExpYear}_{it} = 1\{\text{employed}\}$ and cumulative experience is $\text{CumExp}_{it} = \sum_{\tau \leq t} \text{ExpYear}_{i\tau}$.

OA.2 Additional Descriptive Statistics

This appendix reports descriptive tables on birth timing, marriage and labor-market outcomes that supplement Section 2.4. Variables and samples are defined in Appendix OA.1.

OA.2.1 Birth Timing and Education

Table OA.3 reports the share of women whose first birth falls in each model period, by completed education.⁶¹ Early motherhood is concentrated among the least educated: at ages 14–17 the first-birth share is 41% for high-school dropouts, 14% for high-school graduates and 2% for college graduates.

OA.2.2 Early Births and Marriage

Childbearing outside marriage can lower later marriage prospects through economic circumstances and the returns to partner search (Becker, 1991).⁶² Table OA.4 reports the probability of ever marrying for women whose first birth precedes first marriage (Panel A) and for women with no premarital birth (Panel B). In Panel A the rate rises with completed education and falls with the age at the premarital birth; in Panel B it is uniformly high. The association of premarital fertility with lower marriage rates, strongest for the least educated, is consistent with selection and with marriage-market penalties to premarital childbearing.

⁶¹Entries are shares of all women in the education group. Only the displayed age bins are reported, so columns need not sum to one; the residual is first births after the last bin or no observed first birth.

⁶²Bronars and Grogger (1994) document that women with unplanned births are less likely to be married when their children are young.

Table OA.3. Timing of the First Birth by Completed Education (share of all women in the group, %)

	HS Dropout	HS Graduate	College Graduate
14–17	41	14	2
18–21	32	31	6
22–25	12	19	15
26–29	3	9	23
30–33	1	5	14
34–37	1	2	6
38–41	0	1	2

Notes: Entries report the share (%) of women in each education group whose first birth occurred in the indicated model period (four-year age bins) (column percentages).

Table OA.4. Probability of Ever Marriage: Premarital Birth vs. No Premarital Birth

	Education Outcome		
Age at first birth	HS Dropout	HS Graduate	College Graduate
<i>Panel A: Premarital First Birth</i>			
14–17	67	81	83
18–21	58	75	72
22–29	46	59	69
<i>Panel B: No Premarital Birth</i>			
All ages	94	96	98

Notes: Panel A conditions on first birth occurring before first marriage; probabilities shown by age at first birth. Panel B conditions on no birth prior to first marriage (includes women who never marry). Entries report share (%) ever married during the survey window.

OA.2.3 Education, Experience, and Labor-Market Outcomes

Table OA.5 reports wage levels, wage growth and experience by ability quartile and age, with wage statistics computed among employed women as defined in Appendix OA.1. The wage gap between the top and bottom quartiles widens from \$3,354 at age 20 to more than \$35,000 at age 40 (Panel A); returns to experience are steeper at higher ability (Panel B); and higher-ability women accumulate more experience by a given age (Panel C) and see faster annualized wage growth (Panel D). Log wage changes after a non-work spell are weak or negative for lower-ability women and modestly positive for higher-ability women, and time out of the labor force after the first birth rises with ability (Panel E): high-ability women face both steeper returns to continuous experience and larger career costs from early childbearing.

Table OA.5. Descriptive Statistics by Ability: Labor Market Outcomes

Outcome	Ability Quartile			
	Q1 (Low)	Q2	Q3	Q4 (High)
<i>Panel A: Wage at Given Age (2016 \$, workers only)</i>				
Age 20	19,688	21,554	22,811	23,042
Age 25	23,954	27,850	32,250	38,412
Age 30	27,778	33,689	38,978	49,126
Age 40	30,382	40,112	46,392	65,713
<i>Panel B: Return to Experience (log wage growth, %)</i>				
5 years potential exp.	24	35	45	57
10 years potential exp.	39	52	65	78
15 years potential exp.	47	61	75	90
<i>Panel C: Cumulative Work Experience (years)</i>				
At age 25	1.85	3.20	3.85	3.99
At age 30	3.60	6.04	7.20	7.51
At age 40	7.94	12.65	14.64	14.44
<i>Panel D: Annualized Log Wage Growth Rate (%)</i>				
Average	2.69	3.12	3.53	4.33
<i>Panel E: Wage Changes Around Non-Work Spells</i>				
Δ log wage after 1-year gap	0.02	-0.01	0.00	0.04
Δ log wage after 3-year gap	-0.02	-0.03	0.05	0.05
Δ log wage after 5-year gap	-0.01	-0.12	0.02	0.07
Time out after first child (years)	0.31	0.50	0.56	0.60

Notes: “Work” defined as ≥ 20 hours/week for ≥ 26 weeks at $\geq$ minimum wage. Panel A: average wages for workers at each age. Panel B: $\ln w_{t+x} - \ln w_t$ where t is first working year; potential experience cumulates only working years. Panel C: cumulative years meeting work definition. Panel E: log wage change between last working year and 1/3/5 years after gap; “time out” = weeks not working in 5 years post-birth $\div$ 52.

OA.3 What the Unintended-Birth Moment Measures

The two wantedness blocks—the four per-woman shares and the eight composition cells of panel C of Table 6—carry 23 and 26 percent of the nested-test statistic (Table 5). This appendix describes what the survey’s wantedness question records, how the moment is built on both sides, and the per-woman counts behind Figure 1.

The survey question. Each woman’s first birth is matched to its own pregnancy record (Online Appendix OA.1): of the 4,580 women in the structural sample with a first birth at ages 14–41, 4,136 are matched and 4,124 have a classifiable status at conception (the twelve without one are dropped here; ten of them answer the wantedness question and are kept in the targeted moment). That status (Q9-78 to Q9-80) falls into five groups: the method in use failed; she had stopped using a method, not in order to conceive; she had used none since the previous interview and was not trying to conceive; she had stopped in order to conceive; she was using nothing because she wanted a birth. The wantedness question (Q9-81) is asked of the first three groups only, so every mistimed or unwanted birth belongs to one of them.

Three facts follow. *First, unintended first births are mostly births to non-users.* 73.2 percent of mistimed or unwanted first births were conceived with no method in use, 85.1 percent in the lowest quartile and 53.0 percent in the highest, and 54 percent by women who had used no method since the previous interview; method failures account for 27 percent. *Second, the ability gradient is in non-use, not in failure.* The unintended share of first births falls by 16.0 percentage points from the lowest to the highest quartile. Births to women who had never used a method contribute 21.6 points of that fall, births after stopping 1.8, and failures -7.4: failures are a larger share of first births in the top quartile (16.6 percent) than in the bottom (8.5), because top-quartile women use methods more. *Third, the gradient is a timing composition.* Among teenage first births the unintended share lies between 63 and 70 percent in every quartile, and adding single-year indicators for age at first birth moves the top-quartile difference in the unintended share from -0.160 (s.e. 0.021) to $+0.0385$ (s.e. 0.0218). Low-ability women report more of their first births unwanted because they have them earlier, when first births in every quartile are more often unwanted.

The moment on both sides. A birth while a method is in use is a failure, the smallest of the three components and the only one whose gradient runs the other way, so a moment built on failures would ask the model to reproduce through failures a gradient the data place in non-use. The data moment is instead built from the first birth’s own report (a person-level flag from the pregnancy listed first at each wave disagrees with it for 14.7 percent of women) and counts women rather than births. In the model a first birth is unintended if, at the start of its period, $\Delta V + \sigma_\omega \varepsilon^\omega < 0$ at the realised value of the transitory component, whatever method the woman chose (Section 3.4); a woman with a negative realised value who uses nothing because the cost of a method and her taste shock outweigh the expected cost of a birth is the survey’s “never used, not wanting” respondent, unintended on both sides. The sign of ΔV , with no band of indifference around zero, is a fair counterpart of the survey question because the answer “didn’t matter” is rare, 3.7 percent of first births.

The estimated scale of the transitory component is $\sigma_\omega = 0.140$, in the units of ΔV . At age 18, among women childless and at risk, the systematic part of ΔV has a standard deviation of 0.027 across all women and at most 0.012 within a quartile, so the shock is about 5 times the dispersion the state produces and 12 times the within-quartile dispersion. The ability gradient is in the means, -0.015 in the lowest quartile against -0.069 in the highest at 18, a spread of 0.060, with shares below indifference of 0.53 and 0.69: for an individual woman the classification is largely the draw, for a quartile it is the mean. The restricted model needs a larger shock, $\sigma_\omega = 0.325$, and a wider spread of means (-0.091 to -0.234) to reach a gradient of the wrong shape (Section 5.3).

Table OA.6. First births per woman, wanted and unwanted, by AFQT quartile

	Q1	Q2	Q3	Q4	All	Q4 minus Q1
<i>First birth by age 17 (% of women)</i>						
Any	26.7	15.6	8.3	3.2	13.4	-23.6 (1.3)
unintended	16.7	10.8	5.8	2.2	8.8	-14.5 (1.1)
intended	10.0	4.9	2.5	1.0	4.5	-9.0 (0.8)
<i>First birth by age 21 (% of women)</i>						
Any	61.0	45.8	29.8	17.3	38.2	-43.7 (1.7)
unintended	32.5	25.3	17.0	10.2	21.1	-22.3 (1.5)
intended	28.5	20.5	12.9	7.1	17.1	-21.4 (1.4)
<i>First birth by age 25 (% of women)</i>						
Any	75.1	64.6	48.1	33.9	55.2	-41.3 (1.7)
unintended	37.0	31.5	23.2	15.8	26.8	-21.2 (1.6)
intended	38.1	33.1	24.9	18.1	28.4	-20.0 (1.7)
<i>First birth by age 41 (% of women)</i>						
Any	86.3	80.9	71.3	70.0	77.1	-16.3 (1.6)
unintended	40.1	35.6	27.8	21.3	31.2	-18.8 (1.7)
intended	46.2	45.3	43.5	48.7	45.9	2.5 (1.9)
Women	1,326	1,335	1,348	1,347	5,356	

Notes: Percent of women with a first birth by the age shown, split by the wantedness the woman reported for that pregnancy (unintended: “not at that time” or “not at all”; intended: “yes”, “didn’t matter”, or stopped contraception to conceive). Sample: every woman in the structural sample; a woman whose first birth by the age shown has no wantedness record is dropped for that age, so “any” is the sum of the two rows on a common sample of 5,356 women and differs slightly from the first-birth moments of Section 4, which use all 5,820 (by 21: 61.0 against 63.1 percent in the lowest quartile, 17.3 against 18.5 in the highest). Last column: Q4 minus Q1, robust standard error in parentheses.

Per woman. The unintended share of births divides by births, which move with ability too; the object the paper needs counts women, and Table OA.6 does so on the same women throughout. Unwanted births are 116 percent of the lifetime gap in first births, because wanted births run slightly the other way; they are 62 percent of the gap by 17 and 51 percent by 21. The gradient holds within race and ethnicity: the top-to-bottom-quartile difference in unwanted first births by 21 is -15.9 percentage points among non-Black non-Hispanic women (s.e. 2.4), -28.9 among Black women (s.e. 4.0) and -10.5 among Hispanic women (s.e. 3.6), while wanted first births by 41 do not differ significantly by ability within any of the three groups. The targeted moment is the per-woman share at the age where the gap is: the share of each quartile’s women whose first birth came by 21 and was unintended, 32.5 percent in the lowest quartile and 10.2 percent in the highest. The eight composition cells of panel C of Table 6 are the second target; Section 4.2 describes what each block identifies and Section 5.3 how the two models fit them.

Table OA.7. Use of Any Contraceptive Method by AFQT Quartile

	(1) Pooled	(2) + race FE	(3) Non-Black non-Hisp.	(4) Black	(5) Hispanic
AFQT Q2	0.065*** (0.012)	0.059*** (0.012)	0.076*** (0.020)	0.049** (0.019)	0.060** (0.026)
AFQT Q3	0.107*** (0.012)	0.096*** (0.013)	0.119*** (0.020)	0.081*** (0.023)	0.070** (0.032)
AFQT Q4	0.147*** (0.013)	0.132*** (0.014)	0.138*** (0.020)	0.139*** (0.034)	0.194*** (0.035)
Observations	25,009	25,009	14,084	6,966	3,959

Notes: LPM. Unit: woman $\times$ wave, 1984–1992, women aged 18–35 not reporting a pregnancy they had wanted at that interview. Outcome: reports using any contraceptive method. Controls: education group, parity, age at the wave, marital status at the previous interview, window length and wave fixed effects. Columns (3)–(5) estimate the same regression within each NLSY79 screener group. Standard errors clustered by respondent. Reference: AFQT Q1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA.3.1 The Use Margin

The decomposition of Section 2.3 needs the use margin measured on women at risk rather than on births. Table OA.7 estimates a linear probability model with the indicator for using any method as the outcome, the controls of the birth-hazard regressions of Section 2.3, and the woman–wave panel of the seven contraception waves. The item is asked of women who have had intercourse and are not pregnant, and it begins in 1984, so the sample is 25,009 woman–waves.

The raw share reporting a method rises from 62.1 percent in the lowest AFQT quartile to 75.9 percent in the highest. With controls the top-quartile difference is 0.147 (s.e. 0.013), and unlike the within-method gradient it survives race: 0.132 (s.e. 0.014) with race fixed effects, 0.138 (s.e. 0.020) among non-Black non-Hispanic women, 0.139 (s.e. 0.034) among Black women and 0.194 (s.e. 0.035) among Hispanic women. Two limits remain. The item records whether a method was used, not how consistently, so it cannot separate a woman who chose not to contracept from one who meant to and did not manage it. And it conditions on reported intercourse, which itself rises with ability at these ages (Online Appendix OA.6.4).

OA.4 Education and the Ability Gradient

This appendix defines the controls of Table 1 and reports the two further pieces of evidence summarized in Section 2.3: a prospective design that uses only what was measured in 1979, before the births, and the within-education regressions on adult women.

Definitions for Table 1. The unit is a childless woman $\times$ annual interview t , 1979–1993, and the outcome is a first child born in calendar year $t + 1$; women whose first birth has no pregnancy record are left out of the unintended and intended columns. All columns include age and year fixed effects. Enrollment status at t takes four values: in high school, in college, out of school without a diploma, out of school with one; in columns 3–5 every girl is enrolled in high school, so only the highest grade completed at t enters. The 1979 expectations are the schooling the respondent expected to complete (SCHOOL-31) and the number of children she expected (FER-3), entered with the Rotter locus-of-control score and parents’ highest grade, each with a missing indicator.

OA.4.1 Girls followed from 1979

Table OA.8 follows the 1,803 women who were 14–16 and childless at the 1979 interview, with every control measured at that interview. Column (2) holds fixed enrollment status, whether the girl was on track for her age or behind, and the schooling she expected to complete (SCHOOL-31: “as things now stand, what is the highest grade or year you think you will actually complete?”). The gradient in first births by 18 falls from -21.9 to -17.8 percentage points, while the gradient in unintended first births by 18 barely moves, from -14.9 to -14.2 . With expected fertility, the Rotter score and parents’ schooling added (column 3), the gradient in unintended first births by 21 is -20.1 (s.e. 3.3), against -5.3 (s.e. 2.9) for intended ones. The same holds within expected schooling: among girls who expected to finish high school and go no further, 30 percent in the lowest quartile and 13 percent in the highest had an unintended first birth by 21; among girls who expected a college degree, 32 and 5 percent. Race and ethnicity (column 4) reduce the unintended gradient by 21 to -14.5 (s.e. 3.4).

OA.4.2 Adults within completed education

Among adult women, whose schooling is largely complete, I estimate a linear probability model for the probability of a birth separately within each education group:

$$B_{it} = \alpha_e + \sum_{q=2}^4 \beta_{eq} \mathbf{1}[\text{AFQT}_i = q] + \mathbf{x}'_{it} \boldsymbol{\gamma}_e + \delta_t + \varepsilon_{it}, \quad \text{estimated within education } e, \quad (\text{OA.1})$$

where $B_{it} \in \{0, 1\}$ equals one if woman i had a live birth in the 12-month window following snapshot wave t , $\mathbf{1}[\text{AFQT}_i = q]$ are AFQT quartile indicators (quartile 1 omitted), $\mathbf{x}_{it}$ contains parity at the snapshot year (0/1/2+ prior live births), age-at-wave indicators and marital status at the previous annual interview (never married, married, separated, divorced

Table OA.8. First Births by 18 and 21 Among Girls Childless at 14–16 in 1979

	% of women		Q4 minus Q1, percentage points			
	Q1	Q4	(1)	(2)	(3)	(4)
First birth by 18	26.4	4.8	-21.9 (2.4)	-17.8 (2.8)	-15.0 (2.9)	-13.6 (3.1)
unintended	17.6	2.7	-14.9 (2.0)	-14.2 (2.5)	-13.3 (2.5)	-10.2 (2.6)
First birth by 21	50.7	14.5	-36.4 (3.0)	-29.9 (3.6)	-25.4 (3.8)	-23.1 (4.1)
unintended	29.5	7.2	-22.5 (2.6)	-21.7 (3.1)	-20.1 (3.3)	-14.5 (3.4)
intended	21.3	7.4	-14.0 (2.4)	-8.3 (2.7)	-5.3 (2.9)	-8.7 (3.3)
Schooling in 1979: enrollment, grade for age, expected				✓	✓	✓
Expected children, Rotter, parents' schooling					✓	✓
Race and ethnicity						✓
Women				1,803		

Notes: LPM, one observation per woman. Sample: NLSY79 women aged 14–16 and childless at the 1979 interview. Outcomes: first birth by the age shown, and first birth reported unintended or intended on its own pregnancy record, as in Figure 1; a woman whose first birth by that age has no record is left out for that outcome. Entries in columns (1)–(4): top minus bottom AFQT quartile, percentage points, robust standard errors in parentheses. All columns control for age at the 1979 interview. Column (2) adds enrollment status, grade for age (on track, one year behind, two or more) and expected schooling (fewer than 12 years, 12, 13–15, 16 or more), all from 1979; column (3) adds the number of children expected (FER-3), the Rotter score and parents' highest grade; column (4) adds race and ethnicity.

or widowed), and δ_t are snapshot-year fixed effects; these are the controls of the within-method regressions (Section 2.3). Standard errors are clustered by respondent. The sample is all NLSY79 women aged 18–35 not trying to conceive, pooled across the seven survey waves of Section 2.1. If AFQT matters only through education, $\beta_{eq} = 0$ within each group. Figure OA.1 plots the predicted birth probability by quartile within each education group and Table OA.9 reports the coefficients.

The gradient holds within both non-college groups and is absent among college graduates: the top-quartile coefficient is -4.5 percentage points among high-school dropouts ($p < 0.05$) and -2.9 among high-school graduates ($p < 0.01$), while among college graduates the Q3 and Q4 coefficients are positive with standard errors of about 2.5 pp. Panel B of Table OA.9 shows the same pattern for unintended births, those the woman reported as having come too soon or not wanted at all. Pooling the education groups, the top quartile has 1.1 fewer unintended births per hundred women a year (s.e. 0.3), about half the raw rate (1.6 against 3.3); marital status accounts for most of the difference from the gap in births of any kind, because married women have more births but fewer unintended ones. These estimates are unchanged by attrition re-weighting and strengthen when women whose AFQT was measured after a first birth are dropped (Additional Results, Section S.3). Family background does not account for them: adding parents' highest grade, log 1979 family income and 1979 poverty status leaves the top-quartile coefficient at -0.047 (s.e. 0.019) among dropouts and -0.024 (s.e. 0.006) among high-school graduates. A race control cuts both by more than half: with race added in its place the two coefficients are -0.019 (s.e. 0.018) and -0.012 (s.e. 0.006), and with the two blocks together -0.026 and -0.010 . Under the fuller specification

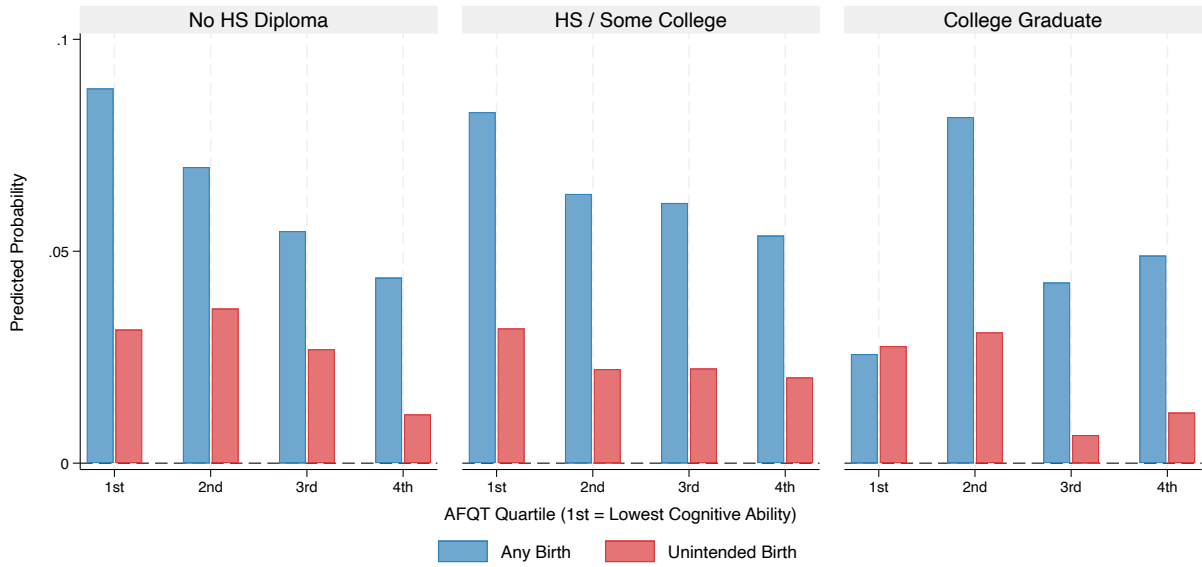


Figure OA.1. Birth probability by AFQT quartile and education group, unconditional on contraceptive method. Denominator: NLSY79 women aged 18–35 not actively trying to conceive, pooled across seven survey waves. Blue bars: any live birth in the 12-month window. Red bars: a live birth in the same window that the woman reported, on its pregnancy record at the next interview that has one, as having come too soon or not wanted at all. Births with no pregnancy record at either of the next two interviews (21 percent; 26 in the lowest quartile, 15 in the highest) are left out of the red bars rather than counted as wanted. LPM-predicted probabilities from Equation (OA.1), evaluated at AFQT quartile means; controls set at sample averages within each education group.

of Table OA.11, which also holds enrollment and the age-adjusted score, both coefficients are -0.004 (s.e. 0.021 and 0.007). Online Appendix OA.5.2 separates the two blocks across every sample and outcome and decomposes the movement variable by variable. The first-birth timing gradient on which the model is estimated is present and monotone within every race and ethnicity group (Online Appendix OA.5.5), though not of the same size in each; the residual 18–35 hazard gradient is concentrated among Hispanic women and, less precisely, Black women. The adult gradient within education is thus smaller than the teen gradient of Table 1 and, unlike it, does not survive a race control: most of the ability gap in first births is realized before schooling is, and it is there that the gradient survives family background and race together (columns 4–5 of Table 1).

Table OA.9. Within-Education Birth Probability by AFQT Quartile (LPM, 1982–1992)

	(1) No HS Diploma	(2) HS / Some College	(3) College
<i>Panel A: Any birth</i>			
AFQT Q2	-0.019* (0.010)	-0.019*** (0.005)	0.056** (0.028)
AFQT Q3	-0.034** (0.015)	-0.021*** (0.005)	0.017 (0.025)
AFQT Q4	-0.045** (0.018)	-0.029*** (0.006)	0.023 (0.024)
Observations	5,251	22,342	4,498
<i>Panel B: Unintended birth</i>			
AFQT Q2	0.005 (0.007)	-0.010*** (0.003)	0.003 (0.024)
AFQT Q3	-0.005 (0.011)	-0.009*** (0.003)	-0.021 (0.021)
AFQT Q4	-0.020* (0.011)	-0.012*** (0.004)	-0.016 (0.021)
Observations	5,149	22,023	4,476

Notes: LPM estimated separately within each education group. Unit: woman $\times$ snapshot wave (1982–1992). Panel A outcome: any live birth within a fixed 12-month window (July of t through June of $t + 1$). Panel B outcome: any birth reported as unwanted or mistimed. Sample: women not trying to conceive, unconditional on contraceptive method. Controls: parity, age-at-wave dummies, marital status at the previous interview, and snapshot-year FE. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA.5 Robustness of the Within-Method Gradient

This appendix checks the within-method gradient of Section 2.3 against additional controls and sampling weights in equations (OA.1) and (2) (Appendix OA.5.1), separates the family-background controls from the race controls (Appendix OA.5.2), checks the classification of the failure outcome (Appendix OA.5.4), and estimates the gradient within race and ethnicity (Appendix OA.5.5). The Additional Results (not for publication, Section S.3) add the descriptive relationship between ability and family background, the timing of the AFQT relative to first births, attrition by ability, and the birth hazard without conditioning on method.

OA.5.1 Additional Controls and Sampling Weights

Table OA.10 reports the baseline within-method regressions of Section 2.3, whose coefficients the main text quotes.

Table OA.10. Within-Method Birth Probability by AFQT Quartile (LPM, 1982–1992): Baseline Estimates

	(1) Pooled	(2) Pill	(3) Barrier	(4) Behavioral
AFQT Q2	-0.013*** (0.004)	-0.011** (0.005)	-0.024* (0.013)	-0.006 (0.019)
AFQT Q3	-0.011** (0.004)	-0.011** (0.005)	-0.024* (0.013)	0.015 (0.021)
AFQT Q4	-0.014*** (0.004)	-0.015*** (0.005)	-0.025** (0.013)	0.017 (0.023)
Observations	16,087	11,299	3,828	960

Notes: Estimates of equation (2). Linear probability of a birth in the twelve months after the wave among women using contraception, 1982–1992 waves, by AFQT quartile (Q1 omitted). A birth counts only if reported unwanted at the time (Online Appendix OA.1.7). Controls: education, parity, age, marital status at the previous interview and wave fixed effects; the pooled column adds method fixed effects. Standard errors clustered by woman.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.11 varies the controls of the baseline regressions cumulatively, from a specification without age and marital status (column 1) through enrollment status and age-adjusted AFQT quartiles (columns 3–4), unweighted (Panel A) and with the NLSY79 cross-sectional weights (Panel B). The last three columns add the 1979 family-background variables alone (column 5), race and ethnicity alone (column 6), and both (column 7); Appendix OA.5.2 takes up what each block does. Table OA.12 reports the baseline and the fully controlled specification by method. The unconditional counterpart is in Appendix OA.5.6; the within-education and unintended-birth counterparts are in the Additional Results, Section S.3.

The demographic controls leave the within-method gradient where it is: the top-quartile coefficient is -0.016 without age and marital status, -0.014 at the baseline and -0.011 with enrollment and the age-adjusted score. So does family background: adding parents' highest grade, log 1979 family income and 1979 poverty status leaves it at -0.011 (s.e. 0.004) in column (5). The race and ethnicity indicators, added in their place, take it to 0.000 (s.e. 0.004) in column (6), and the two blocks together to -0.001 (s.e. 0.005) in column (7). They are not interchangeable, and it is race that does the work; Appendix OA.5.2 separates them across every sample and outcome. By method (Table OA.12), the pill gradient goes from -0.015 to -0.003 (s.e. 0.005) under the full specification; the barrier and behavioral samples are too small to say. The within-education and unintended-birth gradients behave the same

way. These are statements about the annual hazard at ages 18–35 conditional on parity and marital status, which leaves out the teenage years where most of the gap in unwanted first births arises. Appendix [OA.5.5](#) takes up what a race control does in these regressions and shows that the first-birth timing gradient the model is estimated on holds within every race and ethnicity group, as does the gradient in unwanted first births by 21 counted per woman (Online Appendix [OA.3](#)).

I report unweighted estimates and use the weights as a specification test. The NLSY79 design is stratified on race, ethnicity and 1978 family income, covariates rather than outcomes, so unweighted least squares is consistent for a correctly specified conditional mean and more efficient than weighted least squares (DuMouchel and Duncan, 1983; Winship and Radbill, 1994; Wooldridge, 1999); a gap between the two estimates signals heterogeneity across strata, not a reason to prefer the weighted estimate (Solon et al., 2015). The moments of the structural model are computed unweighted. The last row of each table reports the DuMouchel and Duncan (1983) test that the weighted and unweighted AFQT coefficients are equal.

Within method, weighting leaves the baseline gradient at -0.012 (s.e. 0.005), and the test does not reject in any column ($p = 0.67$ at the baseline). Births unconditional on method (Appendix [OA.5.6](#)) are the exception: the weighted baseline coefficient is -0.019 (s.e. 0.006) against -0.030 unweighted, and the test rejects in every column until race is added. The heterogeneity is the one Appendix [OA.5.5](#) documents, a gradient concentrated among Hispanic and, less precisely, Black women, whom the weights down-weight; Solon et al. (2015) recommend reporting it rather than averaging it away. The object the structural model targets is not affected: the ratio of first births by age 21 between the lowest and highest quartiles is 4.8 with the 1979 weights against 3.7 unweighted.

OA.5.2 Family Background or Race?

The specification that removes the adult gradient adds two blocks of controls at once: the woman’s race and ethnicity, and her family’s circumstances in 1979—her parents’ highest grade completed, the family’s income and whether it was below the poverty line. Reported together, as they are in column (7) of Table [OA.11](#), they invite the reading that socioeconomic status rather than ability accounts for the gradient. The two blocks do different work. Table [OA.13](#) separates them and collects the top-quartile coefficient across the samples and outcomes the paper uses: column (2) adds family background alone, column (3) race and ethnicity alone, and column (4) both.

Family background is not what absorbs the gradient. Within method the top-quartile coefficient is -0.014 (s.e. 0.004) at the baseline and -0.015 (s.e. 0.005) with family background added; among pill users -0.015 and -0.013 ; for first births of any kind -0.030

Table OA.11. Within-Method Birth Probability by AFQT Quartile: Additional Controls and Sampling Weights (LPM, Pooled)

	(1) Previous	(2) Baseline	(3) + Enrol.	(4) Age-adj.	(5) + Family	(6) + Race	(7) + Both
<i>Panel A: Unweighted</i>							
AFQT Q2	-0.014*** (0.004)	-0.013*** (0.004)	-0.012*** (0.004)	-0.008** (0.004)	-0.008** (0.004)	-0.004 (0.004)	-0.004 (0.004)
AFQT Q3	-0.013*** (0.004)	-0.011** (0.004)	-0.010** (0.004)	-0.009** (0.004)	-0.010** (0.004)	-0.001 (0.004)	-0.002 (0.004)
AFQT Q4	-0.016*** (0.004)	-0.014*** (0.004)	-0.013*** (0.004)	-0.011*** (0.004)	-0.011** (0.004)	0.000 (0.004)	-0.001 (0.005)
<i>Panel B: NLSY79 sampling weights</i>							
AFQT Q2	-0.014*** (0.005)	-0.013** (0.005)	-0.013** (0.005)	-0.007 (0.005)	-0.008* (0.005)	-0.003 (0.005)	-0.004 (0.005)
AFQT Q3	-0.011** (0.005)	-0.010* (0.005)	-0.009* (0.005)	-0.007 (0.005)	-0.007 (0.005)	-0.001 (0.005)	-0.001 (0.005)
AFQT Q4	-0.013** (0.005)	-0.012** (0.005)	-0.011** (0.005)	-0.008* (0.005)	-0.007 (0.005)	-0.000 (0.005)	-0.000 (0.005)
Test weighted = unweighted, p-value	0.500	0.665	0.637	0.803	0.623	0.998	0.992
Observations	16,087	16,087	16,087	16,087	16,087	16,087	16,087

Notes: Sample and outcome as in Table OA.10, column (1); all columns include method-group FE. Column (1): education group at the wave, parity, window length and snapshot-year FE. Column (2), the baseline: (1) plus age-at-wave dummies and marital status at the previous interview. (3) adds enrollment status at the wave; (4) replaces the AFQT quartiles by quartiles of AFQT residualized on age-at-test dummies; (5) adds mother's and father's highest grade, log 1979 family income and 1979 poverty status (with missing indicators); (6) adds the race and ethnicity indicators instead; (7) adds both blocks. Panel B weights every column by the NLSY79 cross-sectional weight for the wave. Test row: p -value of the [DuMouchel and Duncan \(1983\)](#) test that the weighted and unweighted AFQT coefficients are equal (joint test on the normalized weight $\times$ AFQT interactions in the unweighted regression augmented with the weight and its interaction with every regressor). SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and -0.026 ; for unintended first births -0.011 and -0.009 ; among high-school dropouts -0.045 and -0.047 ; among high-school graduates -0.029 and -0.024 . In two of the six rows the coefficient is larger in absolute value with family background than without it, and in none of them does it lose significance. The race and ethnicity indicators do what the two blocks do together: column (3) is within a thousandth of column (4) in four of the six rows, and neither column leaves a coefficient significant at the five percent level, apart from the high-school-graduate row in column (3).

The [Gelbach \(2016\)](#) decomposition reaches the same conclusion from the other direction, by assigning the movement between the baseline and the fully controlled specification to the individual covariates (Table OA.14). In the unconditional sample the Black and Hispanic indicators account for 110 percent of the movement in the top-quartile coefficient, and within method for 116 percent—more than all of it in both, because the four family-background

Table OA.12. Within-Method Birth Probability by AFQT Quartile and Method: Baseline vs. Full Controls

	Pill		Barrier		Behavioral	
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Full	Baseline	Full	Baseline	Full
AFQT Q2	-0.011** (0.005)	-0.004 (0.004)	-0.024* (0.013)	-0.012 (0.011)	-0.006 (0.019)	0.013 (0.019)
AFQT Q3	-0.011** (0.005)	-0.002 (0.005)	-0.024* (0.013)	-0.014 (0.011)	0.015 (0.021)	0.035* (0.021)
AFQT Q4	-0.015*** (0.005)	-0.003 (0.005)	-0.025** (0.013)	-0.007 (0.012)	0.017 (0.023)	0.031 (0.024)
Observations	11,299	11,299	3,828	3,828	960	960

Notes: Sample and outcome as in Table OA.10, columns (2)–(4). “Baseline” is Table OA.10. “Full” is column (7) of Table OA.11, Panel A: enrollment status, age-adjusted AFQT quartiles, race and family background added; unweighted. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

variables contribute with the opposite sign. Parents’ schooling contributes next to nothing—mother’s and father’s high-school completion carry coefficients of 0.004 and -0.003 , neither distinguishable from zero—and 1979 family income contributes with the wrong sign: conditional on ability, women from higher-income families have *more* births in a given wave, so controlling for income widens the ability gap rather than narrowing it.

What a race control does here is mechanical. Forty-seven percent of lowest-quartile women are Black and seven percent of top-quartile women are, so a race indicator removes the between-group component of AFQT by construction. The AFQT is measured before labor-market entry, and the premarket-skill literature reads that between-group component as skill rather than as a confounder (Carneiro and Heckman, 2003; Carneiro et al., 2005; Neal and Johnson, 1996); footnote 12 states why the baseline excludes race on those grounds. Whether the ability gradient is a race gradient in disguise is a separate question, and it is answered the way the paper answers the same question for education—within groups. Appendix OA.5.5 does so: the first-birth timing gradient on which the model is estimated is present and monotone within every race and ethnicity group, as is the gradient in unwanted first births by age 21 counted per woman (Online Appendix OA.3). The within-method failure gradient is the exception, and Appendix OA.5.5 reports it as one.

OA.5.3 Cognitive or Non-Cognitive Skill?

Heckman et al. (2006) find that non-cognitive skill matters at least as much as cognitive skill

Table OA.13. Family Background and Race Separately: Top-Quartile Coefficient by Sample and Outcome (LPM)

	(1) Baseline	(2) + Family background	(3) + Race	(4) + Both
Within method, pooled	-0.014*** (0.004)	-0.015*** (0.005)	-0.004 (0.005)	-0.005 (0.005)
Oral contraceptive	-0.015*** (0.005)	-0.013*** (0.005)	-0.004 (0.005)	-0.004 (0.005)
Any first birth	-0.030*** (0.005)	-0.026*** (0.005)	-0.010* (0.005)	-0.010* (0.006)
Unintended first birth	-0.011*** (0.003)	-0.009*** (0.003)	0.001 (0.003)	0.001 (0.004)
No HS diploma	-0.045** (0.018)	-0.047** (0.019)	-0.019 (0.018)	-0.026 (0.019)
HS / some college	-0.029*** (0.006)	-0.024*** (0.006)	-0.012** (0.006)	-0.010 (0.006)
Parents' schooling, 1979 income, poverty		✓		✓
Race and ethnicity			✓	✓

Notes: Each cell is the AFQT top-quartile coefficient from a separate linear probability model, with the standard error below. Rows 1–2 are the within-method failure hazard of equation (2); rows 3–4 the unconditional first-birth and unintended-first-birth hazards; rows 5–6 the within-education first-birth hazard of equation (OA.1) inside the two non-college groups. Every column holds the baseline controls of the row's own equation, so that column (1) reproduces the estimate reported in the main text or in Appendix OA.4. Column (2) adds mother's and father's highest grade, log 1979 family income and 1979 poverty status, with missing indicators; column (3) adds the Black and Hispanic indicators instead; column (4) adds both. SE clustered by respondent. Reference: AFQT Q1; non-Black non-Hispanic.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

for teenage pregnancy, which raises the question of whether the AFQT quartile stands in for traits the ASVAB does not measure. The NLSY79 fields two non-cognitive scales, and it fields both before the outcome windows of this paper open: the [Rotter \(1966\)](#) locus-of-control score in 1979 and the [Rosenberg \(1965\)](#) self-esteem score in 1980, against contraception and wantedness modules that begin in 1982. Neither scale can therefore be an outcome of the fertility behaviour it is used to explain.

Table OA.15 adds them to the baseline specification of each row, one block at a time, and repeats the race column of Table OA.13 as a scale. The non-cognitive blocks do almost nothing. Within method the top-quartile coefficient is -0.014 at the baseline and -0.013 with both scales; for first births of any kind -0.030 and -0.029 ; for unintended first births -0.011 and -0.010 ; among high-school dropouts -0.045 and -0.047 . Across the six rows the two scales together leave between 89 and 105 percent of the baseline coefficient standing, and in two rows the coefficient is larger in absolute value with them than without. The race

Table OA.14. Gelbach Decomposition of the Change in the Top-Quartile Coefficient (Unconditional Sample)

	Contribution to Q4 gap	Share of gap
Mother completed HS	0.0010	-5.7%
Father completed HS	-0.0017	9.2%
Log family income 1979	0.0038	-20.8%
In poverty 1979	-0.0019	10.6%
Black	-0.0153	83.9%
Hispanic	-0.0048	26.2%
Family background (four rows above)	0.0012	-6.5%
Race and ethnicity	-0.0201	110.1%
Total (baseline minus controlled)	-0.0189	100%

Notes: Each row is $\hat{\gamma}_k \hat{\beta}_k$, where $\hat{\gamma}_k$ is the top-quartile coefficient in a regression of background variable k on the AFQT quartile dummies and the baseline controls, and $\hat{\beta}_k$ its coefficient in the fully controlled specification (Gelbach, 2016). Rows sum to the difference between the top-quartile coefficients with and without the background block, up to the missing-indicator terms. The within-method version is in the Additional Results, Section S.3.

indicators, in the same table and on the same samples, leave 26 percent of the within-method coefficient and -7 percent of the unintended-birth coefficient.

Two qualifications keep this from being more than it is. The scales are coarse—four item pairs and ten items—so classical measurement error attenuates how much of the gradient they can absorb, and the comparison is between controls rather than between the latent factors a factor model would recover. What the table rules out is that the ability gradient documented here is an artifact of the non-cognitive traits the NLSY79 happens to measure. It does not rule out a non-cognitive component in the adherence parameter itself, and the paper does not claim otherwise: Section 3.4 reads $c_{\theta g}$ as *fertility-control capacity* rather than cognition alone, which is why the counterfactuals of Section 6 remove the need for adherence instead of raising skill.

OA.5.4 Measuring the Failure Outcome

The failure outcome rests on the classification of each window birth by the pregnancy record at the next interview (Online Appendix OA.1.7), and two of its conventions are choices rather than facts. Table OA.16 reports the within-method gradient under four conventions, from the most inclusive, column (a), to the one the paper uses, column (d).

The conventions differ mainly in the treatment of births whose pregnancy has no record: 16 percent of window births, 18 percent in the lowest AFQT quartile against 13 percent in the highest, because the women who miss an interview or leave the block blank are disproportionately of low measured ability. Since the outcome is rare, assigning those births

Table OA.15. Non-Cognitive Skill: Top-Quartile Coefficient by Sample and Outcome (LPM)

	(1) Baseline	(2) + Rotter	(3) + Rosenberg	(4) + Both	(5) + Race
Within method, pooled	-0.014*** (0.004)	-0.013*** (0.004)	-0.014*** (0.004)	-0.013*** (0.004)	-0.004 (0.005)
Oral contraceptive	-0.015*** (0.005)	-0.013*** (0.005)	-0.015*** (0.005)	-0.014*** (0.005)	-0.004 (0.005)
Any first birth	-0.030*** (0.005)	-0.030*** (0.005)	-0.029*** (0.005)	-0.029*** (0.005)	-0.010* (0.005)
Unintended first birth	-0.011*** (0.003)	-0.010*** (0.003)	-0.010*** (0.003)	-0.010*** (0.003)	0.001 (0.003)
No HS diploma	-0.045** (0.018)	-0.047** (0.018)	-0.045** (0.019)	-0.047** (0.019)	-0.019 (0.018)
HS / some college	-0.029*** (0.006)	-0.028*** (0.006)	-0.028*** (0.006)	-0.028*** (0.006)	-0.012** (0.006)
Rotter locus of control, 1979		✓		✓	
Rosenberg self-esteem, 1980			✓	✓	
Race and ethnicity					✓

Notes: Each cell is the AFQT top-quartile coefficient from a separate linear probability model, with the standard error below; samples, outcomes and baseline controls are those of Table OA.13, so column (1) is identical in the two tables. Column (2) adds the Rotter locus-of-control score fielded in 1979, column (3) the Rosenberg self-esteem score fielded in 1980, column (4) both, and column (5) the Black and Hispanic indicators in their place. Both scores are standardised over the estimation sample, so their coefficients are per standard deviation; a missing score is set to zero and flagged with an indicator, the convention used for parents' schooling in Table OA.13. SE clustered by respondent. Reference: AFQT Q1; non-Black non-Hispanic.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

to the numerator moves the estimate: the top-quartile coefficient is -0.022 (s.e. 0.005) in column (a), which counts them as failures, and -0.018 (s.e. 0.005) in column (b), which treats them as missing. Column (c) also stops counting the births the woman reports she wanted, and column (d) adds the rule for “didn’t matter”: -0.014 (s.e. 0.004).

Neither end of the ladder is free of assumption. Column (a) requires that an unrecorded pregnancy was avoided, and column (d) that unrecorded births are missing at random within the cells of the regression, which the quartile difference above makes unlikely in the direction of the gradient. The two therefore bracket the gradient, between -0.014 and -0.022, significant at either end. The same classification defines the eight within-method failure moments, so the model is estimated on column (d).

Table OA.16. The Within-Method Gradient Under Four Conventions for the Failure Outcome

	(1) (a) no record = failure	(2) (b) no record missing	(3) (c) + wanted not a failure	(4) (d) + ambivalence rule
AFQT Q2	-0.018*** (0.005)	-0.015*** (0.004)	-0.014*** (0.004)	-0.013*** (0.004)
AFQT Q3	-0.019*** (0.005)	-0.014*** (0.005)	-0.012*** (0.004)	-0.011** (0.004)
AFQT Q4	-0.022*** (0.005)	-0.018*** (0.005)	-0.015*** (0.004)	-0.014*** (0.004)
Failures	530	391	368	347
Mean failure rate	0.0327	0.0243	0.0229	0.0216
Observations	16,226	16,087	16,087	16,087

Notes: LPM, equation (2), pooled across method groups with method fixed effects and the controls of Table OA.10. Column (a): a window birth is a failure unless the pregnancy record at the next interview reports it planned, and a birth with no record is a failure. (b): births with no record are missing. (c): births the woman reports she wanted are not failures. (d): “didn’t matter” is a failure when a method was in use and not when none was; the classification used in the paper and in the estimation. Failures and the mean failure rate are computed over the estimation sample, so they fall short of the counts in Appendix OA.1.7 by the woman-waves dropped for missing covariates or singleton cells. Standard errors clustered by respondent. Reference: AFQT Q1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA.5.5 The Ability Gradient Within Race and Ethnicity

Table OA.17 reports the share of women with a first birth by ages 17, 19, 21 and 25 by AFQT quartile separately for non-Black non-Hispanic, Black and Hispanic women, using the pooled quartile cut-points of the main text, and for the largest group also with quartiles drawn within the group. The timing gradient is present, large and monotone in every group: the ratio of first births by 21 between the lowest and highest quartiles is 3.7 among non-Black non-Hispanic women, 3.2 among Hispanic women and 2.3 among Black women. Among non-Black non-Hispanic women, 18.8% of the lowest quartile have a first birth by 17 against 2.1% of the highest, 15.1% and 1.5% with quartiles drawn within the group. The first-birth timing moments the structural model is estimated on are not a race gradient in disguise.

Table OA.18 does the same for the reduced-form hazards of Section 2, where the picture differs. The 12-month birth hazard at ages 18–35, conditional on education, parity, age, marital status and wave, has a top-quartile coefficient of -0.010 with race fixed effects and 0.002 (s.e. 0.007) among non-Black non-Hispanic women, against -0.011 and -0.032 among Black and Hispanic women; within contraceptive method the coefficients are -0.004, 0.003 (s.e. 0.006), -0.008 and -0.003. Two things reconcile this with Table OA.17. First, the hazard regressions condition on education, parity and marital status at ages 18–35, by which time, for the majority group, the ability gradient has already played out through schooling, marriage and the first birth itself; the timing gradient at 14–21 is what those controls absorb. Second, a race indicator removes the between-group part of the variation in ability along

Table OA.17. First-Birth Timing by AFQT Quartile Within Race and Ethnicity

	Q1 (lowest)	Q2	Q3	Q4 (highest)	Q1/Q4	Women
<i>Non-Black non-Hispanic (pooled quartiles)</i>						
First birth by 17	0.188	0.110	0.046	0.021	8.9	405/ 747/1,074/1,276
First birth by 19	0.398	0.265	0.148	0.066	6.0	
First birth by 21	0.560	0.426	0.270	0.150	3.7	
First birth by 25	0.726	0.637	0.472	0.331	2.2	
<i>Black (pooled quartiles)</i>						
First birth by 17	0.224	0.144	0.074	0.049	4.6	693/ 466/ 242/ 103
First birth by 19	0.433	0.294	0.178	0.155	2.8	
First birth by 21	0.623	0.451	0.302	0.272	2.3	
First birth by 25	0.765	0.642	0.521	0.417	1.8	
<i>Hispanic (pooled quartiles)</i>						
First birth by 17	0.157	0.070	0.045	0.000	–	375/ 270/ 177/ 111
First birth by 19	0.365	0.200	0.209	0.108	3.4	
First birth by 21	0.549	0.363	0.266	0.171	3.2	
First birth by 25	0.757	0.656	0.418	0.315	2.4	
<i>Non-Black non-Hispanic (quartiles drawn within the group)</i>						
First birth by 17	0.151	0.062	0.040	0.015	10.1	876/ 875/ 876/ 875
First birth by 19	0.332	0.195	0.112	0.048	6.9	
First birth by 21	0.490	0.343	0.216	0.123	4.0	
First birth by 25	0.688	0.535	0.425	0.293	2.4	

Notes: NLSY79 women with a valid AFQT score. Share with a first birth by the age shown, by AFQT quartile and NLSY79 race/ethnicity screener group. The first three panels use the pooled quartile cut-points of the main text; the last panel draws quartiles within the non-Black non-Hispanic group. Q1/Q4: ratio of the two shares. Women: number of women in Q1/Q2/Q3/Q4.

with the racial difference in timing, and AFQT accounts for most of that difference: 74 percent of the Black–white gap in first births by 21 (from 20.3 to 5.2 percentage points) and all of the Hispanic–white gap (from 10.9 to -1.5), the pattern [Neal and Johnson \(1996\)](#) and [Carneiro et al. \(2005\)](#) document for wages.⁶³ Restricting the sample to white women, as much of the NLSY79 literature does ([Francesconi, 2002](#); [Sheran, 2007](#); [Van der Klaauw, 1996](#)), or to the cross-sectional sample ([Heckman et al., 2006](#)), would leave the timing and unwanted-birth moments unchanged and remove the within-method gradient.

⁶³Within-method failure (Panel C) is the exception. Black and Hispanic women using the same method have annual failure rates of 3.9 and 2.3 percent against 1.4 for non-Black non-Hispanic women, as [Sundaram et al. \(2017\)](#) find in the National Survey of Family Growth, and AFQT accounts for little of that gap (from 2.1 to 2.1 percentage points for Black women, 0.5 to 0.5 for Hispanic women). The within-method evidence of Section 2.3 is therefore a pooled fact, as are the eight within-method failure moments of the estimation, which carry 6 percent of the nested test (Section 5.3); the timing and unwanted-birth moments hold within every group (Table OA.17 and Online Appendix OA.3).

Table OA.18. Reduced-Form Birth Hazards by AFQT Quartile Within Race and Ethnicity (LPM)

	(1) Pooled	(2) Pooled + race FE	(3) Non-Black non-Hispanic	(4) Black	(5) Hispanic
<i>Panel A: Any birth, unconditional on method (eq. 1)</i>					
AFQT Q2	-0.019*** (0.005)	-0.011** (0.005)	0.001 (0.007)	-0.011 (0.007)	-0.019* (0.011)
AFQT Q3	-0.024*** (0.005)	-0.010** (0.005)	0.003 (0.007)	-0.010 (0.009)	-0.028** (0.012)
AFQT Q4	-0.030*** (0.005)	-0.010* (0.005)	0.002 (0.007)	-0.011 (0.014)	-0.032** (0.015)
Observations	32,091	32,091	18,187	8,764	5,140
<i>Panel B: Unintended birth, unconditional on method (eq. 1)</i>					
AFQT Q2	-0.006** (0.003)	-0.001 (0.003)	0.007* (0.004)	-0.004 (0.005)	-0.009 (0.007)
AFQT Q3	-0.009*** (0.003)	-0.001 (0.003)	0.006 (0.004)	-0.001 (0.007)	-0.006 (0.009)
AFQT Q4	-0.011*** (0.003)	0.001 (0.003)	0.006* (0.004)	0.006 (0.011)	-0.007 (0.009)
Observations	31,648	31,648	18,020	8,580	5,048
<i>Panel C: Birth within contraceptive method (eq. 2)</i>					
AFQT Q2	-0.013*** (0.004)	-0.008* (0.004)	0.001 (0.006)	-0.014** (0.007)	-0.012 (0.009)
AFQT Q3	-0.011** (0.004)	-0.003 (0.005)	0.003 (0.006)	-0.002 (0.010)	-0.005 (0.011)
AFQT Q4	-0.014*** (0.004)	-0.004 (0.005)	0.003 (0.006)	-0.008 (0.012)	-0.003 (0.012)
Observations	16,087	16,087	9,708	4,076	2,303

Notes: Panels A and B: sample and controls as in the unconditional hazard regressions of Appendix OA.5.6, i.e. all NLSY79 women aged 18–35 not trying to conceive; Panel C: as in Table OA.10, column (1). Column (1) pooled; column (2) adds race/ethnicity fixed effects; columns (3)–(5) estimate separately within group, keeping the pooled AFQT quartile cut-points. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA.5.6 Unconditional Birth Hazard by Ability

The within-method test conditions on the method to isolate the per-use effectiveness margin. A complementary question is whether the *unconditional* ability gradient in births—

Table OA.19. Birth Probability Unconditional on Contraceptive Method (LPM, 1982–1992)

	(1) Any birth	(2) Unintended birth
AFQT Q2	-0.019*** (0.005)	-0.006** (0.003)
AFQT Q3	-0.024*** (0.005)	-0.009*** (0.003)
AFQT Q4	-0.030*** (0.005)	-0.011*** (0.003)
Observations	32,091	31,648

Notes: LPM. Denominator: all NLSY79 women aged 18–35 not actively trying to conceive, pooled across seven snapshot cohorts ($t \in \{1982, 1984, 1985, 1986, 1988, 1990, 1992\}$). Unconditional on method choice; includes users and non-users. Column 1: any live birth in a fixed 12-month window (July of t through June of $t + 1$). Column 2: any birth reported as unwanted or mistimed. Controls: education group, parity, age-at-wave dummies, marital status at the previous interview, and snapshot-year FE. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

combining both the method-selection and per-use margins—is quantitatively important. I estimate a version of equation (2) on the full population of NLSY79 women aged 18–35 who were not actively trying to conceive, pooled across the same survey waves, without conditioning on the method used. The outcome is again a live birth in the fixed 12-month window. In a separate specification, I restrict the outcome to births that the woman reports as unwanted or mistimed, isolating conceptions that represent failures of fertility control.

Table OA.19 reports the results. The unconditional birth gradient is declining in AFQT: Q4 women are 3.0 percentage points less likely to have a birth than Q1 women ($p < 0.01$). The unintended-birth column shows a gradient of the same shape (−1.1 pp for Q4, $p < 0.01$), which is what a difference in how reliably methods are used would produce and is harder to reconcile with a difference in how many births were planned. The unconditional gradient is larger than the within-method one (−1.4 pp), but the two are estimated on different populations: the unconditional sample includes women using no method. The comparison that isolates method selection holds the sample fixed. Among method users, adding method fixed effects moves the Q4 coefficient from −1.2 to −1.4 pp, so method selection compresses rather than generates the ability gradient. The results are consistent with a genuine per-use effectiveness channel: higher-ability women implement a chosen method more consistently and correctly. The structural model developed in Section 3 formalizes this channel as ability-dependent adherence with a method of externally fixed efficacy.

OA.6 Alternative Channels

This appendix takes up the channels other than adherence that could produce an ability gradient in births among contraceptive users: the failure-rate gradient that the method mix alone would predict at published typical-use rates (Appendix OA.6.1); the first-birth rate among women using no method (Appendix OA.6.3) and the ability gradient in coital frequency (Appendix OA.6.4), which test the identifying restriction on baseline birth risk; and a bound on differential abortion (Appendix OA.6.5). The method-selection multinomial logit of Section 2.3 is in Appendix OA.6.2; the ability gradient in men’s first parenthood and marital sorting are in the Additional Results (not for publication, Section S.4).

OA.6.1 Method Effectiveness and the Method Mix

Table OA.20 lists every method in the NLSY79 contraception battery with its first-year typical-use and perfect-use failure rate (Trussell, 2011) and the share of women in the bottom and top AFQT quartiles reporting it. Among the reversible methods with appreciable use in this cohort the pill has the lowest typical-use failure rate, and it is used more by low-ability women. Assigning each woman the typical-use failure rate of her most effective reported method and averaging by quartile, the method mix alone predicts a Q1/Q4 failure ratio of 1.06, against an observed birth-rate ratio of 2.72: method choice accounts for a few percent of the gap, and the rest is realized within method. Among all women asked the method question, pill use is nearly the same across quartiles (45% vs. 47%) because bottom-quartile women are far more often sterilized (22% vs. 6%); among users of reversible methods, the population of Section 2.3, the pill share is higher at the bottom.

OA.6.2 Method-Selection Multinomial Logit

Table OA.21 reports predicted method-choice probabilities from the multinomial logit of Section 2.3. Predicted probabilities are averaged over the sample distribution of education, parity, and wave. The pill-use probability declines from 76.5% in the lowest AFQT quartile to 62.5% in the highest, while the barrier-method probability rises from 15.5% to 29.4%. Behavioral method use is roughly constant across quartiles. Figure OA.2 plots these probabilities.

OA.6.3 Baseline Birth Risk Among Non-Users

The model attributes the within-method ability gradient to adherence because baseline birth risk p_{be}^0 is common within an education–age cell: in equation (3) an ability gradient in p^0 and one in $c_{\theta g}$ are observationally equivalent given the method. The moment that separates them

Table OA.20. Contraceptive Methods in the NLSY79: Failure Rates and Use by AFQT Quartile

NLSY79 method	Tier	First-year failure (%)		Share reporting method (%)		
		Typical use	Perfect use	Q1 (lowest)	Q4 (highest)	All
Female sterilization	Excluded	0.5	0.5	22.4	5.8	13.5
Male sterilization (partner)	Excluded	0.15	0.10	1.7	4.4	3.4
IUD	Excluded	0.8	0.6	5.2	2.0	3.3
Norplant (implant)	Excluded	0.05	0.05	0.0	0.0	0.0
Depo-Provera (injectable)	Excluded	6	0.2	0.0	0.0	0.0
Oral contraceptive pill	Pill	9	0.3	44.9	46.5	47.0
Diaphragm	Barrier	12	6	1.6	9.5	4.8
Cervical cap	Barrier	16	9	0.1	0.1	0.1
Sponge	Barrier	12	9	0.4	1.9	1.1
Condom	Barrier	18	2	7.0	14.4	10.8
Foam	Barrier	28	18	1.8	2.1	2.0
Jelly / cream	Barrier	28	18	0.8	0.5	0.6
Suppository	Barrier	28	18	1.0	1.3	1.2
Natural family planning	Behavioral	24	0.4	0.3	1.2	0.7
Rhythm / calendar	Behavioral	24	5	1.5	3.7	2.7
Withdrawal	Behavioral	22	4	1.5	3.4	2.8
Douching	Behavioral	85	85	2.1	0.2	0.9
Abstinence	Behavioral	0	0	1.3	3.8	2.3
Any method		85 (no method)		87.9	88.4	88.6
Woman-waves				5,787	6,216	24,331

Notes: Failure rates are the percentage of women experiencing an unintended pregnancy within the first year of use, from [Trussell \(2011\)](#), Table 1 (cervical cap from Trussell 2004; douching is not a method and is assigned the no-method rate). Shares: NLSY79 women aged 18–35 asked the method question in 1982–1992, woman-wave observations; multiple methods may be reported. “Excluded” methods are LARC and sterilization, which the within-method analysis omits (Section 2.1).

is the first-birth rate among childless women using *no* method, who face p_{be}^0 whatever their adherence, so that any unmodelled correlate of ability in fecundity or exposure must show up there. The sample is the universe of the contraception screening question (FFER-139), childless women aged 18–33 who were sexually active and not pregnant, which is also the universe behind the contraception moments of Section 4; the method battery behind the within-method regressions is asked only of users. The outcome is a first birth before the next observed wave.

Table OA.22 reports the result. Among non-users the first-birth rate over the interval to the next wave (1.8 years on average) is 4.2 percent in the lowest quartile and 4.7 in the highest, 2.4 and 2.5 percent a year annualized; panel B of Table 2 reports the same rate for the lowest quartile and the regression-adjusted differences of column (2). The top-quartile gradient within education, age and wave is +0.7 pp (s.e. 1.2 pp), a zero with the sign opposite to the one a confound would need; among users over the same waves and controls

Table OA.21. Predicted Method-Choice Probabilities by AFQT Quartile (1982–1992)

AFQT Quartile	P(Pill)	P(Barrier)	P(Behavioral)
Q1 (Lowest)	0.765	0.155	0.080
Q2	0.718	0.193	0.089
Q3	0.687	0.227	0.086
Q4 (Highest)	0.625	0.294	0.081
Observations	15,095		

Notes: Predicted probabilities from multinomial logit estimated on NLSY79 women reporting reversible contraceptive use, pooled across seven snapshot waves (1982–1992). Base outcome: pill. Probabilities averaged over the sample distribution of education group, parity, and wave. LARC and sterilization excluded. The estimation sample is 15,095 woman-waves, against 16,087 in the within-method linear probability model of Table OA.10. The difference is mainly that the logit leaves out woman-waves at which the respondent reported a pregnancy she had wanted, as the unconditional sample does, since her method report then says little about avoiding a birth; the within-method regressions keep them and count a planned birth in the window as no failure.

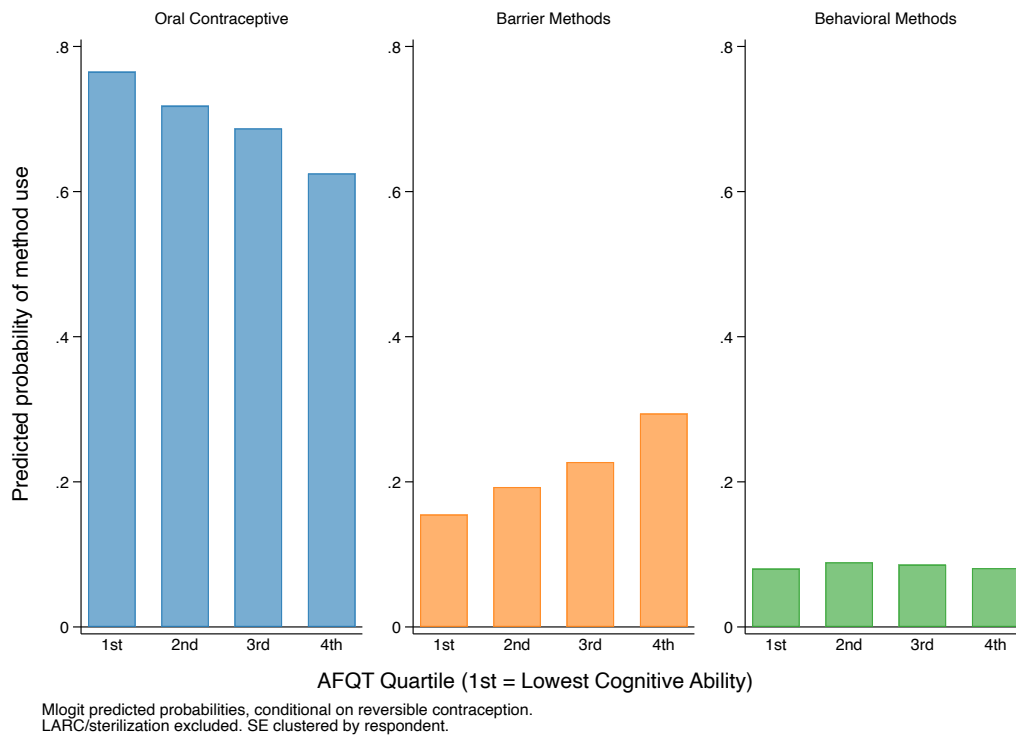


Figure OA.2. Predicted method-choice probability by AFQT quartile. Multinomial logit estimated on NLSY79 women reporting reversible contraceptive use across seven survey waves (1982–1992); base outcome = pill. Predicted probabilities averaged over the sample distribution of education, parity, and wave. LARC/sterilization excluded.

it is -1.9 pp. The gradient appears where a method is in use and not where none is.

The bound is not tight. The non-user cells hold 745 to 938 person-years, so the ninety-

Table OA.22. First-birth rate by AFQT quartile, among non-users and users

	(1) No method, raw	(2) No method, controls	(3) Users, controls
AFQT Q2	-0.0002 (0.0105)	-0.0008 (0.0108)	-0.0192 (0.0120)
AFQT Q3	0.0021 (0.0101)	0.0018 (0.0107)	-0.0126 (0.0117)
AFQT Q4	0.0053 (0.0103)	0.0071 (0.0122)	-0.0187 (0.0118)
Observations	3,372	3,372	5,819
R-squared	0.000	0.008	0.010

Notes: Linear probability models. Unit: woman $\times$ screening wave. Outcome: a first birth between the interview and the next observed wave. Sample: childless women aged 18–33 who were sexually active and not currently pregnant, the universe of the contraception screening question (FFER-139 in 1984–1990, Q9-64G/H from 1992). Columns (1)–(2) are women reporting no method, column (3) women reporting a method. Controls in (2) and (3): education group, age at wave, wave, and exposure length. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

five percent interval around the non-user gradient, from about -1.7 to $+3.1$ pp, contains a baseline-risk gradient as large as the within-method one, -1.4 pp. What the check establishes is that the one moment that can contradict the restriction does not, and that its point estimate has the wrong sign for the alternative, as does the ability gradient in coital frequency (Online Appendix OA.6.4). The mirror-image model, in which ability moves baseline risk instead of adherence and everything else is re-estimated on the same moments, is estimated alongside the model of Section 3 and reported with the results.

OA.6.4 Coital Frequency and the Exposure Margin

Adherence $c_{\theta g}$ is the share of *exposures* on which a method is used as intended, while the moment that identifies it is an *annual* failure rate; the two differ by the number of exposures per year, which the model does not contain. If low-ability women have intercourse more often, they fail more often per year at identical per-exposure adherence, and the estimator books the difference as adherence.

The NLSY79 asks women how many times they had intercourse in the past month in 1984 and 1985 (FFER-137), two of the snapshot waves behind equation (2): 4,764 person-waves and 140 failures, ages 19 to 29, mean frequency 8.3 times a month. Table OA.23 shows that the premise does not hold. Frequency rises with ability: the top-to-bottom gap is 1.92 acts a month (s.e. 0.44, $t = 4.41$) with the baseline controls of Table OA.10, 1.59 (s.e. 0.41) with marital status at the wave added, and it is present within each marital state.

Table OA.23. Coital frequency by AFQT quartile

	Times per month		log(1 + times)	
	(1)	(2)	(3)	(4)
	Baseline	+ Marital at wave	Baseline	+ Marital at wave
AFQT Q2	0.9522** (0.3927)	0.7297* (0.3886)	0.1320*** (0.0394)	0.1015*** (0.0386)
AFQT Q3	1.4903*** (0.4150)	1.1397*** (0.4115)	0.1883*** (0.0405)	0.1409*** (0.0396)
AFQT Q4	1.9197*** (0.4356)	1.5900*** (0.4300)	0.2685*** (0.0433)	0.2242*** (0.0423)
Observations	4,764	4,764	4,764	4,764

Notes: Dependent variable: times of intercourse in the past month (FFER-137), levels in columns (1)–(2) and log(1 + times) in (3)–(4). Unit: woman $\times$ snapshot wave. Sample: the within-method sample of Table OA.10 restricted to the 1984 and 1985 waves. Controls in (1) and (3): the baseline of Table OA.10 (education group at the wave, parity, age-at-wave dummies, marital status at the previous interview, method group and wave); columns (2) and (4) add marital status at the wave. Exposure length is constant at one year on these waves and is omitted. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.24, on identical rows in every column, shows that frequency barely moves failures: one additional act per month adds 0.00028 to the annual failure probability (s.e. 0.00035), and controlling for it moves the top-quartile gradient slightly away from zero, from -0.0224 to -0.0239 with frequency-quintile fixed effects. No coefficient on these rows is individually significant; the columns establish the direction of the movement, not its precision. Attributing the whole top-quartile gradient to exposure would require top-quartile women to have about 81 fewer acts per month than bottom-quartile women, ten times the sample mean, rather than the 1.92 more they report.

The same two objects give a sharper statement. With n exposures per year and p the annual within-method failure rate, the per-exposure rate is $r = 1 - (1 - p)^{1/n}$, the object the technology in equation (3) parameterizes. At quartile means the annual rate falls from 5.0 to 1.9 percent between the bottom and top quartiles, a ratio of 0.391, while the per-exposure rate falls from 6.2 to 1.8 per ten thousand acts, a ratio of 0.294: the targeted moment understates the ability gradient in the parameter it identifies, because high-ability women reach a lower annual failure rate on roughly thirty percent more exposures.

Three limits remain. The measure exists only for 1984 and 1985, so the exercise speaks to model periods $t = 3$ and $t = 4$ and not to the teens, where the counterfactual of Section 6 does its work; a one-month recall window is a proxy for the rate of exposure; and frequency is self-reported, although overturning the result would require high-ability women to *over-report*.

Table OA.24. Within-method birth probability, with and without coital frequency

	(1) No control	(2) Linear	(3) Quadratic	(4) Quintile FE	(5) + Marital at wave
AFQT Q2	-0.0217** (0.0087)	-0.0219** (0.0087)	-0.0219** (0.0087)	-0.0223** (0.0087)	-0.0224*** (0.0087)
AFQT Q3	-0.0158* (0.0096)	-0.0162* (0.0096)	-0.0162* (0.0096)	-0.0165* (0.0096)	-0.0167* (0.0096)
AFQT Q4	-0.0224** (0.0094)	-0.0229** (0.0094)	-0.0229** (0.0094)	-0.0239** (0.0095)	-0.0240** (0.0095)
Times per month		0.0003 (0.0003)	0.0003 (0.0005)		
Observations	4,764	4,764	4,764	4,764	4,764

Notes: Linear probability models for a first birth in the year following the snapshot wave, the outcome of Table OA.10. Every column is estimated on the same 4,764 rows, the within-method sample restricted to 1984 and 1985. Column (1) is the baseline specification of Table OA.10: education group, parity, age-at-wave dummies, marital status at the previous interview, method group and wave. Column (2) adds times per month linearly, (3) adds its square, (4) replaces it by frequency-quintile fixed effects, and (5) adds marital status at the wave to (4). SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA.6.5 A Bound on Differential Abortion

Let b_q be the share of quartile- q women with a first birth by age A and A_q their abortions per woman by A . Abortion can generate the birth gradient only if high-ability women abort more: the conception gradient is zero when $A_4 - A_1 = b_1 - b_4$. The NLSY79 records roughly 40% of abortions (Jones and Forrest, 1992), so reported counts a_q are scaled by $s = 2.5$. Table OA.25 reports, by age 20, 22 and 25, the levels, the gaps and the ratio A_4/A_1 that abortion would need to explain the entire birth gap. By age 25 the birth gap is 0.417 and the reported abortion gap -0.028; explaining the birth gap would require top-quartile women to abort 2.04 times as many pregnancies as bottom-quartile women, against an observed ratio of 1.18. Even assigning every unreported abortion to the top quartile leaves a conception gap of 0.104. Table OA.26 reports the regression version.

Table OA.25. Births, Abortions, and the Bound on the Abortion Channel, by AFQT Quartile

	By age 20	By age 22	By age 25
<i>Panel A: Levels by AFQT quartile</i>			
First birth (share), Q1 (lowest)	0.493	0.633	0.752
First birth (share), Q2	0.348	0.488	0.642
First birth (share), Q3	0.216	0.336	0.474
First birth (share), Q4 (highest)	0.120	0.203	0.336
Reported abortions per woman, Q1 (lowest)	0.082	0.116	0.161
Reported abortions per woman, Q2	0.106	0.143	0.187
Reported abortions per woman, Q3	0.103	0.143	0.179
Reported abortions per woman, Q4 (highest)	0.102	0.144	0.189
Any reported abortion (share), Q1 (lowest)	0.069	0.088	0.115
Any reported abortion (share), Q2	0.091	0.112	0.137
Any reported abortion (share), Q3	0.091	0.120	0.138
Any reported abortion (share), Q4 (highest)	0.088	0.116	0.144
Any pregnancy: birth or abortion (share), Q1 (lowest)	0.520	0.655	0.766
Any pregnancy: birth or abortion (share), Q2	0.398	0.531	0.684
Any pregnancy: birth or abortion (share), Q3	0.283	0.406	0.537
Any pregnancy: birth or abortion (share), Q4 (highest)	0.192	0.286	0.421
<i>Panel B: Q1 minus Q4 gaps</i>			
Birth gap, $b_1 - b_4$	0.373	0.430	0.417
Reported abortion gap, $a_1 - a_4$	-0.020	-0.028	-0.028
Any-pregnancy gap	0.328	0.369	0.345
Conception gap, reported abortions ($s = 1$)	0.353	0.403	0.388
Conception gap, uniform under-reporting ($s = 2.5$)	0.323	0.361	0.346
Conception gap, under-reporting in Q4 only ($s_1 = 1, s_4 = 2.5$)	0.200	0.187	0.104
<i>Panel C: Could abortion alone generate the birth gap?</i>			
Required A_4/A_1 for a zero conception gap, uniform s	2.82	2.48	2.04
Required A_4/A_1 if Q1 fully reported ($s_1 = 1$)	5.54	4.70	3.59
Observed a_4/a_1	1.24	1.24	1.18
Women	5,939		

Notes: The 5,939 NLSY79 women with a valid AFQT score, quartiles at the estimation-sample cut-points (Online Appendix OA.1.3). Abortions are dated reports from the 1984–2000 fertility modules (the 1984 module asks about lifetime abortions; later modules since the last interview), de-duplicated by month and year. s is the under-reporting factor applied to reported counts. “Required A_4/A_1 ” is $(s a_1 + b_1 - b_4)/(s a_1)$, the ratio of true top- to bottom-quartile abortions at which conceptions would be equal across the two quartiles. The adversarial row assigns all unreported abortions, $(s - 1)$ times the sample mean per woman, to the top quartile.

Table OA.26. Any Abortion by Age 25 on AFQT Quartile (LPM)

	(1) Any abortion by 25	(2) + race, cohort	(3) + family background	(4) Abortions by 25 (count)
AFQT Q2	0.022* (0.012)	0.026** (0.012)	0.021 (0.013)	0.028 (0.019)
AFQT Q3	0.023* (0.012)	0.029** (0.013)	0.017 (0.013)	0.016 (0.020)
AFQT Q4	0.030** (0.012)	0.037*** (0.014)	0.023 (0.015)	0.030 (0.022)
Women	5,939	5,939	5,939	5,939

Notes: One observation per woman. Column (4) outcome is the number of reported abortions by age 25. Robust SE.
Reference: AFQT Q1.

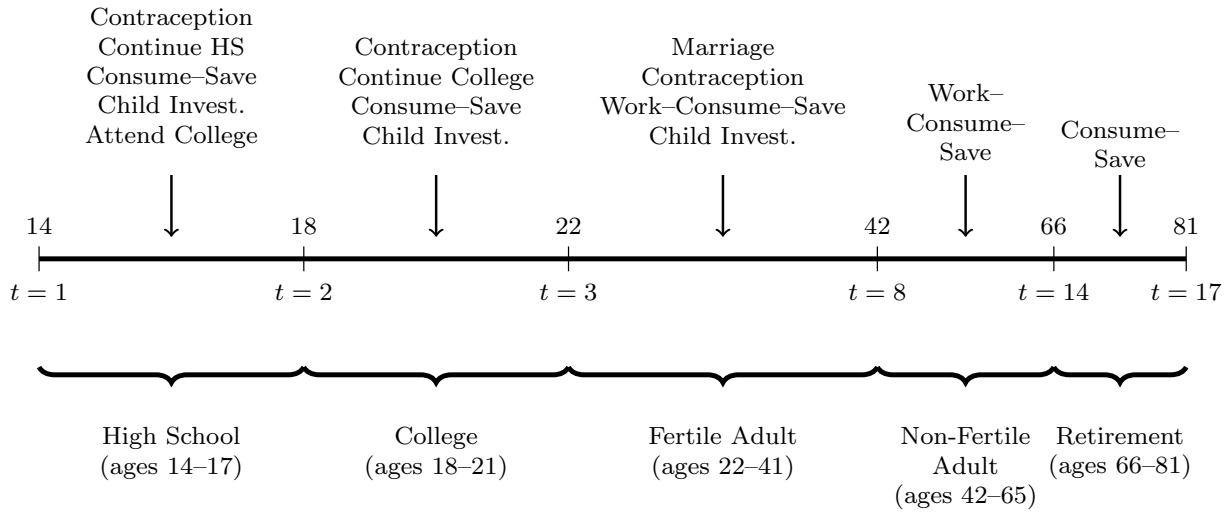
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA.7 The Model

OA.7.1 Environment, timing, and state space

Section 3 of the paper gives the state, the choices and the young-adult recursion; this appendix states the schooling stages, the post-fertile stages and the transitory wantedness shock. Figure OA.3 shows the life cycle of a woman who attends college; Figures OA.4 and OA.5 show the within-period timing in young adulthood and in the college period (the teen period is analogous).

Figure OA.3. Women Attending College Life Cycle

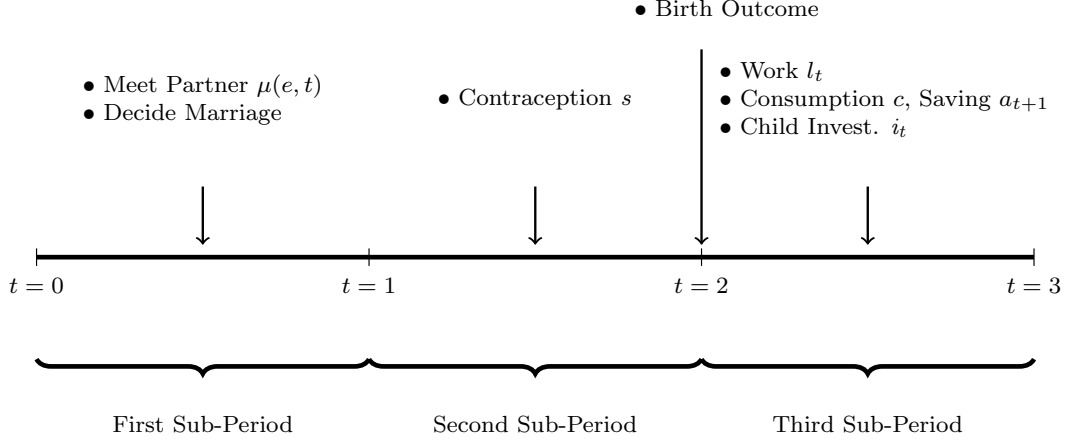


Notes: The figure shows the life cycle of a woman who attends college. A model period is four years and age = $10 + 4t$, so $t = 1$ covers ages 14–17 and $t = 17$ ages 78–81. The life cycle is divided into five stages: (i) high school ($t = 1$), (ii) college age ($t = 2$), (iii) fertile adult ($t = 3-7$, ending at 41), (iv) non-fertile working adult ($t = 8-13$), and (v) retirement ($t = 14-17$). Ages are shown above the timeline and period indices below it; the horizontal axis is not drawn to scale, since the two schooling periods carry most of the decisions. Above the timeline the figure lists the decisions taken in each stage.

The two schooling stages differ from young adulthood in two respects. There is no marriage sub-stage while a woman is enrolled: women who leave school after $t = 1$ meet partners from $t = 2$ (ages 18–21), college women from $t = 3$. And the schooling continuation decision—continue high school or drop out at $t = 1$, graduate from college or drop out at $t = 2$ —is taken *after* the fertility outcome, so that a birth can raise the cost of continuing; high-school graduates also face the college-entry decision at the end of $t = 1$. Women not enrolled in college at $t = 2$ follow the young-adult timing of Figure OA.4. Sections OA.7.4–OA.7.7 give the Bellman equations stage by stage in the notation fixed below (Section OA.7.1).

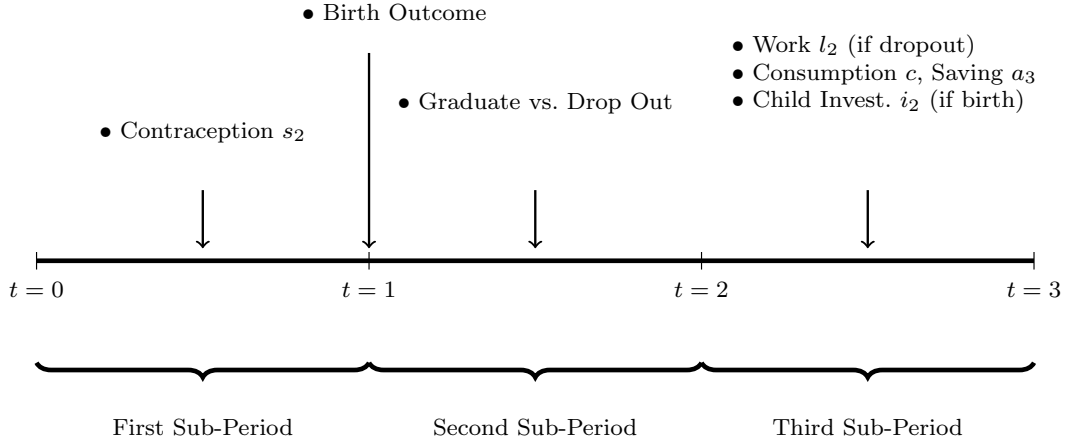
Time is discrete in four-year periods, $t \in \{1, \dots, T\}$, with decisions made for $t =$

Figure OA.4. Childless Women Between Ages 22–41: Within-Period Timing



Notes: Each period is divided into three sub-periods: (i) marriage (if single), (ii) contraception (if childless and fertile), and (iii) labor supply, consumption–saving, and (if a birth occurs) child investment.

Figure OA.5. College-Track Women (Ages 18–21): Within-Period Timing



Notes: Within-period timing at $t = 2$ (ages 18–21) for women enrolled in college. Sub-period (i): choose a contraceptive method n_2 , then a stochastic birth may occur. Sub-period (ii): after observing the birth outcome, choose whether to graduate or drop out; having a child raises the psychic cost of continuing ($\kappa_{k,C}$). Sub-period (iii): consumption–saving and child investment (if birth). College students do not work ($l_2 = 0$) and receive the student allowance w_C net of tuition TC ; dropouts enter the labor market immediately and choose l_2 . Women not enrolled in college at $t = 2$ follow the young-adult timing in Figure OA.4.

$1, \dots, T - 1$ and terminal period $T = 17$ (age 78), in which agents consume all remaining resources and die. Fertility is feasible through ages 14–41, i.e. for $t \leq T_F = 7$; women can work through age 65 and retire from age 66 onward. Each woman can have at most one child, and the child resides with the household for one period only, so child investment is a one-time choice made in the birth period. The state $\Omega_{it} = \{a_t, \theta_i, e_t, x_t, m_t, k_t, m_k\}$ is defined in Section 3.1; V_t^ℓ denotes the value function in period t and within-period sub-stage $\ell \in \{1, 2, 3\}$, and in non-fertile and retirement periods, which have a single stage, ℓ is suppressed. Choice variables are next-period assets $a_{t+1} \in [0, \bar{a}]$, consumption $c_t \geq 0$, female labor supply $l_t \in \{0, 1\}$,⁶⁴ child investment $i_t \geq 0$ (only if $k_t = 2$), and the contraceptive method $n_t \in \{0, 1, 2\}$ (only in fertile periods when $k_t = 1$).

OA.7.2 Income, taxes/transfers, equivalence scales, and experience

Disposable resources. Gross annual household non-asset income, $\tilde{y}_t^0(\Omega_{it}, l_t)$, includes female earnings when working and spousal earnings when married. Disposable annual income follows the approximation to the U.S. tax-and-transfer system in Daruich and Fernández (2024),

$$y_t^a(\Omega_{it}, l_t) = \lambda(\tilde{y}_t^0(\Omega_{it}, l_t))^{1-\tau} + T(m_t),$$

and model-period resources aggregate annual resources:

$$y_t(\Omega_{it}, l_t) = 4 y_t^a(\Omega_{it}, l_t).$$

Budget constraint. The within-period budget constraint is

$$\phi_c(m_t, \mathbb{1}\{k_t = 2\}) c_t + a_{t+1} = (1 + r)a_t + y_t(\Omega_{it}, l_t) - \mathbb{1}\{k_t = 2\} i_t,$$

where $\phi_c(\cdot)$ is the equivalence scale, which depends on marital status and the presence of a newborn.

Experience accumulation. Experience evolves according to

$$x_{t+1} = x_t + \mathbb{1}\{l_t = 1\}.$$

⁶⁴I follow Attanasio et al. (2008) in modeling female labor supply.

OA.7.3 Discrete choices and taste shocks

Discrete alternatives (labor supply, schooling continuation, college entry) carry i.i.d. Type-I extreme value taste shocks. For a generic discrete choice $d \in \mathcal{D}$ with shocks $\varepsilon_t(d)$, scale $\sigma_{\mathcal{D}}$ and choice-specific value net of shocks $v_t(\Omega, d)$, the ex-ante value is

$$V_t(\Omega) = \mathbb{E}_{\varepsilon} \left[\max_{d \in \mathcal{D}} \{v_t(\Omega, d) + \sigma_{\mathcal{D}} \varepsilon_t(d)\} \right] = \gamma_E \sigma_{\mathcal{D}} + \sigma_{\mathcal{D}} \log \sum_{d \in \mathcal{D}} \exp \left(\frac{v_t(\Omega, d)}{\sigma_{\mathcal{D}}} \right),$$

where γ_E is the Euler–Mascheroni constant.

OA.7.4 Working, non-fertile and retired households (ages 42–81; $t = 8$ –17)

After age 41 ($t \geq 8$) fertility risk is absent and no child is present. For $t = 8, \dots, 13$ (ages 42–65) the household chooses whether the woman works, $l_t \in \{0, 1\}$, consumption and next-period assets. It draws taste shocks $\{\varepsilon_t(0), \varepsilon_t(1)\}$ for labor supply, so that with $v_t(\Omega_{it}, l)$ the choice-specific value net of shocks, $V_t(\Omega_{it}) = \mathbb{E}_{\varepsilon} [\max_{l \in \{0, 1\}} \{v_t(\Omega_{it}, l) + \sigma_l \varepsilon_t(l)\}]$.

Conditional on l , the choice-specific problem is

$$\begin{aligned} v_t(\Omega_{it}, l) = & \max_{a_{t+1} \geq 0, c_t \geq 0} \{u(c_t) + \psi_l \mathbb{1}_{\{l=1\}} + \beta \mathbb{E}[V_{t+1}(\Omega_{i,t+1}) \mid \Omega_{it}, l, a_{t+1}]\} \\ \text{s.t.} \quad & \phi_c(m_t) c_t + a_{t+1} = (1+r)a_t + y_t(\Omega_{it}, l), \\ & x_{t+1} = x_t + \mathbb{1}_{\{l=1\}}, \end{aligned}$$

where gross annual household income is

$$\tilde{y}_t^0(\Omega_{it}, l) = \mathbb{1}_{\{l=1\}} w(\Omega_{it}) + \mathbb{1}_{\{m_t=1\}} w^h(\Omega_{it}),$$

and disposable model-period resources are

$$y_t(\Omega_{it}, l) = 4 \left[\lambda (\tilde{y}_t^0(\Omega_{it}, l))^{1-\tau} + T(m_t) \right].$$

Here $w(\Omega_{it})$ is the woman’s wage as a function of education, experience and the other state variables, and spousal labor income $w^h(\Omega_{it})$ is received only when married.⁶⁵

From age 66 onward ($t = 14, \dots, 16$) the household is retired: labor supply is zero, the only intertemporal choice is savings, and gross annual non-asset income is the woman’s own Social Security benefit plus the spousal benefit when married,

⁶⁵The husband’s earnings are a reduced-form function of the woman’s observed characteristics, as in [Adda et al. \(2017\)](#); [Sheran \(2007\)](#); [Van der Klaauw \(1996\)](#).

$\tilde{y}_t^0 = ss_t(e_t) + \mathbb{1}_{\{m_t=1\}} ss_t^h(e_t, m_k)$, both depending on education. The problem is the one above with $l_t = 0$ and ψ_l absent; in the terminal period $t = T = 17$ agents consume all remaining resources and die.

OA.7.5 Young adulthood (ages 22–41; $t = 3$ –7)

In young adulthood schooling is complete (e_t is fixed) and marriage-market and fertility risk are active until $t = T_F = 7$. The three sub-stages—marriage if single, contraception if childless, then labor supply, consumption–saving and child investment—are stated in Section 3.2, with $j \in \{k, nk, ok\}$ indexing the child-status outcome ($k_t = 2, 1, 3$).

The transitory wantedness shock. Writing $\Delta V_t = V_t^{3,k}(\Omega_{it}) - V_t^{3,nk}(\Omega_{it})$ and $\tilde{\Delta V}_t = \Delta V_t + \sigma_\omega \varepsilon_{it}^\omega$ for its realised value, the contraception sub-stage value is $V_t^{3,nk}(\Omega_{it}) + \mathbb{E}_{\varepsilon^\omega} [\sigma \log \sum_{n=0}^2 \exp\{(p_n(\theta_i, g(t), e_t) \tilde{\Delta V}_t - \kappa_n)/\sigma\}]$, where the outer expectation is computed by seven-point Gauss–Hermite quadrature and the choice probabilities are the logit shares of Section 3.4 evaluated at $\tilde{\Delta V}_t$. In simulation each woman draws her own ε_{it}^ω , chooses a method at the realised value, and a first birth in the period is recorded as unintended if $\tilde{\Delta V}_t < 0$. With $\sigma_\omega = 0$ the expectation is the integrand. With the exposure shares of Section 4, the woman is at risk with probability $a_{\theta t}$ and the value is $(1 - a_{\theta t})V_t^{3,nk} + a_{\theta t} V_t^2$.

Never having a child. In the last fertile period $t = T_F = 7$, a reduced-form utility shifter $\phi_{nk}(e)$ for remaining childless, by education, matches the observed mass of women who never have children; it enters the no-birth continuation value inside the contraception sub-stage:

$$V_7^{3,nk}(\Omega_{i7}) + \mathbb{1}_{\{k_7=1\}} \phi_{nk}(e_7).$$

OA.7.6 College age (ages 18–21; $t = 2$)

Women who do not enroll in college at $t = 2$ follow the young-adult problem of Section OA.7.5. This subsection states the college track: enrolled women do not work, receive the student allowance w_C , pay direct schooling costs TC and, after observing the fertility outcome, choose between graduating and dropping out into the labor market as high-school graduates; a child raises the psychic cost of continuing by $\kappa_{k,C}$.

Sub-stage 3: consumption–saving and (if a birth occurs) child investment. Let $j \in \{k, nk\}$ denote the fertility outcome in $t = 2$ and $d \in \{G, CD\}$ the education decision

from sub-stage 2 (graduate vs. drop out). For $d = G$, the household solves

$$\begin{aligned} v_2^{3,j}(\Omega_{i2}; G) &= \max_{a_3 \geq 0, c_2 \geq 0, i_2 \geq 0} \left\{ u(c_2) + \mathbb{1}_{\{j=k\}} u_k(i_2) - \mathbb{1}_{\{j=k\}} \kappa_{k,C} + \beta V_3^1(\Omega_{i3}) \right\} \\ \text{s.t.} \quad & \phi_c(m_2, \mathbb{1}_{\{j=k\}}) c_2 + a_3 = (1+r)a_2 + (w_C a_\theta - TC) - \mathbb{1}_{\{j=k\}} i_2. \end{aligned}$$

For $d = CD$, the household chooses labor supply $l_2 \in \{0, 1\}$ and solves

$$\begin{aligned} v_2^{3,j}(\Omega_{i2}; CD) &= \max_{l_2 \in \{0,1\}, a_3 \geq 0, c_2 \geq 0, i_2 \geq 0} \left\{ u(c_2) + \psi_l^j \mathbb{1}_{\{l_2=1\}} + \mathbb{1}_{\{j=k\}} u_k(i_2) + \beta V_3^1(\Omega_{i3}) \right\} \\ \text{s.t.} \quad & \phi_c(m_2, \mathbb{1}_{\{j=k\}}) c_2 + a_3 = (1+r)a_2 + y_2(\Omega_{i2}, l_2) - \mathbb{1}_{\{j=k\}} i_2, \\ & x_3 = x_2 + \mathbb{1}_{\{l_2=1\}}. \end{aligned}$$

In the dropout branch, disposable non-asset income is

$$y_2(\Omega_{i2}, l_2) = 4 \left[\lambda (\tilde{y}_2^0(\Omega_{i2}, l_2))^{1-\tau} + T(m_2) \right], \quad \tilde{y}_2^0(\Omega_{i2}, l_2) = \mathbb{1}_{\{l_2=1\}} w(\Omega_{i2}),$$

while in the student branch disposable resources are the student allowance net of direct schooling costs, $w_C - TC$.⁶⁶

Sub-stage 2: continue college vs. drop out. After observing j , college women choose $d \in \{G, CD\}$ with Type-I extreme value shocks. A graduation cost ϕ_{grad} , incurred only upon completion, matches observed college dropout. Then

$$V_2^{2,j}(\Omega_{i2}) = \max_{d \in \{G, CD\}} \left\{ v_2^{3,j}(\Omega_{i2}; d) + \mathbb{1}_{\{d=G\}} (\phi_{grad} + \mathbb{1}_{\{j=k\}} \phi_{k,D}) + \sigma_{CD} \varepsilon_2(d) \right\}.$$

Sub-stage 1: contraception. At the start of $t = 2$, childless women in the college track choose a method as in Section OA.7.5, with $V_2^{2,k}$ and $V_2^{2,nk}$ as the continuation values: writing $\Delta V_2 = V_2^{2,k}(\Omega_{i2}) - V_2^{2,nk}(\Omega_{i2})$,

$$V_2^1(\Omega_{i2}) = (1 - a_{\theta 2}) V_2^{2,nk}(\Omega_{i2}) + a_{\theta 2} \left[V_2^{2,nk}(\Omega_{i2}) + \sigma \log \sum_{n=0}^2 \exp \left\{ \frac{p_n(\theta_i, g(2), e_2) \Delta V_2 - \kappa_n}{\sigma} \right\} \right].$$

OA.7.7 Teenage (ages 14–17; $t = 1$)

At $t = 1$ all women are in high school. The within-period timing is: (i) contraception, (ii) after observing the fertility outcome, continue high school vs. drop out, and (iii) consumption–

⁶⁶In school periods the student allowance net of schooling costs is a non-taxable, transfer-like resource and enters the budget constraint directly; the tax/transfer mapping applies to labor-market income when working.

saving and, if a birth occurs, child investment. Teens who remain in school receive an allowance w_{HS} ; dropouts enter the labor market and begin accumulating experience.

Sub-stage 3: consumption–saving, child investment, and college entry at $t = 2$.

Let $j \in \{k, nk\}$ denote the fertility outcome in $t = 1$. Conditional on the schooling decision $d \in \{HSG, HSD\}$ (complete high school vs. drop out) from sub-stage 2, teens solve

$$\begin{aligned} v_1^{3,j}(\Omega_{i1}; d) = & \max_{a_2 \geq 0, c_1 \geq 0, i_1 \geq 0} \left\{ u(c_1) + \mathbb{1}_{\{j=k\}} u_k(i_1) + \mathbb{1}_{\{d=HSG\}} \mathbb{1}_{\{j=k\}} \phi_{k,HSD} \right. \\ & \left. + \beta \left[\mathbb{1}_{\{d=HSG\}} V_2^{CD,j}(\Omega_{i2}) + \mathbb{1}_{\{d=HSD\}} V_2^1(\Omega_{i2}) \right] \right\} \\ \text{s.t.} \quad & \phi_c(m_1, \mathbb{1}_{\{j=k\}}) c_1 + a_2 = (1+r)a_1 + y_1(\Omega_{i1}; d) - \mathbb{1}_{\{j=k\}} i_1, \\ & x_2 = x_1 + \mathbb{1}_{\{d=HSD\}}. \end{aligned}$$

Resources satisfy $y_1(\Omega_{i1}; d) = w_{HS}$ if $d = HSG$, while if $d = HSD$ the teen works as a dropout and

$$y_1(\Omega_{i1}; HSD) = 4 \left[\lambda (\tilde{y}_1^0(\Omega_{i1}))^{1-\tau} + T(m_1) \right], \quad \tilde{y}_1^0(\Omega_{i1}) = w(\Omega_{i1}).$$

At the end of $t = 1$, teens who complete high school ($d = HSG$) draw a Type-I extreme value shock and choose whether to enroll in college at $t = 2$, $d_C \in \{C, NC\}$. With $v_2^1(\cdot)$ the beginning-of-period value at $t = 2$ given the education choice,

$$V_2^{CD,j}(\Omega_{i2}) = \max_{d_C \in \{C, NC\}} \{v_2^1(\Omega_{i2}; d_C) + \mathbb{1}_{\{d_C=C\}} \kappa_C(\theta, j) + \sigma_C \varepsilon_2(d_C)\}.$$

Sub-stage 2: continue high school vs. drop out. After observing j , teens choose $d \in \{HSG, HSD\}$ with Type-I extreme value shocks:

$$V_1^{2,j}(\Omega_{i1}) = \max_{d \in \{HSG, HSD\}} \{v_1^{3,j}(\Omega_{i1}; d) + \sigma_{HS} \varepsilon_1(d)\}.$$

Sub-stage 1: contraception. At the start of $t = 1$, teens choose a method in the same way, with $\Delta V_1 = V_1^{2,k}(\Omega_{i1}) - V_1^{2,nk}(\Omega_{i1})$ and the teen exposure share $a_{\theta 1}$:

$$V_1^1(\Omega_{i1}) = (1 - a_{\theta 1}) V_1^{2,nk}(\Omega_{i1}) + a_{\theta 1} \left[V_1^{2,nk}(\Omega_{i1}) + \sigma \log \sum_{n=0}^2 \exp \left\{ \frac{p_n(\theta_i, 1, e_1) \Delta V_1 - \kappa_n}{\sigma} \right\} \right].$$

OA.8 Method Choice: Choice Probabilities and Expected Conception Risk

This appendix states the fertility-control sub-problem of Section 3.4 in closed form. The stage-2 objective for a childless woman is

$$V_t^{(2)} = V_t^{\text{no child}} + \mathbb{E}_\varepsilon \max_{n \in \{0,1,2\}} \{p_n \Delta V - \kappa_n + \sigma \varepsilon_n\}, \quad \Delta V \equiv V_t^{\text{child}} - V_t^{\text{no child}},$$

with p_n from equation (3), $\kappa_0 = 0$, and ε_n i.i.d. type-I extreme value. Writing $v_n = p_n \Delta V - \kappa_n$,

$$P(n) = \frac{e^{v_n/\sigma}}{\sum_{n'} e^{v_{n'}/\sigma}}, \quad \mathbb{E} \max_n \{v_n + \sigma \varepsilon_n\} = \sigma \log \sum_{n'} e^{v_{n'}/\sigma} + \sigma \gamma_E,$$

where γ_E is Euler's constant and is dropped as an additive constant. The expected probability of a first birth is $\bar{p} = \sum_n P(n)p_n$, the share using any method is $1 - P(0)$, and the pill share among users is $P(2)/(1 - P(0))$.

Comparative statics. For $\Delta V < 0$, v_n is increasing in the efficacy of n and decreasing in κ_n ; $P(0)$ is increasing in ΔV , so participation falls continuously as the value of a birth rises and reaches the no-method corner only as $\Delta V \rightarrow +\infty$. A higher $c_{\theta g}$ lowers p_1 and p_2 in proportion to $(\rho_n^0 - \rho_n)$, which lowers the failure rate within each method and, by raising $|v_n - v_0|$, raises the probability of using one. The within-method failure moments respond to c and ρ_n^0 but not to κ ; the any-method use moments respond to both.

Units and normalisations. ΔV , κ_n and σ are in utility units. Because the efficacy ratios ρ_n are fixed externally and the non-adherent rates ρ_n^0 are common across women, $c_{\theta g}$ has a scale: $c_{\theta g} = 1$ is perfect use and $c_{\theta g} = 0$ the failure rate of a user who never adheres. The counterfactuals of Section 6 use this scale. Equalising adherence to the top quartile changes p_n at unchanged κ_n ; a uniform price cut lowers κ_1 and κ_2 by the same amount for everyone; and a long-acting reversible method replaces the pill with $\rho_{\text{LARC}} = 0.005/0.85$ (IUD, 0.2–0.8% per year; Trussell, 2011), a failure rate that does not depend on c , at the pill's cost κ_2 .

Smoothness. $P(n)$, $\bar{p}$ and the log-sum are differentiable in every parameter, and the estimation averages these probabilities across simulated women rather than the realized method draws (Appendix OA.11.5). The method-share and within-method-failure moments are therefore smooth in the parameter vector, as the sandwich standard errors and the nested test require.

OA.9 Interpretation and Scope of the Fertility-Control Channel

The non-adherence anchors. The anchor ρ_n^0 of equation (3) is estimated rather than imported from Trussell (2011) because the NLSY79 failure measure of Section 2.3 is a first-birth rate per year of *reported* use, which includes misreported use, discontinuation between interviews and births after stopping: 3.3 percent a year in the lowest quartile against the 12.6 percent the observed method mix implies at published typical-use rates (Online Appendix OA.6.1). Estimating the anchors also leaves the ratio of pill to low-tier failure among imperfect users to the data. With a non-adherent user fully unprotected ($\rho_n^0 = 1$) that ratio would lie between 0.78 and 0.99 over the estimated range of adherence, against about 0.5 in the data in every quartile. At $\hat{\rho}_1^0 = 0.993$ and $\hat{\rho}_2^0 = 0.463$ the implied ratio runs from 0.46 at the lowest estimated adherence to 0.38 at the highest, in the range of the data.

What $c_{\theta g}$ absorbs. Adherence is identified from within-method birth rates (Section 3.4), so it carries whatever produces an ability gradient in births among users of a given method other than the method itself. Three sources enter. The first is pregnancy resolution: the model maps method use into live births, not into conceptions, so $c_{\theta g}$ absorbs any ability-correlated difference in abortion. Because abortion is more common at the top of the ability distribution in this cohort, ignoring it inflates the estimated adherence gradient by roughly a sixth; it cannot generate the gradient, which would take a top-to-bottom abortion ratio of 2.04 against 1.18 observed (Appendix OA.6.5), so the estimates carry a small upward correction rather than being a lower bound. The second is behavior around the method. The acts adherence requires—taking a pill daily, using a barrier method at every occasion, keeping appointments, negotiating with a partner—are the ones whose lapses the adherence literature ties to the absence of a routine and to not understanding the method’s instructions (Rosenberg et al., 1995), to school performance (Wu et al., 2016), and to executive function and working memory (Insel et al., 2006; Stilley et al., 2010); they include the partner’s conduct (Ashraf et al., 2014), and the reported-use measure includes lapses and discontinuation between interviews. Their common level goes to ρ_n^0 , which is the same for every woman; any ability gradient in them goes to $c_{\theta g}$, which is the reliability of use as practised, not of the method as prescribed. The third is the trait: these acts are the ones non-cognitive skills govern, and Heckman et al. (2006) show that cognitive and non-cognitive skills are correlated and that both predict teen pregnancy, so $c_{\theta g}$ is the fertility-control capacity of the skill bundle the AFQT proxies. The estimates condition on measured AFQT and do not depend on which trait does the work; the interpretation of the policy lever does, which is why Section 6 studies interventions that remove the need for adherence rather than interventions that raise

cognitive skill.

The exception is the youngest age group. No method report exists before 1982, when the youngest respondents were already 18, so $c_{\theta,14-17}$ is identified residually, from the teen first-birth hazards and the wantedness of teen first births given the channels pinned down at later ages; since the teen gradient is the largest, it is the parameter on which the central claim leans hardest.

Scope. The model tracks only the first birth, and a child imposes its cost for one four-year period; the moments that identify fertility control are first-birth hazards, so later births add states without identifying variation. Both restrictions bias the welfare results in a known direction. A child cost lasting several periods raises the lifetime cost of an early birth most for women with the steepest wage profiles, so the one-period cost understates the gains from improving fertility control. Higher-parity births work the same way for the lowest-ability women, whose completed fertility is highest, and the opposite way for high-ability women, who delay and then compress childbearing; on net they widen the ability gap in the cost of early motherhood. Neither restriction affects the identification of adherence, which rests on first-birth hazards at ages where second births are rare.

What an adherence-proof technology removes. The NLSY79 cohort, born 1957–1964, was fertile from the mid-1970s to the late 1990s, when effectiveness depended overwhelmingly on adherence; long-acting methods diffused in the United States after the cohort had largely completed its fertility. That environment is why the channel is visible, and it bounds what the estimates license. An adherence-proof method removes the adherence margin, so a lowest-quartile woman who can choose one today bears a smaller burden than the model attributes to her, and the level of the gradient is a statement about this cohort rather than a constant of the population. The counterfactual of Section 6 is the model’s answer to what removing that margin would have done for this cohort under universal take-up, an upper bound in both senses, not a forecast for a current cohort. What transfers is qualitative: where effectiveness depends on user behavior, differences in fertility-control capacity generate differences in birth timing that the price of contraception cannot undo, so the policy implication concerns which kind of instrument reaches which women, not a magnitude.

OA.10 Wage Process Estimation

This appendix describes the estimation of the earnings profiles that parameterize the model’s earnings opportunities: conditional mean annual earnings by age, education, cognitive ability and experience for women, and by age and the wife’s education for husbands. The estimated

coefficients and the retirement-income process are in the Additional Results, Section S.5.

OA.10.1 Wage measures and estimation samples

Women. Let w_{it} denote real annual wage-and-salary income (2016 dollars). The sample is the woman-years that meet the employment definition of Appendix OA.1 (minimum weeks, hours and annual earnings). The dependent variable is in levels, as the model is parameterized.

Husbands/partners. Let w_{it}^m denote the partner's real annual wage-and-salary income and Age_{it}^m his age. The sample is the years in which the woman is married and partner earnings exceed the earnings threshold used for women. Education in the husband regressions is the woman's, $\text{Educ}_i^f \in \{\text{HSD}, \text{HSG}, \text{COL}\}$, so that the coefficients reflect the partners that women of each education level attract through assortative mating.

OA.10.2 Baseline specification: women

I estimate:

$$\begin{aligned}
w_{it} = & \alpha_t + \beta_1 \text{Age}_{it}^f + \beta_2 \left(\text{Age}_{it}^f \right)^2 + \delta_0 \text{CumExp}_{it} \\
& + \sum_{e \in \{\text{HSG}, \text{COL}\}} \left(\gamma_e + \delta_e \text{CumExp}_{it} \right) \mathbb{1}\{\text{Educ}_i^f = e\} \\
& + \sum_{q=2}^4 \left(\gamma_q + \delta_q \text{CumExp}_{it} \right) \mathbb{1}\{\text{Ability}_i = q\} \\
& + \sum_{e \in \{\text{HSG}, \text{COL}\}} \sum_{q=2}^4 \left(\gamma_{eq} + \delta_{eq} \text{CumExp}_{it} \right) \mathbb{1}\{\text{Educ}_i^f = e\} \mathbb{1}\{\text{Ability}_i = q\} + \varepsilon_{it}.
\end{aligned} \tag{OA.2}$$

where α_t are calendar-year fixed effects. The interactions $\text{CumExp} \times \text{Educ}^f \times \text{Ability}$ let returns to experience vary across education and ability quartiles.

OA.10.3 Baseline specification: husbands/partners

For husbands/partners I estimate:

$$\begin{aligned}
w_{it}^m = & \alpha_t^m + \theta_{\text{HSG}} \mathbf{1}\{\text{Educ}_i^f = \text{HSG}\} + \theta_{\text{COL}} \mathbf{1}\{\text{Educ}_i^f = \text{COL}\} + \beta_1^m \text{Age}_{it}^m + \beta_2^m (\text{Age}_{it}^m)^2 \\
& + \sum_{e \in \{\text{HSG}, \text{COL}\}} \left(\beta_{1e}^m \text{Age}_{it}^m + \beta_{2e}^m (\text{Age}_{it}^m)^2 \right) \mathbf{1}\{\text{Educ}_i^f = e\} \\
& + \lambda^m \mathbf{1}\{\text{MarryBeforeBirth}_i = 1\} \\
& + \sum_{e \in \{\text{HSG}, \text{COL}\}} \kappa_e^m \mathbf{1}\{\text{Educ}_i^f = e\} \mathbf{1}\{\text{MarryBeforeBirth}_i = 1\} + u_{it}.
\end{aligned} \tag{OA.3}$$

with year fixed effects α_t^m ; the indicator $\mathbf{1}\{\text{MarryBeforeBirth}_i = 1\}$ captures differences in spouse earnings associated with marrying before the first birth.

OA.10.4 Model inputs and aggregation to four-year periods

The estimated coefficients generate predicted annual earnings by (age, education, ability quartile, cumulative experience). A model period is four years, so the predicted annual earnings at the period's start-of-bin age are the period's annual earnings opportunity, and cumulative experience is updated by the model's accumulation rule. In retirement, labor earnings are replaced by an education-specific deterministic retirement income constructed from the pre-retirement predicted wage arrays.

OA.10.5 Sensitivity of the fertility-control estimates to the wage process

The exercise. Selection into employment, measurement error in AFQT and the earnings floor all attenuate the ability gradient that least squares recovers in (OA.2), and an understated gradient understates the opportunity cost of an early birth, leaving more of the within-education birth gradient of Section 2 for the adherence channel to explain. The adverse direction is therefore a gradient steeper than the estimate, and the sensitivity is run only in that direction. Let κ multiply every coefficient in (OA.2) that varies with the ability quartile—the level terms γ_q , the experience interactions δ_q and the education interactions γ_{eq} and δ_{eq} —holding the Q1 profile, the education gradient and the age profile fixed, so that κ scales the Q2–Q4 premia and $\kappa = 1$ is the estimate; a proxy reliability of 0.9 implies a measurement-error correction of $\kappa \approx 1.11$. At $\kappa \in \{1.11, 1.5, 2\}$ I re-estimate the twelve parameters of the fertility-control technology, the three age-group intercepts a_g and the nine ability increments, holding every other parameter at its estimated value. Adherence is then

Table OA.27. Wage-process sensitivity: adherence gradient as the ability gradient in earnings is scaled

κ	Earnings gap Q4–Q1		Adherence gap		Criterion	Unwanted-birth gradient (pp)
	\$000	rel. to $\kappa = 1$	14–17	18–25		
1.00	7.1	1.00	0.480	0.320	382.3	21.3
1.11	7.9	1.11	0.461	0.321	388.1	20.9
1.25	8.9	1.25	0.461	0.320	394.5	20.9
1.50	10.7	1.50	0.461	0.320	409.3	21.0
2.00	14.3	2.00	0.451	0.321	452.7	20.6

Notes: κ scales every ability-varying coefficient of the female earnings function (OA.2), holding the Q1 profile, the education gradient and the age profile fixed; $\kappa = 1$ is the estimate. “Earnings gap” is the predicted Q4–Q1 annual earnings of a high-school graduate at ten years of experience and age 30, in thousands of 2016 dollars. “Unwanted-birth gradient” is the model’s share of women whose first birth came by age 21 and was unwanted, lowest quartile minus highest, in percentage points (data: 22). At each κ the twelve fertility-control parameters are re-estimated with every other parameter held at its estimated value.

the only margin that can absorb the change in wages, so the design overstates how far it moves relative to a full re-estimation. Table OA.27 reports, at each κ , the Q4–Q1 earnings gap of a high-school graduate at ten years of experience, the adherence gaps, the criterion and the model’s ability gradient in unwanted first births by age 21.

Results. The adherence gaps barely move up to $\kappa = 1.5$, which raises the earnings gap from \$7.1k to \$10.7k: the 14–17 gap goes from 0.48 to 0.46 and the 18–25 gap stays at 0.32. At $\kappa = 2$ (\$14.3k) the 14–17 gap is 6 percent below the estimate and the 18–25 gap is unchanged. The criterion rises monotonically in κ , from 382.3 at the estimate to 452.7 at $\kappa = 2$: a steeper earnings gradient does not substitute for the adherence gradient but worsens the fit to the fertility, schooling and labor-supply moments. The mechanism is the identification of Section 4.2. The adherence parameters are pinned by the within-method birth rates, which condition on the method in use and depend on adherence and the externally fixed efficacies, not on wages; scaling the earnings gradient shifts the value of avoiding a birth, method choice, schooling and labor supply, but leaves those moments where they were, and at $\kappa = 1.5$ the search ran 94 generations of a 24-member population over the twelve adherence slots and lowered the criterion by 0.14. The last column shows the same separation on the margin the nested test turns on (Section 5.3): the unwanted-birth gradient is 21.3 points at the estimate and 20.6 at $\kappa = 2$, against 22 in the data. Wantedness is the sign of $\Delta V + \sigma_\omega \varepsilon^\omega$, and κ holds the lowest quartile’s profile fixed by construction, so a steeper gradient makes early births rarer at the top but cannot make the bottom quartile’s births unwanted, and the gradient is measured at the bottom.

OA.11 Estimation Details

This appendix reports the externally set parameters, the weighting matrix, the standard-error calculation, the distribution theory for the nested test of Section 4.2, and the solution, simulation and optimization algorithms. Table OA.28 maps each object of the fertility-control technology to the moments that carry its identifying variation (Section 4.2). The model is overidentified: 138 targeted moments against 76 structural parameters.

Table OA.28. Fertility-Control Parameters and the Moments That Identify Them

Object	Moments	Variation used
λ_{be} (baseline odds)	first-birth shares by age $\times$ quartile (28), jointly with the schooling block	level and age profile of first births, given the education composition of each quartile
ρ_n (method efficacy)	external: Trussell (2011)	perfect-use failure rates
$c_{\theta g}$ (adherence)	birth rate within pill / within low tier, by quartile (8)	ability gradient in failure holding the method fixed
ρ_1^0, ρ_2^0 (non-adherent failure)	the same 8 moments	level and pill/low-tier ratio of failure, common across quartiles
$\kappa_1, \kappa_2, \sigma$ (method costs)	any-method use by education $\times$ age $\times$ ability half (25)	participation given ΔV ; the pill share among users (8) is untargeted
$\omega_{0,\theta}$ (child preference)	unintended first birth by 21, share of women, by quartile (4); first-birth timing	first births by 21 to women with $\Delta V < 0$, given adherence and method costs
σ_ω (transitory wantedness)	unintended share of first births by age at first birth, Q1 and Q4 (8)	dispersion of wantedness within a quartile at a given age at birth
ΔV (opportunity cost)	wages (external); employment, marriage, schooling blocks	ability gradient in wages and experience returns

Over-identifying: the ability *gradient* in the 28 first-birth shares—their variation across quartiles at each age, as opposed to their level and age profile—must be reproduced by the three channels jointly.

Notes: Each row lists an object of the fertility-control technology, the moment block that carries most of its identifying variation, and the variation that moves it; counts are numbers of moments (full list in Table OA.30), and “External” marks an object fixed outside the estimation. All moments enter one criterion, so the mapping says which variation is informative in the sense of [Andrews et al. \(2017\)](#), not that each block identifies its object alone.

OA.11.1 Externally set parameters

Table OA.29 lists the parameters fixed outside SMM. The discount factor and the coefficient of relative risk aversion are [Adda et al.’s \(2017\)](#) estimates; tuition, taxes and transfers use established calibrations; and the bounds on conception probabilities rule out both perfect control and deterministic fecundity.

Table OA.29. Externally Set Parameters

Parameter	Value	Source / Interpretation
Discount factor β_a	0.959 (annual)	Estimate, Adda et al. (2017, Table 4)
Risk aversion γ	1.98	CRRA; estimate, Adda et al. (2017, Table 4)
Risk-free rate r_a	0.04 (annual)	Real return (Adda et al., 2017)
College tuition TC	\$10,200	Annual tuition, 2016 \$ (Vandenbroucke, 2023)
Tax progressivity τ	0.18	(Darulich and Fernández, 2024)
Tax scale λ	0.85	
Transfer floor T_S (single)	\$8,634	Annual floor, 2016 \$ (Darulich and Fernández, 2024)
Transfer floor T_C (couple)	\$12,943	
Conception bounds $(\underline{\lambda}, \bar{\lambda})$	(0.05, 0.80)	Imperfect control bounds (Trussell, 2004)
Method efficacy ρ_1, ρ_2	0.03/0.85, 0.003/0.85	Perfect-use failure relative to no method (Trussell, 2011)
LARC efficacy ρ_{LARC}	0.005/0.85	Counterfactual only; IUD 0.2–0.8%/yr (Trussell, 2011)
Equivalence scale ϕ_{ca}, ϕ_{ck}	0.8, 0.5	Second adult, child

Notes: Monetary values in 2016 dollars per year. Annual flows converted to four-year model periods as described in text.

OA.11.2 The weighting matrix

W is diagonal and fixed, with $W_{jj} = 1/\hat{\sigma}_j^2$ and $\hat{\sigma}_j$ the bootstrap standard error of data moment j from 1,000 replications resampling women, with a single draw applied to all four analysis samples so that cross-block covariances are not forced to zero. I do not use the efficient $W = \Omega^{-1}$: [Altonji and Segal \(1996\)](#) show that it carries severe small-sample bias, because $\hat{\Omega}$ is correlated with the sample moments, and a diagonal W is standard in structural life-cycle work ([Adda et al., 2017](#); [Keane and Wolpin, 1997](#)); and $\hat{\Omega}$ is numerically singular here, with rank 136 of 138.

OA.11.3 Standard errors

With W fixed, $\hat{\Theta}$ is a minimum-distance estimator with asymptotic variance

$$\widehat{\text{Var}}(\hat{\Theta}) = \left(1 + \frac{1}{S}\right) (G'WG)^{-1} G'W \Omega WG (G'WG)^{-1}, \quad G = \frac{\partial m^{sim}(\Theta)}{\partial \Theta'} \Big|_{\hat{\Theta}},$$

where Ω is the bootstrap variance of the data moments and S is the ratio of the simulated to the estimation sample, the simulation-noise inflation of [McFadden \(1989\)](#). The inflation factor refers to the criterion the estimator minimized, evaluated on 10,000 simulated women, so $S = 1.68$ and $1 + 1/S = 1.59$; the Jacobian is differenced on 160,000 women, because finite-difference noise of order σ_{sim}/h in each column enters $G'WG$ as a positive-definite term that overstates the curvature of the criterion and understates the standard errors. G is computed by two-sided finite differences with a step of one percent of each parameter's box width and again at half that step; a parameter whose standard error moves by more than a quarter between the two steps is reported as unstable. Three kinds of parameter carry no standard error: a parameter at a corner of its box, where the estimator is not asymptotically

normal (none in the estimated model; the 26–33 adherence intercept and the dropout continuation wedge $\phi_{k,HSD}$ in the restricted one); a direction along which the weighted Jacobian is numerically flat (the three adherence increments at 26–33 in the estimated model, where adherence is essentially complete in every quartile, and ψ_{lk} for dropouts in the restricted model); and a parameter fixed at a nesting point by construction. The condition number of $G'WG$ over the remaining directions is reported as a check that the flat directions do not contaminate their neighbours.

OA.11.4 Distribution theory for the nested test

The restricted model imposes $c_{\theta g} = c_g$ by fixing the nine adherence increments at zero and is otherwise identical to the unrestricted one: same technology, moments, weights and bounds. The difference in minimized criteria, $D = L_R - L_U$, is therefore a test statistic. Two departures from χ^2_9 run in opposite directions. Because $W \neq \Omega^{-1}$, D is asymptotically a weighted sum of chi-squares; I therefore also report the mean-matched statistic $D \cdot 9 / \text{tr}(P_r \Sigma)$, whose first moment equals that of a χ^2_9 under the null, with P_r and Σ computed from the Jacobian and Ω at $\hat{\Theta}$. Because the null fixes the nine increments at the boundary of their parameter space, the true null distribution is stochastically smaller than χ^2_9 and the asymptotic p -value is conservative. A re-estimation bootstrap of the critical value would cost roughly two hundred short global searches; given the size of the statistic in Section 5 relative to the one-percent critical value of 21.7, it cannot change the verdict, and I do not run it.

Are L_U and L_R global minima? D is informative only if L_R is the restricted model’s minimum, and every counterfactual rests on L_U . I therefore estimated each model from several starting points with the same differential-evolution search.⁶⁷ once warm-started from the previous estimate, 10 times from a vector drawn uniformly over the whole parameter box under its own seed (24 hours per run), and 5 times from a vector drawn within ± 0.15 box widths of the incumbent (22 hours for the restricted model, 16 for the unrestricted), the incumbent being kept out of every initial population. *Restricted model.* The far starts ended between 665.8 and 2,928.5 (median 911.2). The lowest, polished for a further 28 hours with the full population, reached 569.7, whereas the warm-started run had ended at 925.9, 356.2 points (62.5 percent) higher. The near starts, drawn around that lowest far start, ended between 586.0 and 728.5, none below the polished value, and the median displacement of a parameter from it was at most 0.05 box widths in any run. *Unrestricted model.* The

⁶⁷The search was run on the moment vector of the previous data build; the estimation sample was unified afterwards (Appendix OA.1.3), and the paragraph ends with how the two vectors become the reported estimates.

warm-started run, at 415.2, was the lowest of the 16 starts: the far starts ended between 554.1 and 917.6 (median 592.7) and the near starts between 445.5 and 948.7, with a median parameter displacement from the incumbent of at most 0.06 box widths in any run. *Final targets.* Each vector was then polished for 24 hours with the full population on the moment vector of the unified sample. The restricted vector scores 581.1 on the final targets and the polish takes it to 530.7; the unrestricted vector scores 427.5 and is taken to 382.3. These are the criteria reported in the paper. A lower L_U raises D , so the test is conservative with respect to the unrestricted search.

OA.11.5 Solution, simulation, and optimization

The model is a finite-horizon dynamic programme with one continuous state, assets, and discrete states for ability, education, experience, marital status, whether the first birth preceded marriage, and child status. Time is discrete in four-year periods, $\text{age}_t = 10 + 4t$, with $T = 17$. Assets are discretized on a 30-point grid with cubic spacing, which concentrates points near the borrowing constraint; the child-investment and contraception grids also have 30 points, and policies are evaluated off-grid by linear interpolation. The solution is by backward induction: the four retirement periods, a pure consumption–saving problem, are solved by the endogenous grid method; the six non-fertile working periods, in which the discrete labor choice makes the value function non-concave, by DC-EGM with an upper-envelope step; and the seven fertile periods, in which marriage, contraception, schooling and child investment enter as further discrete margins within a three-sub-stage period, by value-function iteration on the grid, with the child-investment first-order condition solved by bisection and the method choice aggregated in closed form as the logit log-sum of Online Appendix OA.8.

Simulation draws 10,000 life histories from pre-drawn uniform random variables with a fixed seed, so that the moments are a deterministic function of the parameter vector. Schooling decisions are evaluated after the fertility draw, conditional on its outcome, as in Online Appendix OA.7. The participation, method-share and within-method-failure moments are averages of predicted probabilities across simulated women, which makes them smooth in the parameters; the birth, marriage and employment moments are averages of realized outcomes under common random numbers. The contraception-use moments cover the ages at which the NLSY79 contraception module was fielded; the within-method-failure moments cover ages 22–38 and exclude the age-18 cell, where the exposure share that converts per-exposure into annual risk differs sharply across education groups.

The criterion of Online Appendix OA.11 is minimized by self-adaptive differential evolution (jDE; Brest et al., 2006), which evaluates each generation’s trial vectors in parallel across the 47 workers of a 48-core node, with a population of 94 started around the warm start (three quarters as local perturbations, one quarter uniform on the bounds). One solve–

simulate-score takes about four seconds on a single-threaded worker, so the search sustains roughly 12 evaluations per second. Each production estimate ran for 36 hours; the final round then re-estimated each model for a further 24 hours from its incumbent on the moment vector of the unified sample, where the incumbent scored 427.5 (estimated model) and 581.1 (restricted) and the polish took them to 382.3 and 530.7, with improvements confined to the sixth decimal over the last hour; the multi-start evidence above is the convergence check. Standard errors use numerical Jacobians on 160,000 simulated women at two finite-difference steps, because the interpolated policies and the upper-envelope comparison make optimal choices move in small jumps as parameters vary, a non-smoothness that more simulated agents do not remove. The full description of the state space, first-order conditions, algorithms and computational requirements is in the Additional Results, Section S.8; the replication package reproduces every table and figure from the estimated vector in under a minute on a laptop.

OA.12 Model Fit

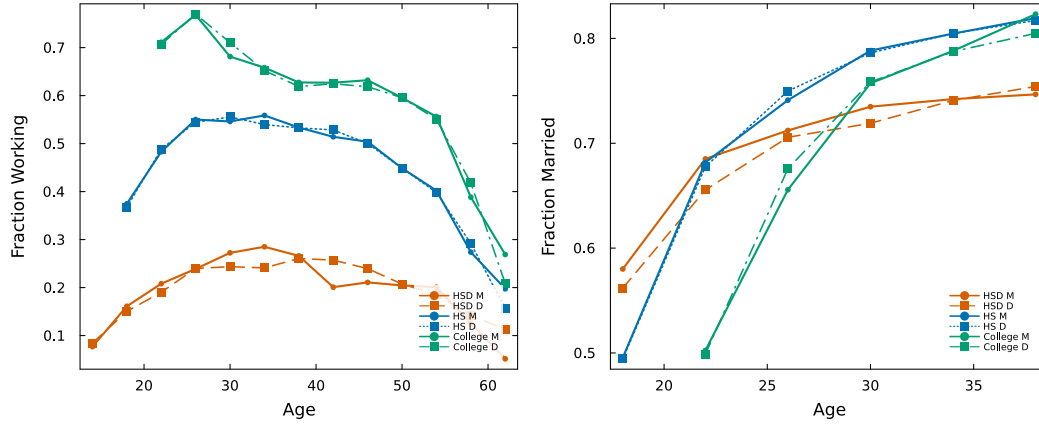
OA.12.1 Targeted Moments

Table OA.30 reports the 138 moments that enter the criterion, each with its data value, its model value and its share of the weighted criterion, in estimation order and grouped into the nine weighted blocks of Section 4. Section 5.1 discusses the first-birth, wantedness and within-method blocks; Figure OA.6 and the paragraphs below cover the rest. The eight zero-weight method-mix moments are in Table OA.31.

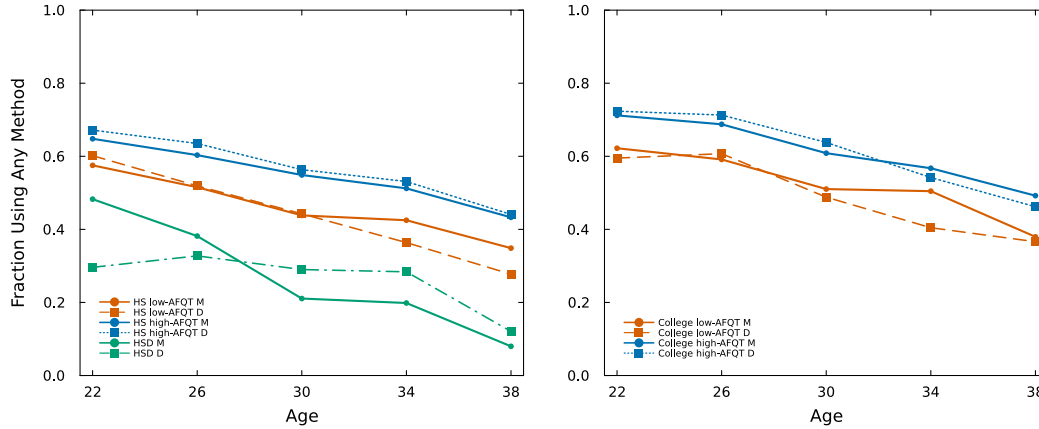
Schooling and child investment. The first ten moments describe the joint distribution of schooling and early fertility. The model matches the association between a teen birth and dropping out, and the ability gradient in college attendance, from 11.5% in Q1 to 68.7% in Q4 in the data and from 11.1% to 67.2% in the model. The one departure is the third quartile, which the model overpredicts; the ability-indexed college allowance of Section 3 matches the level and slope of attendance, and the miss sits in the interior of the distribution, where it does not affect the fertility gradient. The two child-investment moments are the expenditure ratios of [Caucutt and Lochner \(2020\)](#): the model reproduces the college-versus-dropout ratio and overshoots the high-school-versus-dropout ratio. Scaled by their magnitude rather than a bootstrap variance, they carry 0.5 percent of the criterion.

Labor supply. The model reproduces the education gradient in participation and its life-cycle shape, rising in the twenties, flat through the fifties and declining after 58 (Figure OA.6a, left). For high-school graduates the model is within two percentage points at

Figure OA.6. Model Fit: Labor-Force Participation, Marriage, and Contraceptive Use



(a) Labor-force participation (left) and marriage (right) by education



(b) Contraceptive use by education and ability half

Notes: Outcomes by completed education (HSD = dropout, HS = high school, College = college graduate); panel (b) splits the HS and college groups at the median AFQT. Ages index four-year periods: 22 is the period 22–25. Solid lines with circular markers are the model, dashed lines with square markers the NLSY79 data.

every age before 62; the largest misses are at the ends of working life, dropouts at 42 and 62 and college graduates at 62, none above 6 percentage points. Experience is not targeted: the experience profiles that feed wages are disciplined by the 36 employment shares.

Marriage. Marriage rates rise with age in every education group and converge by the early thirties, as in the data (Figure OA.6a, right). No marriage moment departs from the data by more than 2.9 percentage points; the largest miss is dropouts at 22, and college marriage at 26 is slightly underpredicted, closing by 30. The model omits non-pecuniary match quality and the insurance value of a second earner, which is where the remaining discrepancies sit.

Contraception use. The model matches the education gradient in use and its decline with age from 22 on (Figure OA.6b). The largest single discrepancy in the moment vector is dropouts' use at 22, 0.48 in the model against 0.30 in the data, 6.1 percent of the criterion on its own: for a childless dropout the value of avoiding a birth is close to zero, so her use margin is governed by the method costs, which are common across education groups. Among college women the model errs in both directions. Reported use and the model's use margin differ: a woman relying intermittently on a low-tier method may not describe herself as a user, and two users of the same method can differ in how reliably they use it. Birth rates by education and ability are matched throughout.

Table OA.30. Targeted moments: data, model, and share of the criterion

<i>j</i>	Stata <i>j</i>	Moment	Data	Model	Weight share (%)
<i>schooling</i>					
1	1	HS dropout, no birth 14-17	0.070	0.075	0.4
2	2	HS dropout, birth 14-17	0.300	0.306	0.0
3	3	College attendance, no birth 14-17	0.414	0.416	0.0
4	4	College attendance, birth 14-17	0.082	0.085	0.0
5	5	College attendance, AFQT Q1	0.115	0.111	0.0
6	6	College attendance, AFQT Q2	0.247	0.230	0.6
7	7	College attendance, AFQT Q3	0.421	0.448	1.2
8	8	College attendance, AFQT Q4	0.687	0.672	0.5
9	9	College graduation, no birth 18-21	0.619	0.625	0.1
10	10	College graduation, birth 18-21	0.256	0.262	0.0
<i>investment</i>					
11	11	Child investment ratio HS/HSD	1.200	2.946	0.5
12	12	Child investment ratio Coll/HSD	4.600	4.040	0.0
<i>birth_rate</i>					
13	20	Birth rate, AFQT Q1, age 14	0.285	0.321	2.3
14	21	Birth rate, AFQT Q1, age 18	0.631	0.583	3.6
15	22	Birth rate, AFQT Q1, age 22	0.770	0.754	0.5
16	23	Birth rate, AFQT Q1, age 26	0.827	0.811	0.8
17	24	Birth rate, AFQT Q1, age 30	0.858	0.866	0.2
18	25	Birth rate, AFQT Q1, age 34	0.872	0.884	0.5
19	26	Birth rate, AFQT Q1, age 38	0.875	0.900	2.1
20	27	Birth rate, AFQT Q2, age 14	0.166	0.193	1.9
21	28	Birth rate, AFQT Q2, age 18	0.480	0.464	0.4
22	29	Birth rate, AFQT Q2, age 22	0.669	0.651	0.5
23	30	Birth rate, AFQT Q2, age 26	0.752	0.721	2.0
24	31	Birth rate, AFQT Q2, age 30	0.795	0.795	0.0
25	32	Birth rate, AFQT Q2, age 34	0.817	0.822	0.0
26	33	Birth rate, AFQT Q2, age 38	0.825	0.853	2.1
27	34	Birth rate, AFQT Q3, age 14	0.088	0.085	0.1
28	35	Birth rate, AFQT Q3, age 18	0.313	0.310	0.0
29	36	Birth rate, AFQT Q3, age 22	0.504	0.508	0.0
30	37	Birth rate, AFQT Q3, age 26	0.633	0.593	2.7
31	38	Birth rate, AFQT Q3, age 30	0.693	0.677	0.5
32	39	Birth rate, AFQT Q3, age 34	0.722	0.710	0.3
33	40	Birth rate, AFQT Q3, age 38	0.734	0.753	0.7

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<i>j</i>	Stata <i>j</i>	Moment	Data	Model	Weight share (%)
34	41	Birth rate, AFQT Q4, age 14	0.034	0.020	2.0
35	42	Birth rate, AFQT Q4, age 18	0.185	0.227	4.3
36	43	Birth rate, AFQT Q4, age 22	0.366	0.435	7.2
37	44	Birth rate, AFQT Q4, age 26	0.551	0.524	1.1
38	45	Birth rate, AFQT Q4, age 30	0.666	0.628	2.5
39	46	Birth rate, AFQT Q4, age 34	0.707	0.682	1.2
40	47	Birth rate, AFQT Q4, age 38	0.722	0.734	0.3
<i>marriage</i>					
41	48	Share married, educ 1, age 18	0.562	0.580	0.2
42	49	Share married, educ 1, age 22	0.656	0.685	0.6
43	50	Share married, educ 1, age 26	0.706	0.712	0.0
44	51	Share married, educ 1, age 30	0.719	0.735	0.2
45	52	Share married, educ 1, age 34	0.741	0.742	0.0
46	53	Share married, educ 1, age 38	0.754	0.747	0.0
47	54	Share married, educ 2, age 18	0.494	0.496	0.0
48	55	Share married, educ 2, age 22	0.678	0.682	0.1
49	56	Share married, educ 2, age 26	0.750	0.741	0.4
50	57	Share married, educ 2, age 30	0.786	0.788	0.0
51	58	Share married, educ 2, age 34	0.805	0.805	0.0
52	59	Share married, educ 2, age 38	0.817	0.819	0.0
53	60	Share married, educ 3, age 22	0.499	0.502	0.0
54	61	Share married, educ 3, age 26	0.676	0.656	0.6
55	62	Share married, educ 3, age 30	0.758	0.757	0.0
56	63	Share married, educ 3, age 34	0.787	0.788	0.0
57	64	Share married, educ 3, age 38	0.805	0.823	0.6
<i>labor</i>					
58	65	Employment, educ 1, age 14	0.084	0.077	0.1
59	66	Employment, educ 1, age 18	0.151	0.161	0.2
60	67	Employment, educ 1, age 22	0.190	0.208	0.5
61	68	Employment, educ 1, age 26	0.240	0.239	0.0
62	69	Employment, educ 1, age 30	0.244	0.272	0.8
63	70	Employment, educ 1, age 34	0.241	0.285	1.7
64	71	Employment, educ 1, age 38	0.261	0.266	0.0
65	72	Employment, educ 1, age 42	0.257	0.201	2.7
66	73	Employment, educ 1, age 46	0.240	0.211	0.7
67	74	Employment, educ 1, age 50	0.207	0.205	0.0
68	75	Employment, educ 1, age 54	0.185	0.201	0.3
69	76	Employment, educ 1, age 58	0.142	0.126	0.2
70	77	Employment, educ 1, age 62	0.113	0.052	0.7
71	78	Employment, educ 2, age 18	0.367	0.375	0.4
72	79	Employment, educ 2, age 22	0.489	0.483	0.2
73	80	Employment, educ 2, age 26	0.544	0.550	0.2
74	81	Employment, educ 2, age 30	0.556	0.546	0.4
75	82	Employment, educ 2, age 34	0.540	0.559	1.9
76	83	Employment, educ 2, age 38	0.533	0.534	0.0
77	84	Employment, educ 2, age 42	0.529	0.514	1.0
78	85	Employment, educ 2, age 46	0.500	0.504	0.1
79	86	Employment, educ 2, age 50	0.449	0.449	0.0
80	87	Employment, educ 2, age 54	0.398	0.402	0.1
81	88	Employment, educ 2, age 58	0.292	0.274	1.1
82	89	Employment, educ 2, age 62	0.157	0.197	2.1
83	90	Employment, educ 3, age 22	0.706	0.711	0.1
84	91	Employment, educ 3, age 26	0.770	0.768	0.0
85	92	Employment, educ 3, age 30	0.711	0.681	1.7

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<i>j</i>	Stata <i>j</i>	Moment	Data	Model	Weight share (%)
86	93	Employment, educ 3, age 34	0.652	0.658	0.1
87	94	Employment, educ 3, age 38	0.619	0.627	0.1
88	95	Employment, educ 3, age 42	0.625	0.627	0.0
89	96	Employment, educ 3, age 46	0.618	0.632	0.3
90	97	Employment, educ 3, age 50	0.596	0.595	0.0
91	98	Employment, educ 3, age 54	0.551	0.556	0.0
92	99	Employment, educ 3, age 58	0.418	0.388	0.8
93	100	Employment, educ 3, age 62	0.208	0.269	1.1
<i>contraception</i>					
94	101	Contraception use, educ 1, age 22	0.296	0.483	6.1
95	102	Contraception use, educ 1, age 26	0.327	0.382	0.3
96	103	Contraception use, educ 1, age 30	0.290	0.211	0.6
97	104	Contraception use, educ 1, age 34	0.284	0.198	0.5
98	105	Contraception use, educ 1, age 38	0.121	0.080	0.1
99	106	Contraception use, educ 2, AFQT low, age 22	0.602	0.575	0.6
100	107	Contraception use, educ 2, AFQT low, age 26	0.519	0.515	0.0
101	108	Contraception use, educ 2, AFQT low, age 30	0.443	0.438	0.0
102	109	Contraception use, educ 2, AFQT low, age 34	0.364	0.425	1.4
103	110	Contraception use, educ 2, AFQT low, age 38	0.277	0.349	1.5
104	111	Contraception use, educ 2, AFQT high, age 22	0.672	0.648	0.7
105	112	Contraception use, educ 2, AFQT high, age 26	0.635	0.603	0.8
106	113	Contraception use, educ 2, AFQT high, age 30	0.563	0.549	0.1
107	114	Contraception use, educ 2, AFQT high, age 34	0.531	0.512	0.1
108	115	Contraception use, educ 2, AFQT high, age 38	0.441	0.433	0.0
109	116	Contraception use, educ 3, AFQT low, age 22	0.595	0.622	0.1
110	117	Contraception use, educ 3, AFQT low, age 26	0.607	0.591	0.0
111	118	Contraception use, educ 3, AFQT low, age 30	0.488	0.510	0.0
112	119	Contraception use, educ 3, AFQT low, age 34	0.404	0.505	0.6
113	120	Contraception use, educ 3, AFQT low, age 38	0.366	0.380	0.0
114	121	Contraception use, educ 3, AFQT high, age 22	0.724	0.712	0.2
115	122	Contraception use, educ 3, AFQT high, age 26	0.713	0.688	0.9
116	123	Contraception use, educ 3, AFQT high, age 30	0.638	0.609	0.6
117	124	Contraception use, educ 3, AFQT high, age 34	0.542	0.567	0.2
118	125	Contraception use, educ 3, AFQT high, age 38	0.462	0.492	0.2
<i>wantedness</i>					
119	126	Unintended first birth by 21, share of women (own report), AFQT Q1	0.325	0.319	0.1
120	127	Unintended first birth by 21, share of women (own report), AFQT Q2	0.253	0.244	0.1
121	128	Unintended first birth by 21, share of women (own report), AFQT Q3	0.170	0.186	0.6
122	129	Unintended first birth by 21, share of women (own report), AFQT Q4	0.102	0.106	0.1
<i>method_fail</i>					
123	130	Pill annual failure, AFQT Q1, ages 18-33	0.020	0.028	0.6
124	131	Pill annual failure, AFQT Q2, ages 18-33	0.010	0.016	1.4
125	132	Pill annual failure, AFQT Q3, ages 18-33	0.010	0.010	0.0
126	133	Pill annual failure, AFQT Q4, ages 18-33	0.007	0.005	0.2
127	134	Low-tier annual failure, AFQT Q1, ages 18-33	0.050	0.067	0.4
128	135	Low-tier annual failure, AFQT Q2, ages 18-33	0.024	0.039	1.0
129	136	Low-tier annual failure, AFQT Q3, ages 18-33	0.015	0.027	1.3
130	137	Low-tier annual failure, AFQT Q4, ages 18-33	0.013	0.017	0.3
<i>unint_fbage</i>					
139	146	Unintended share of first births at 14-17, AFQT Q1	0.626	0.648	0.2

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j	Stata j	Moment	Data	Model	Weight share (%)
140	147	Unintended share of first births at 14-17, AFQT Q4	0.696	0.840	1.2
141	148	Unintended share of first births at 18-21, AFQT Q1	0.456	0.422	0.5
142	149	Unintended share of first births at 18-21, AFQT Q4	0.564	0.428	4.0
143	150	Unintended share of first births at 22-25, AFQT Q1	0.294	0.359	0.9
144	151	Unintended share of first births at 22-25, AFQT Q4	0.330	0.323	0.0
145	152	Unintended share of first births at 26-41, AFQT Q1	0.256	0.141	2.4
146	153	Unintended share of first births at 26-41, AFQT Q4	0.142	0.199	3.3

Notes: The 138 weighted moments, in estimation order. “Stata j ” is the moment’s index in the data pipeline’s moment file; the gaps are the seven ratios dropped at load as exact functions of moments already in the vector. “Weight share” is the moment’s contribution to the weighted criterion, in percent of the total. The eight zero-weight method-mix moments are in Table [OA.31](#).

OA.12.2 An untargeted check: the composition of method use

The share of the pill among women who use some method carries zero weight in the estimation, so the model’s prediction for it is an untargeted check in the spirit of [Keane and Wolpin \(2007\)](#). If the ability gradient in first births were produced by which method women choose rather than by how well they use it, the pill share would rise steeply with ability. Table [OA.31](#) shows that it does not. Among childless women aged 18–33 the pill share in the data is flat and if anything mildly declining in ability, from 0.835 in the lowest quartile to 0.824 in the highest, with standard errors of about 0.023; the model produces 0.394 to 0.414 across the same cells, flat as well and at a lower level. Composition therefore works against the within-pill gradient of Section 2.3: Online Appendix [OA.6.1](#) prices the observed method mix at published typical-use failure rates and obtains a Q1/Q4 failure ratio of 1.06 against an observed 2.72. The shortfall in the level reflects the two-method technology: with the estimated failure rates, the unintended births among users in the data require more low-tier users than the data have, and the untargeted mix is where the estimator gives. The Additional Results (Section S.6) re-estimate the five method-cost parameters with the eight pill-share moments switched on and everything else held at the estimate, which brings the share to 0.66–0.72 but raises the criterion on the paper’s moments from 382.3 to 686.6, with the loss concentrated in the unintended-birth blocks: the share of the lowest quartile with an unintended first birth by 21 falls from 0.319 to 0.272 against 0.325 in the data. The experiments of Section 6, re-run at that vector, move by little: the long-acting method at Colorado’s take-up lowers first births by 21 by 7.9 percent against 7.0 at the estimate, equalising adherence by 14.5 against 16.8, and the incidence across quartiles is unchanged. The lowest quartile’s welfare gain from the long-acting method is 5.3 percent against 5.8 at the estimate.

Table OA.31. Untargeted: pill share among method users, by AFQT quartile

	Q1	Q2	Q3	Q4
Ages 18–25, data	0.802	0.777	0.769	0.714
model	0.399	0.397	0.410	0.391
Ages 26–33, data	0.756	0.734	0.724	0.647
model	0.377	0.403	0.420	0.409
Ages 18–33, parity 0, data	0.835	0.843	0.845	0.824
(s.e.)	(0.023)	(0.018)	(0.016)	(0.015)
model	0.394	0.399	0.414	0.398

Notes: Share of the pill among childless women using some method, by AFQT quartile. These moments carry zero weight in the criterion. The first two panels are the eight zero-weight moments of the estimation vector; the third is the method composition built from the NLSY79 contraception battery at parity zero, with bootstrap standard errors.

OA.13 Estimated Structural Parameters

Table OA.32 reports the estimated vector with the sandwich standard errors of Section 4, alongside the vector estimated under the restriction $c_{\theta g} = c_g$ of Section 5; parameters are grouped by their role in the model. 3 parameters of the estimated model carry no standard error (Section 4 and the footnote of Section 5.2): the three adherence increments at 26–33, where adherence is essentially complete in every quartile and the criterion is flat in them. In the restricted model the 26–33 adherence intercept and $\phi_{k,HSD}$ sit on bounds of their boxes and ψ_{lk} for dropouts is flat. The row labels map to the symbols of Section 3 as follows: the baseline-fertility rows are the odds λ_{be} of Section 3.4 by education and age group, reported as the college level and the non-negative education increments; in each adherence block “ a , Q1 intercept” is a_g and “ Δ Q2–Q4” are the cumulative increments Δ_{rg} ; the subscripts *low* and *pill* on κ and ρ^0 are the methods $n = 1$ and $n = 2$, $\Delta\kappa_{pill}$ HS and Coll are the education shifters $\Delta\kappa_{2,e}$, and σ_ω is the scale of the transitory component of the value of a birth; in the schooling panel $\phi_{k,BAC}$ is the college-attendance term and $\phi_{k,HSD}$, $\phi_{k,D}$ and ϕ_{grad} are the continuation wedges of Section 3.3. The adherence parameters are in Table 4 of the main text; Figure OA.7 plots them against the restricted model.

Table OA.32. Estimated structural parameters

Parameter	Full model	Restricted: $c_{\theta g} = c_g$
	Estimate (S.E.)	Estimate (S.E.)
A. Fertility-control technology		
<i>Baseline odds of a first birth with no method, λ_{ge}</i>		
College, ages 14–25	5.900 (3.716)	4.849 (1.946)
College, ages 26–33	0.951 (0.434)	1.682 (0.676)

Table OA.32, continued

Parameter	Full model	Restricted: $c_{\theta g} = c_g$
	Estimate (S.E.)	Estimate (S.E.)
College, ages 34–41	0.488 (0.147)	0.021 (0.699)
Δ HS, ages 14–25	2.479 (2.937)	2.337 (2.273) [§]
Δ HS, ages 26–33	0.583 (0.661)	0.695 (0.761)
Δ HS, ages 34–41	0.042 (0.209)	0.658 (0.665)
Δ HSD, ages 14–25	0.377 (4.376) [§]	1.436 (1.737)
<i>Adherence $c_{\theta g}$, ages 14–17 (logit scale)</i>		
a , Q1 intercept	−0.965 (1.359) [§]	3.580 (6.130) [§]
Δ Q2	1.383 (0.873)	0 ^R –
Δ Q3	0.713 (0.418)	0 ^R –
Δ Q4	4.989 (8.360) [§]	0 ^R –
<i>Adherence $c_{\theta g}$, ages 18–25 (logit scale)</i>		
a , Q1 intercept	0.339 (0.584)	−0.032 (1.470)
Δ Q2	0.723 (0.288)	0 ^R –
Δ Q3	0.477 (0.276)	0 ^R –
Δ Q4	0.699 (0.330)	0 ^R –
<i>Adherence $c_{\theta g}$, ages 26–33 (logit scale)</i>		
a , Q1 intercept	4.410 (14.726) [§]	10.000 [†] –
Δ Q2	1.081 [‡] –	0 ^R –
Δ Q3	0.858 [‡] –	0 ^R –
Δ Q4	0.354 [‡] –	0 ^R –
<i>Failure rate at zero adherence</i>		
ρ_{low}^0	0.993 (0.279)	0.215 (0.140) [§]
ρ_{pill}^0	0.463 (0.146)	0.250 (0.178)
<i>Transitory wantedness shock (units of ΔV)</i>		
σ_ω	0.140 (0.031)	0.325 (0.032)
<i>Method costs (units of ΔV)</i>		
κ_{low}	0.002 (0.017)	0.236 (0.065)
κ_{pill}	0.042 (0.036)	0.220 (0.026) [§]
$\Delta\kappa_{pill}$ HS	−0.006 (0.045)	−0.178 (0.025)
$\Delta\kappa_{pill}$ Coll	−0.050 (0.039)	−0.216 (0.026) [§]
σ (method taste shocks)	0.047 (0.013)	0.022 (0.008)
B. Preferences over children		
<i>Utility of a child, $\omega_{0,\theta} + \omega_1 i^{\omega_2}$, with $\omega_{0,\theta} = \omega_0 + \Delta\omega_{0,\theta}$ and $\Delta\omega_{0,Q1} = 0$</i>		
ω_0	−0.076 (0.076)	−0.308 (0.060) [§]
$\Delta\omega_{0,Q2}$	−0.019 (0.018)	−0.165 (0.035)
$\Delta\omega_{0,Q3}$	−0.104 (0.026)	−0.250 (0.049)
$\Delta\omega_{0,Q4}$	−0.081 (0.024)	−0.266 (0.042)
ω_1	0.270 (0.080) [§]	0.267 (0.060) [§]
ω_2	0.024 (0.020)	0.052 (0.023) [§]
<i>Cost of a child, by education</i>		
$\phi_{k,HSD}$	−0.210 (0.050)	−0.001 [†] –
$\phi_{k,D}$	−0.724 (0.373)	−0.564 (0.410) [§]
$\phi_{k,BAC}$	−0.299 (0.135)	−0.185 (0.107)

Table OA.32, continued

Parameter	Full model	Restricted: $c_{\theta g} = c_g$
	Estimate (S.E.)	Estimate (S.E.)
<i>Terminal utility with no child</i>		
ϕ_{nk} HSD	-0.382 (0.208)	0.300 (0.124)
ϕ_{nk} HS	-0.000 (0.066)	-0.187 (0.041)
ϕ_{nk} Coll	-0.042 (0.088)	-0.215 (0.053)
C. Schooling		
<i>College psychic cost, $\xi_C/\theta^{\omega_C} + \xi_\theta(\theta - 1)$</i>		
ξ_C (level)	-0.567 (0.158) [§]	-0.910 (0.148)
ω_C (ability curvature)	0.061 (0.235)	0.142 (0.208)
ξ_θ (ability slope)	0.196 (0.080)	0.352 (0.094)
ϕ_{grad} (college graduation cost)	-0.343 (0.169) [§]	-0.308 (0.130) [§]
<i>Allowances while enrolled</i>		
HS allowance	128.80 (63.54) [§]	117.86 (44.21)
College allowance	103.22 (81.80) [§]	103.69 (46.74)
<i>Scale of the schooling taste shocks</i>		
σ_C (college entry)	0.277 (0.091) [§]	0.499 (0.086)
σ_{CD} (college continuation)	0.383 (0.207) [§]	0.303 (0.178) [§]
σ_{HS} (high-school continuation)	0.086 (0.016) [§]	0.147 (0.017)
D. Marriage: meeting probability μ		
HSD, ages 18–21	0.557 (0.033)	0.525 (0.027)
HSD, ages 22–25	0.265 (0.031)	0.267 (0.027)
HSD, ages 26–33	0.096 (0.028)	0.125 (0.026)
HSD, ages 34–41	0.024 (0.025)	0.053 (0.028)
HS, ages 18–21	0.660 (0.015)	0.665 (0.015)
HS, ages 22–25	0.377 (0.015)	0.389 (0.013)
HS, ages 26–33	0.190 (0.012)	0.167 (0.011)
HS, ages 34–41	0.081 (0.011)	0.113 (0.010)
Coll, ages 22–25	0.520 (0.021)	0.518 (0.023)
Coll, ages 26–33	0.324 (0.024)	0.338 (0.023)
Coll, ages 34–41	0.128 (0.025)	0.109 (0.025)
E. Labor supply		
<i>Disutility of work ψ_l, by education and age</i>		
ψ_l HSD, ages 14–29	-0.0287 (0.0022)	-0.0327 (0.0022)
ψ_l HSD, ages 30–49	-0.0253 (0.0026)	-0.0253 (0.0027)
ψ_l HSD, ages 54–65	-0.0246 (0.0021)	-0.0236 (0.0027)
ψ_l HS, ages 14–29	-0.0066 (0.0005)	-0.0067 (0.0006)
ψ_l HS, ages 30–49	-0.0061 (0.0004)	-0.0062 (0.0005)
ψ_l HS, ages 54–65	-0.0103 (0.0008)	-0.0114 (0.0010)
ψ_l Coll, ages 14–29	0.0013 (0.0011)	0.0011 (0.0033) [§]
ψ_l Coll, ages 30–49	-0.0019 (0.0006)	-0.0015 (0.0008)
ψ_l Coll, ages 54–65	-0.0049 (0.0007)	-0.0051 (0.0009)
ψ_l HSD, ages 50–53	-0.0233 (0.0023)	-0.0243 (0.0022)
ψ_l HS, ages 50–53	-0.0083 (0.0006)	-0.0091 (0.0007)
ψ_l Coll, ages 50–53	-0.0033 (0.0006)	-0.0031 (0.0009)

Table OA.32, continued

Parameter	Full model	Restricted: $c_{\theta g} = c_g$
	Estimate (S.E.)	Estimate (S.E.)
<i>Interaction with a child in the household</i>		
ψ_{lk} HSD	-0.065 (1.014) [§]	-1.897 [‡] –
ψ_{lk} HS	-0.030 (0.006) [§]	-0.070 (0.058)
ψ_{lk} Coll	-0.012 (0.010)	-0.004 (0.011)
<i>Other</i>		
σ_l (labor taste shock)	0.0078 (0.0010)	0.0097 (0.0014)
ψ_{ret} (ages 58+)	-0.0068 (0.0008)	-0.0077 (0.0008)
Estimated parameters	76	67
Criterion $L(\hat{\Theta})$	382.25	530.74
$D = L_R - L_U$	148.5	on 9 restrictions

Notes: Simulated method of moments on the NLSY79. Standard errors in parentheses, from the sandwich of Section 4 with Ω the cluster-bootstrap covariance of the data moments (1,000 replications resampling women) and W its fixed diagonal; the Jacobian is computed by two-sided finite differences on 160,000 simulated women, and the simulation-noise inflation $1 + 1/S = 1.594$ uses the estimation sample of 10,000 (Online Appendix OA.11). Parameters are grouped by their role in the model. The restricted column imposes $c_{\theta g} = c_g$: the nine adherence increments are fixed at zero and every other parameter is re-estimated. ^R restricted to zero by the null. [†] estimate on a bound of its box, where the estimator is not asymptotically normal; no standard error is reported. [‡] the criterion is flat in the parameter (the weighted Jacobian column is numerically zero), so no standard error is defined. [§] the standard error moves by more than a quarter between finite-difference steps h and $h/2$ and is reported as unstable.

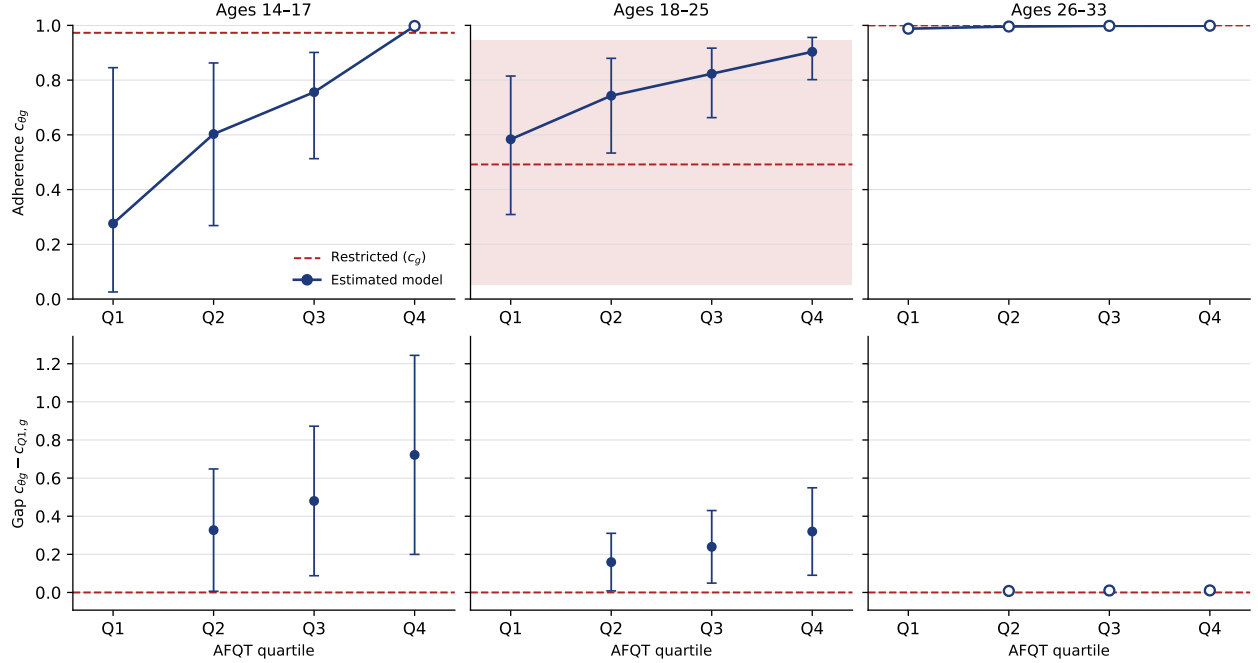


Figure OA.7. Adherence by AFQT quartile, within age group: levels and contrasts
Notes: Top row: estimated adherence $c_{\theta,g}$ (solid, navy) with 95 percent intervals computed on the logit scale, on which it is estimated; the dashed line is the restricted model, in which adherence is common across quartiles, with its 95 percent band shaded. Bottom row: the contrasts $c_{\theta,g} - c_{Q1,g}$ with the standard error of the difference, which nets out the base logit slot that the four levels of an age group share (Panel C of Table 4). The contrasts are what the nested test of Section 5.3 is about, and they are estimated more precisely than the levels above them: a level moves with the base slot, a gradient does not. A hollow marker carries no interval, either because the logistic is saturated near one so that no standard error is defined, or because the logit standard error exceeds three, at which point the interval covers essentially the whole unit interval and reports nothing. The same rule governs the restricted band, which is drawn only at 18–25; at 14–17 the restricted value is saturated (logit standard error 6.1).

OA.14 Counterfactual Outcomes by Ability Quartile

Table OA.33 reports, by AFQT quartile, two outcomes of the experiments of Section 6: first births by age 17, whose by-21 counterpart is in Table 7, and the share of method users on the long-acting method at ages 18–25, zero in every experiment that does not add it to the menu. The incidence matches the aggregate results of the main text: closing the adherence gap acts on the lowest quartile, the price cut on the highest, and take-up of the long-acting method is highest at the bottom, where avoiding a birth is worth most relative to its cost.

Table OA.33. Counterfactual experiments: outcomes by AFQT quartile

			Price cut		LARC menu		
	Baseline	Adherence = Q4	Matched	−10%	Colorado	Pill’s cost	All users
<i>Panel A: First birth by 17, by AFQT quartile</i>							
Q1	0.241	0.156	0.210	0.223	0.212	0.209	0.125
Q2	0.155	0.116	0.123	0.134	0.135	0.133	0.082
Q3	0.072	0.052	0.059	0.065	0.058	0.057	0.022
Q4	0.039	0.039	0.026	0.031	0.035	0.035	0.013
<i>Panel B: LARC share among users at 18–25, by AFQT quartile</i>							
Q1	0.000	0.000	0.000	0.000	0.308	0.343	0.000
Q2	0.000	0.000	0.000	0.000	0.257	0.317	0.000
Q3	0.000	0.000	0.000	0.000	0.279	0.358	0.000
Q4	0.000	0.000	0.000	0.000	0.174	0.323	0.000

Notes: Experiments as defined in Section 6 and columns as in Table 7, run at the estimated parameters with common random numbers, holding each quartile’s 14–17 adherence at its own 18–25 value. Panel A: share of each quartile’s women with a first birth by age 17. Panel B: share of method users at ages 18–25 on the long-acting method, each column taken over its own set of users.

OA.15 Decomposition of the Welfare Gains

Table OA.34 decomposes the ex-ante gains of Table 8 into the windfall to existing users, the saving at the choices made under the experiment, and the remainder. The windfall values the baseline choices at the new costs (Laspeyres); the second saving values the experiment’s choices (Paasche). For a price cut the windfall is a lower bound on the gain by revealed preference, so the remainder is the value of re-optimising: more or better fertility control and everything downstream of it. For the long-acting method the windfall is zero, since nobody pays its cost at baseline, and the saving on new choices is measured against the pill’s cost, the method it nests with. The Additional Results (Section S.7) report the realised-utility version of this measure, and the experiments with 14–17 adherence at its estimated values and the realised-utility version of the measure.

Table OA.34. Ex-ante welfare gains: the windfall to existing users and the remainder

	Total	Windfall to existing users		Total net of windfall	
		at baseline choices	at new choices	at baseline choices	at new choices
<i>Adherence = Q4</i>					
Q1	15.3	0.0	0.0	15.3	15.3
Q2	10.2	0.0	0.0	10.2	10.2
Q3	9.3	0.0	0.0	9.3	9.3
Q4	0.0	0.0	0.0	0.0	0.0
All	8.5	0.0	0.0	8.5	8.5
<i>Uniform cut (matched)</i>					
Q1	24.9	18.4	36.3	4.6	-6.3
Q2	32.4	23.4	48.3	5.9	-7.5
Q3	54.3	33.7	76.6	11.1	-7.6
Q4	49.0	28.9	64.9	11.7	-6.1
All	38.9	25.8	54.7	8.2	-6.9
<i>Uniform cut (-10%)</i>					
Q1	13.2	11.3	16.8	1.6	-2.6
Q2	16.7	14.0	21.7	2.1	-3.4
Q3	26.3	19.6	31.2	4.6	-2.9
Q4	24.1	17.1	26.8	5.1	-1.7
All	19.7	15.4	23.8	3.3	-2.7
<i>LARC (Colorado)</i>					
Q1	5.8	0.0	-1.2	5.8	7.2
Q2	3.8	0.0	-1.2	3.8	5.1
Q3	4.3	0.0	-1.5	4.3	5.9
Q4	1.2	0.0	-0.9	1.2	2.2
All	3.8	0.0	-1.2	3.8	5.1
<i>LARC (free)</i>					
Q1	39.1	0.0	32.5	39.1	3.7
Q2	36.9	0.0	29.0	36.9	4.7
Q3	45.8	0.0	29.6	45.8	9.4
Q4	26.9	0.0	25.3	26.9	1.0
All	36.9	0.0	29.1	36.9	4.6
<i>LARC (all users)</i>					
Q1	24.0	0.0	0.0	24.0	24.0
Q2	20.5	0.0	0.0	20.5	20.5
Q3	28.8	0.0	0.0	28.8	28.8
Q4	12.6	0.0	0.0	12.6	12.6
All	21.2	0.0	0.0	21.2	21.2

Notes: Percent of lifetime consumption, ex ante, by AFQT quartile and for all women, with teen adherence at each quartile's 18–25 level. *Total* is the consumption-equivalent variation of Table 8. *Windfall to existing users* is the expected discounted saving on the methods chosen at baseline when their costs change (at baseline choices) or on the methods chosen under the experiment (at new choices), in the same units; the two *net* columns subtract each from the total. Experiments that change no cost have a windfall of zero.